FastTrack 5.5

XML Request and Result

Contents

1 Introduction

1.1 Scope of the Document

This document contains descriptions of each types of request XML strings that can be passed to the FastTrack API and the respective results produced by FastTrack. This should be used in conjunction with the schemas found in Request.xsd and Result.xsd. With the exception of the Content-Image requests, request queries always return XML strings defined by the Results schema.

This document can be used as a reference for the Project Manager, System Analysis and Developer while implementing the FastTrack modules.

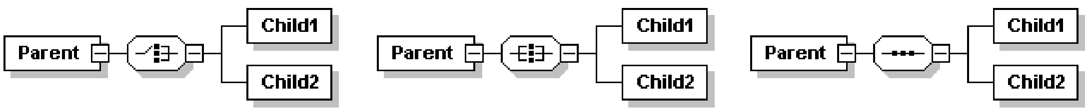
1.2 Reading the diagrams

In the following diagrams, the boxes represent XML elements. Those with a solid border are required, those with a dashed border are optional. Some icons will have a notation such as 1 .. ∞ below them. This indicates that the associated thing can repeat an infinite number of times, but must appear at least once. Inside the box for some elements and below the name is a line that shows the type of that element. These are generally derived datatypes defined in the XML schema documents and based on the built-in XML Schema data types.

There are three kinds of parent-child relationships displayed:

- Choice
- All
- Sequence

These are displayed as follows.



Choice means that only one and any one of the children is selected. All indicates that all of the children are selected. Sequence indicates that all of the children are selected and the order in which they appear is significant.

2 Requests and Results

2.1 Categories of Requests

There are four categories of requests that can be made of the FastTrack Server object.

(1) List

(2) Detail

(3) Content

(4) Interaction

The **List** requests return various lists of entities from the MRC file.

The **Detail** request is used to get the standard and structured information of the entities such as Product, GGPI, GenericItem and so on.

The **Content** requests are used to return documents and images from the MRC file.

The **Interaction** request is used to perform DrugAlert, DuplicateAlert, AllergyAlert and HealthAlert checking.

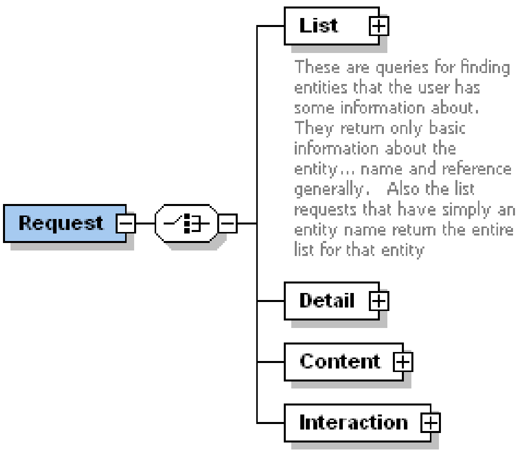


Figure 1 - Request Categories

The Result XML will include a List element, a Detail element, a Content element and an Interaction element depending upon the type of Request.

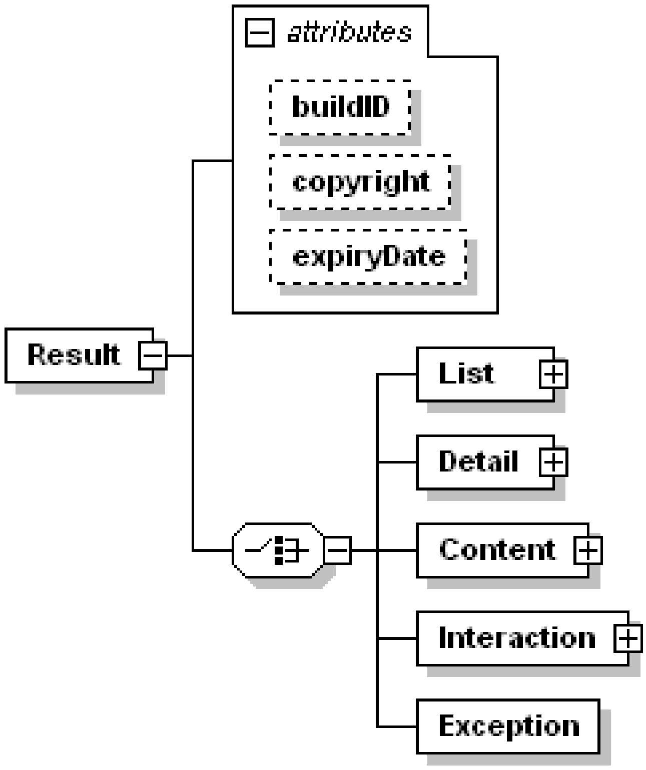


Figure 2 - Result XSD

2.2 Conventions

There are a few conventions for the syntax of FastTrack requests.

2.2.1 ByName

Use the ByName request element to perform a name search. Any entity starting with the search criteria contained in the ByName element will be returned in the result, regardless of case. If the search criteria contains the '%' character, it will be treat as a wildcard, which can be expanded to include any character. For example, '%aspiring%' will return all entities that contain 'aspirin' in the name. If the search criteria is blank, it will be treated as '%', which will return all entities.

ByName request elements are indicated in the diagrams with the type "baseByName".

2.2.2 ByReference

Any ByReference request requires that the reference attribute value be a properly formatted GUID. The ByReference request elements are indicated in the diagrams with the type "baseByReference". The single entity identified by that GUID will be returned in the result. The order of results is dependent upon the request used, but for ByName requests, the results are sorted by the binary representation of the UTF-8 encoded text. For English text, this is ascending alphabetic order.

2.3 Copyright Statement

Every `Result` element returned has the attributes called `copyright`, `buildID`, `expiryDate`. Copyright statement will be something like this "**Powered by MIMS.**

Copyright UBM Medica. All rights reserved." Depending on the region, the copyright statement may be different. This statement **must** be displayed at the appropriate place of the application and all web pages that applied the FastTrack component. BuildID statement shows current build identification number of data file, expired date of this file getting from expiryDate statement.

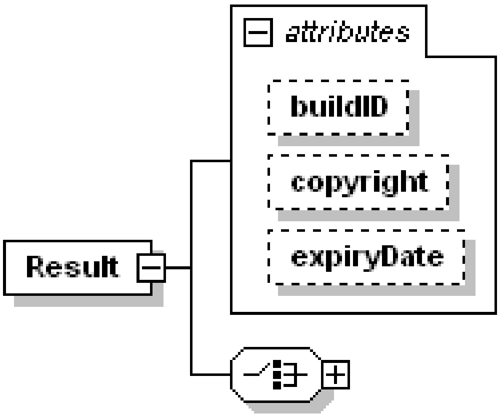


Figure 3 – copyright attribute

```
<Result copyright="Powered by MIMS. Copyright UBM Medica. All rights reserved.">

        …………

</Result>
```

2.4 List

List requests return lists of the respective entity. The Filter element can be added to any single request to limit the number of "records" returned. List requests are often used to return reference values based on information provided by the user, such as, product name, molecule name, etc. List queries may be used to collect data to populate user interface objects like list boxes, check boxes, radio buttons, etc. when developing applications around FastTrack. Each of the sections below describes a list of information that can be retrieved from FastTrack.

```
<Request>
    <List>
        <Product>
            <ByName></ByName>
        </Product>
    </List>
</Request>
```

2.4.1 Product

The list Product request is the only list request that allows more than one value as criteria. Each of the criteria supplied are combined with a logical "and" to produce the final result list of products.

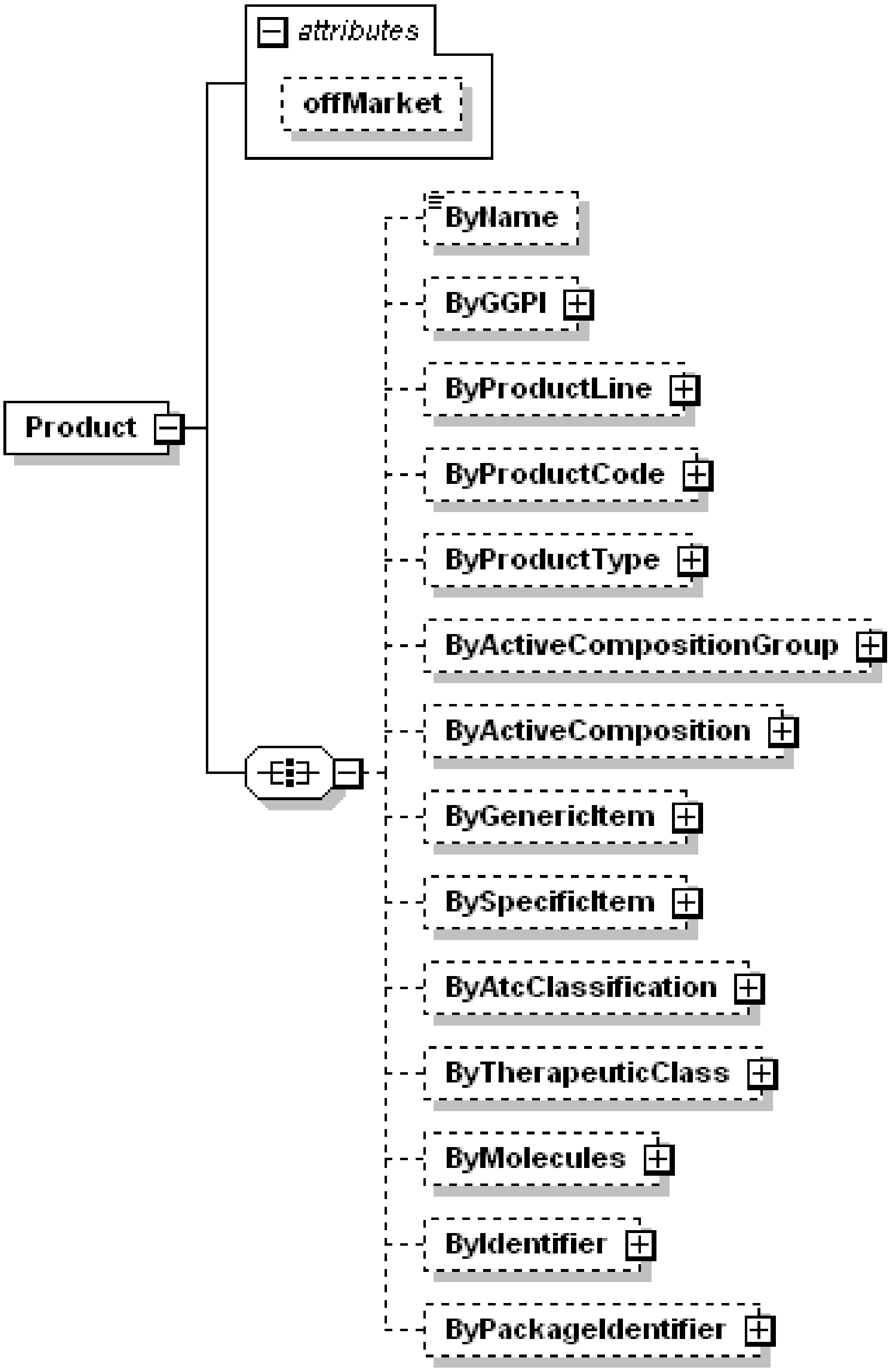


Figure 4 - List Product Request

Although the XML schema is unable to indicate this, at least one of the criteria must be selected in order to make a List Product request.

The following example supplies all of the possible criteria for the list product request. This is not a likely scenario especially because specifying the GGPI, Active Composition, Active Composition Group, Generic Item, and Specific Item are partially redundant. The request can be structured with any one, or any combination of these criteria.

By default, the List-Product request will only return products that are still on the market. The offMarket attribute is used to control the display of off market products. The offMarket attribute only accepts two values, "true" or "false". If the offMarket attribute in the request is "true", all products will be returned. Those products that are off market will have an offMarket attribute with the value "true", while on market products will have the offMarket attribute set to "false". If the offMarket attribute is omitted, only on market products will be returned, but without the offMarket attribute; this is for backward compatibility with older FastTrack versions.

```
<Request>
   <List>
      <Product offMarket="true">
         <ByName>Panadol</ByName>
          <ByProductLine
            reference="{A7E1424F-111D-4674-BA8C-162FA1FBD22C}" />
         <ByActiveCompositionGroup
            reference="{96285061-FA52-497E-8708-F0FF749D4CAE}" />
         <ByActiveComposition
            reference="{52ED3587-A685-4849-92DB-693E6D3D766D}" />
         <ByGenericItem
            reference="{B3E5E9FF-0858-4B5E-E034-080020E1DD8C}" />
         <BySpecificItem
            reference="{2A39437B-437F-45FC-83D9-1FCC9845CC58}" />
         <ByAtcClassification code="N02BE01" />
         <ByTherapeuticClass code="1111" />
         <ByMolecules combination="OR">
            <ByMolecule
            reference="{B714D242-956B-4F26-80A6-A94E3EBCAA74}" />
            <ByMolecule
            reference="{AC340A53-6611-425A-8A75-70E82F00B79C}" />
         </ByMolecules>
      </Product>
   </List>
</Request>
```

2.4.1.1 ByName

This functions just as all other ByName list requests. The supplied text is matched with the content of the Product's name.

```
<Request>
  <List>
    <Product>
      <ByName>%eye drops%\%</ByName>
    </Product>
  </List>
</Request>
```

If the entire list of Products is desired, this element can be included with no text to match all Product names.

```
<Request>
  <List>
    <Product>
      <ByName> </ByName>
    </Product>
  </List>
</Request>
```

2.4.1.2 Other element to list product

List of other element to list product:

ByGGPI, ByProductLine, ByProductCode, ByProductType, ByActiveCompositionGroup, ByActiveCompostion, ByGenericItem, BySpecificItem, By TherapeuticClass, ByIdentifier and ByPackageIdentifier.

These are all single values of criteria supplied to limit the resulting product list to those that match the criteria supplied. For entities such as Active Composition where the Product may be associated to multiple values, as long as the Product is associated with the one supplied in the request, it will be returned regardless of how many other values it is associated with. For the identifier criteria, both the Identifier itself and Identifier Type reference must be supplied. One Identifier may identify only one product or it may identify several products.

2.4.1.3 ByAtcClassification

The ATC code supplied in the request is used to retrieve all the products associated with that ATC code or any descendant of the given ATC code. For instance, if the ATC code N02 is supplied, then products associated with N02, N02AB, or N02BB05 are returned. Supplying an ATC code less than one digit long, greater than seven digits long, or exactly 6 digits long will result in an error. The input code should follow the ATC coding standard which is either one or three or four or five or seven characters. E.g. N02BB05 can search by N or N02 or N02B or N02BB or N02BB05.

2.4.1.4 ByMolecules

Deprecation Warning

The ByMolecules element will be removed in a future version of FastTrack. Developers are advised to avoid its use.

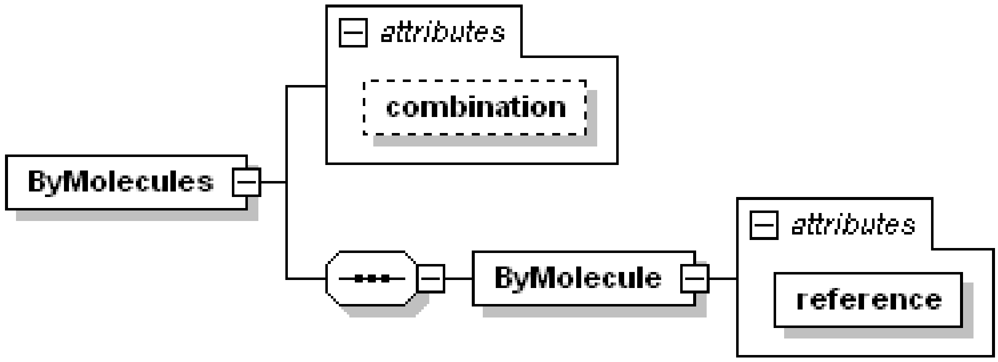


Figure 5 - List ByMolecules Structure

The "ByMolecules" structure is used in several list requests and behaves as follows. A list of molecule references is supplied, one each in the reference attribute of each "ByMolecule" element. The "ByMolecules" element itself has an optional attribute called combination which can have any of the four values AND, OR, XAND, XOR. If the attribute is omitted in the request, it defaults to "AND" behavior.

AND: Those products that contain all of the molecules listed are returned. Products which contain these molecules in addition to other molecules are also returned.

OR: Those products that contain at least one of the molecules listed are returned. Products which contain additional molecule are also returned.

```
<Request>
  <List>
    <Product>
      <ByMolecules combination="OR">
        <ByMolecule reference="{E04944CC-C949-439E-9507-FE28D9965D39}" />
        <ByMolecule reference="{6316B321-CE0D-4F9A-95F3-802DB1E778C2}" />
      </ByMolecules>
    </Product>
  </List>
</Request>
```

XAND: The matching products must contain all of the molecules listed and no additional molecules.

XOR: Those products that contain any one molecule from the list are returned. Products containing more than one molecule, even molecules not listed in the request, are not returned.

2.4.1.5 Result of product list

The result is the typical structure for List results. It contains only a reference and name for each element returned.

```
<Result buildID="" copyright="" expiryDate="1/1/2015">
   <List total="18" from="0" to="18">
      <Product reference="{92648F7E-66AA-4BB8-9904-B1FF2B0B69CF}"
         name="Panadol Caplets" />
      <Product reference="{E5791840-2711-47BA-BD8E-463D35CA03AB}"
         name="Panadol Caplets" />
      <Product reference="{2A476AEB-98BC-47D0-88CE-BAFDE022962E}"
         name="Panadol Gel Caps Tablets" />
      <Product reference="{744BB577-89FD-40E7-B08A-13BEDC7012C2}"
         name="Panadol Gel Caps Tablets" />
      <Product reference="{AE85C343-C895-4916-94F9-A4410DEED348}"
         name="Panadol Gel Caps Tablets" />
      <Product reference="{A0DFBE60-7600-68D1-E034-080020E1DD8C}"
         name="Panadol Gel Tabs" />
      <Product reference="{A0DFD74A-C387-6E0A-E034-080020E1DD8C}"
         name="Panadol Gel Tabs" />
      <Product reference="{A0DFD74A-C38C-6E0A-E034-080020E1DD8C}"
         name="Panadol Mini Caps Tablets" />
      <Product reference="{A0DFD74A-C391-6E0A-E034-080020E1DD8C}"
         name="Panadol Mini Caps Tablets" />
      <Product reference="{A0DFDC2A-EFB8-6E66-E034-080020E1DD8C}"
         name="Panadol Mini Caps Tablets" />
      <Product reference="{A0DFDC2A-EFBD-6E66-E034-080020E1DD8C}"
         name="Panadol Mini Caps Tablets" />
      <Product reference="{BB8927BF-03F2-2B88-E034-080020E1DD8C}"
         name="Panadol Rapid Caplets" />
      <Product reference="{BB8927BF-03F4-2B88-E034-080020E1DD8C}"
         name="Panadol Rapid Caplets" />
      <Product reference="{160136CD-D52A-43FD-9D66-61513EC69DA7}"
         name="Panadol Tablets" />
      <Product reference="{204E02C9-A2F0-4508-B4E3-9329F7041754}"
         name="Panadol Tablets" />
      <Product reference="{5F23B5BB-6A70-4FBD-90D4-371DA3466FE8}"
         name="Panadol Tablets" />
      <Product reference="{8D60DC86-8FF6-47FA-82C5-668F1B97849F}"
         name="Panadol Tablets" />
      <Product reference="{CA059893-D38E-416E-B718-C9DF58F4184B}"
         name="Panadol Tablets" />
   </List>
</Result>
```

2.4.2 Package

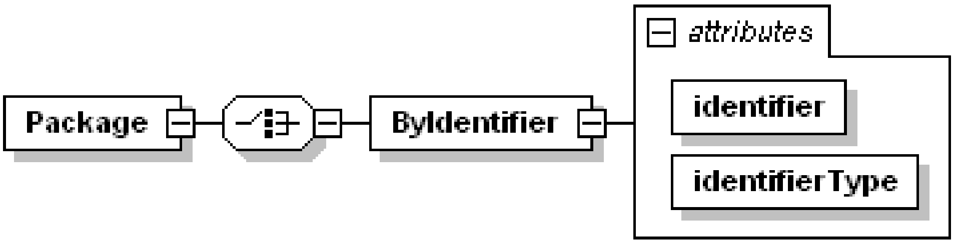


Figure 6 - List Package Request

The List Package By Identifier request allows the user to retrieve references for all the Packages that are assigned a particular Identifier. The types of identifiers supported are listed in the List Package Identifier Type request. When supplying an Identifier, its Type must also be specified.

```
<Request>
   <List>
      <Package>
         <ByIdentifier identifier="456-78-K24T"
            identifierType="{D01A3FF2-AD83-4347-8419-F31EA55E226F}"
            />
      </Package>
   </List>
</Request>
```

2.4.2.1 Result of Package List

Since Packages do not have names, the result is slightly different than other List requests. Only the Package references are returned.

```
<Result buildID="" copyright="" expiryDate="1/1/2015">
   <List total="3" from="0" to="3">
      <Package reference="{A7E9F69F-0686-4ABF-9904-C48832A5DA6B}" />
      <Package reference="{114DCE52-CEFD-477D-8BAC-338C6081296B}" />
      <Package reference="{3140ED87-C6B1-4ED8-9E19-A2C8FCDECB81}" />
   </List>
</Result>
```

2.4.3 ProductLine

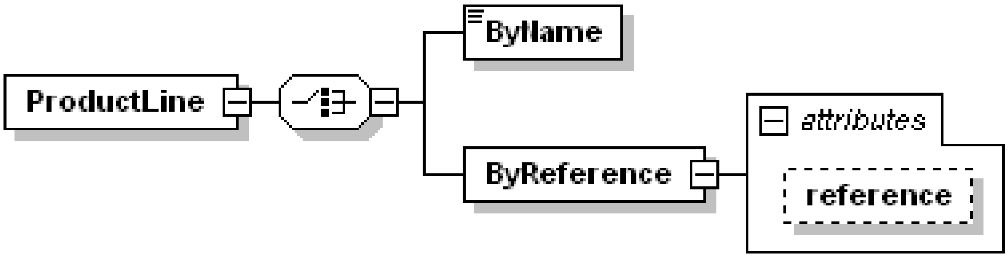


Figure 7 - List ProductLine Request

2.4.3.1 ByName

This is the typical ByName request.

2.4.3.2 ByReference

This is the typical ByReference request.

2.4.3.3 Result of ProductLine

The result is the typical structure for List results.

```
<Result buildID="" copyright="" expiryDate="1/1/2015">
   <List total="2" from="0" to="99999999">
       <ProductLine reference="{5780DAE2-1EBC-4FCB-93F9-0910FA2E00D3}"
name="Tylenol Cold & Flu" />
       <ProductLine reference="{3E8408AA-0BD4-4D47-BEB9-F147B80620F6}"
name="Tylenol Cold & Flu Non-Drowsy" />
   </List>
</Result>
```

2.4.4 Molecule

The Molecule request returns a list of Molecules with their name, references, language codes and name source codes based on the search criteria in the request XML. This will potentially list more than one result for a given molecule if that molecule is supplied with synonyms. In any case, exactly one name for each molecule in the database will be indicated with a primaryName="True" attribute. This name is the one that is used in all other FastTrack results, (except Product Details), and is the name that can be used as criteria for any “ByMoleculeName” request.

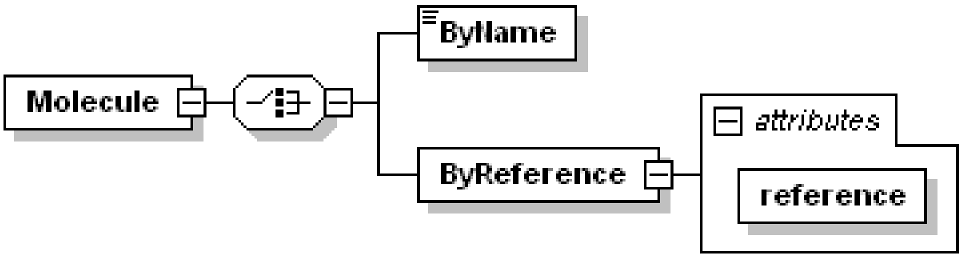


Figure 8 - List Molecule Request

2.4.4.1 ByName

This is like other ByName requests. The results, however, are somewhat different. When a ByName request is executed, all Molecule synonyms are searched, so one Molecule reference may appear in the results of different Molecule ByName requests, each time with a different name.

2.4.4.2 ByReference

This request will list all synonyms for a given Molecule reference. Exactly one XML element in this result will be marked with the primaryName="True" attribute.

2.4.4.3 Result of Molecule List

```
<Result buildID="“ copyright="“ expiryDate="1/1/2015”>
    <List total="4" from="0" to="4">
        <Molecule
            reference="{AC340A53-6611-425A-8A75-70E82F00B79C}"
            name="acetaminophen"
            languageCode="en"
            nameSourceCode="USP"/>
        <Molecule
            reference="{AC5DD1F1-6FEA-4D45-E034-080020E1DD8C}"
            name="acetanilidum"
            languageCode="la"
            nameSourceCode=""/>
        <Molecule
            reference="{48F34BC5-FF9B-488E-95CF-F1D0A4293E80}"
            name="acetazolamide"
            languageCode="en"
            nameSourceCode="INN"/>
            primaryName="True"/>
        <Molecule
            reference="{6DF0221F-30DE-4B3A-8AB5-98FCD6CF75D4}"
            name="acetazolamide sodium"
            languageCode="en"
            nameSourceCode="USP"
            primaryName="True"/>
    </List>
</Result>
```

2.4.5 ActiveCompositionGroup

This request returns Active Composition Group information based on the Molecule name, the Molecule reference or a list of contained Molecules.

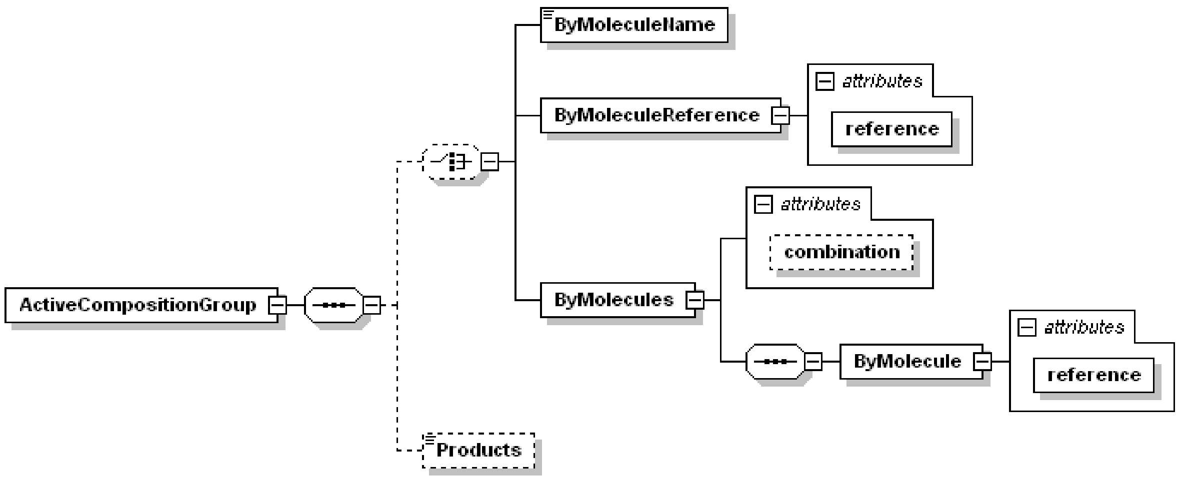


Figure 9 - List ActiveCompositionGroup Request

2.4.5.1 ByMoleculeName

The molecule name, or contained characters of a name, is passed into the request as the value of the ByMoleculeName element. A list of Active Composition Groups is returned where each one contains a Molecule whose primary name (See List Molecule request) matches the supplied criteria.

```
<Request>
   <List>
      <ActiveCompositionGroup>
         <ByMoleculeName>%theophy</ByMoleculeName>
      </ActiveCompositionGroup>
   </List>
</Request>
```

2.4.5.2 ByMoleculeReference

The molecule reference is passed into the request as the value of the reference attribute of the ByMoleculeReference element. A list of the Active Composition Groups that contain the specified Molecule is returned. Each ActiveCompositionGroup element contains a Molecule element for each Molecule contained by that Active Composition Group.

```
<Request>
   <List>
      <ActiveCompositionGroup>
         <ByMoleculeReference
         reference="{07222880-CA28-483A-86F3-E61C52D0990C}"/>
      </ActiveCompositionGroup>
   </List>
</Request>
```

2.4.5.3 ByMolecules

This is the same behavior as for the List Product ByMolecules request. See the List Product request for details.

2.4.5.4 Blank criteria

If neither ByMoleculeName nor ByMoleculeReference nor ByMolecules is specified, then all Active Composition Groups in the file will be returned in one result.

```
<Request>
   <List>
      <ActiveCompositionGroup />
   </List>
</Request>
```

2.4.5.5 Products Option

If the "Products" tag is included in the request, then the result will include the products which contain an item associated with each Active Composition Group in addition to the basic result which is the molecules contained by the Active Composition Group. This option can be combined with any of the criteria above. One example is shown below.

```
<Request>
   <List>
      <ActiveCompositionGroup>
         <ByMoleculeName>theophy</ByMoleculeName>
         <Products/>
      </ActiveCompositionGroup>
   </List>
</Request>
```

2.4.5.6 Result of ActiveCompositionGroup List

The Active Composition Group is described by it's reference value and a list of the Molecules that make up the Active Compositions contained by this group. This example shows an Active Composition Group with just one Molecule. If there were several, then the Molecule element would be repeated. The Molecule elements are returned in the alphabetic order of their names.

```
<Result buildID="" copyright="" expiryDate="1/1/2015">
   <List total="1" from="0" to="1">
      <ActiveCompositionGroup
         reference="{C0EF176B-E4D6-49F0-AD42-4F0D434AD7D1}"
            name="fluoxetine hydrochloride">
         <Molecules count="1">
            <Molecule
            reference="{07222880-CA28-483A-86F3-E61C52D0990C}"
            name="fluoxetine hydrochloride" />
         </Molecules>
      </ActiveCompositionGroup>
   </List>
</Result>
```

With the Products option defined a sample result would look like the following

```
<Result buildID="" copyright="" expiryDate="1/1/2015">
   <List total="1" from="0" to="1">
      <ActiveCompositionGroup
         reference="{D945B0A6-73FF-4318-9A28-A3BF64859394}"
            name="theophylline">
         <Molecules count="1">
            <Molecule
            reference="{3A036BCF-4862-4B2B-A108-2CDBEEE9F9C0}"
            name="theophylline" />
         </Molecules>
         <Products count="5">
            <Product
            reference="{8F1029E1-4D39-4AEB-B422-407E61B0CA06}"
            name="Nuelin SR Tablets" />
            <Product
            reference="{6B593507-A167-4FAE-80B7-2FD545F97F97}"
            name="Nuelin SR Tablets" />
            <Product
            reference="{054DE614-553A-4ED4-806A-ACD9832922CA}"
            name="Nuelin SR Tablets" />
            <Product
            reference="{82ECF141-7F3F-41A4-BC33-C494D11D2720}"
            name="Nuelin Syrup" />
            <Product
            reference="{8E266787-2AEE-4FA3-A8EA-346C5C5F7D5C}"
            name="Nuelin Tablets" />
         </Products>
      </ActiveCompositionGroup>
   </List>
</Result>
```

2.4.6 ActiveComposition

This request returns Active Compositions and has the same possible criteria as the List ActiveCompositionGroup request.

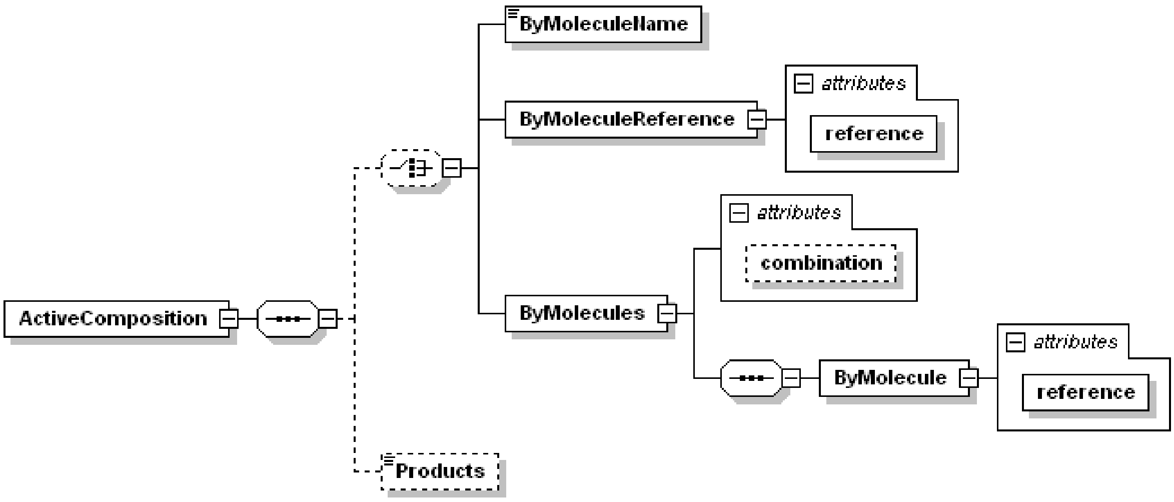


Figure 10 - List ActiveComposition Request

2.4.6.1 ByMoleculeName

The molecule name, or contained characters of a name, is passed into the request as the value of the ByMoleculeName element. A list of Active Compositions is returned where each one contains a Molecule whose primary name (See List Molecule request) matches the supplied criteria.

```
<Request>
   <List>
      <ActiveComposition>
         <ByMoleculeName>theophy</ByMoleculeName>
      </ActiveComposition>
   </List>
</Request>
```

2.4.6.2 ByMoleculeReference

The molecule reference is passed into the request as the value of the reference attribute of the ByMoleculeReference element. A list of the Active Compositions that contain the specified Molecule is returned. Each ActiveComposition element contains a Molecule element for each Molecule contained by that Active Composition.

```
<Request>
   <List>
      <ActiveComposition>
         <ByMoleculeReference
            reference="{07222880-CA28-483A-86F3-E61C52D0990C}"/>
      </ActiveComposition>
   </List>
</Request>
```

2.4.6.3 ByMolecules

This is the same behavior as for the List Product ByMolecules request. See the List Product request for details.

2.4.6.4 Blank criteria

If neither ByMoleculeName nor ByMoleculeReference nor ByMolecules is specified, then all Active Compositions in the file will be returned in one result.

```
<Request>
   <List>
      <ActiveComposition />
   </List>
</Request>
```

2.4.6.5 Products Option

This option has the same XML request syntax and behaves exactly as it does for the List Active Composition Group request. The results are also structured similarly.

2.4.6.6 Result of ActiveComposition List

The Active Composition is described by it's reference value and a list of the Molecules that make up the composition with each Molecule's strength/concentration information. There are a very few Active Compositions where a Molecule's strength is measured in a range. In those cases, the strength attribute indicates the lower bound and the maximumStrength indicates the upper bound of that strength. There are also a very few Active Compositions that have missing strength information. These are used to identify Products where that information is not known. The strength, strengthUnit, perVolume and perVolumeUnit attributes can be empty strings if that information is not available from the country. Also, the perVolume and perVolumeUnit attributes will be non-empty only for Active Compositions used to describe Products where the drug Item is measured by weight or volume. In rare cases, there will be free text data with additional information about the active composition that appears as text contained by the Molecule element.

```
<Result buildID="" copyright="" expiryDate="1/1/2015">
   <List total="1" from="0" to="1">
      <ActiveComposition
      reference="{00BB52CE-8B85-4F70-ACB5-6228DD747F08}">
         <Molecules count="1">
            <Molecule
            reference="{07222880-CA28-483A-86F3-E61C52D0990C}"
            name="fluoxetine hydrochloride" strength="20"
            maximumStrength="" strengthUnit="mg"
            perVolume="" perVolumeUnit="" />
          </Molecules>
      </ActiveComposition>
   </List>
</Result>
```

With the Products option defined a sample result would look like the following

```
<Result buildID=““ copyright=““ expiryDate=“1/1/2015”>
  <List total="3" from="0" to="3">
    <ActiveComposition
    reference="{3D3A6914-D6EF-40F7-BCA3-7F323FA61F74}">
      <Molecules count="1">
        <Molecule
        reference="{AC340A53-6611-425A-8A75-70E82F00B79C}"
        name="paracetamol" strength="250" maximumStrength=""
        strengthUnit="mg" perVolume="" perVolumeUnit=""/>
      </Molecules>
      <Products count="2">
        <Product
          reference="{AAEF6D44-EAC6-5DA1-E034-080020E1DD8C}"
          name="Arfen supp 250 mg"/>
        <Product
          reference="{B693E087-DB11-48F4-E034-080020E1DD8C}"
          name="Poro supp 250 mg"/>
      </Products>
    </ActiveComposition>
    <ActiveComposition
    reference="{52ED3587-A685-4849-92DB-693E6D3D766D}">
      <Molecules count="1">
        <Molecule
        reference="{AC340A53-6611-425A-8A75-70E82F00B79C}"
        name="paracetamol" strength="500" maximumStrength=""
        strengthUnit="mg" perVolume="" perVolumeUnit=""/>
      </Molecules>
      <Products count="6">
        <Product
          reference="{AAEF6D44-EAC8-5DA1-E034-080020E1DD8C}"
          name="Arfen supp 500 mg"/>
        <Product
          reference="{AAEF6D44-D68E-5DA1-E034-080020E1DD8C}"
          name="Dumin tab 500 mg"/>
        <Product
          reference="{AAEF6D44-CC14-5DA1-E034-080020E1DD8C}"
          name="Panadol tab 500 mg"/>
        <Product
          reference="{AAEF6D44-CC2E-5DA1-E034-080020E1DD8C}"
          name="Partamol tab 500 mg"/>
        <Product
          reference="{AAEF6D44-DF5A-5DA1-E034-080020E1DD8C}"
          name="Serimol film-coated tab 500 mg"/>
        <Product
          reference="{AAEF6D44-CC26-5DA1-E034-080020E1DD8C}"
          name="Uphamol tab 500 mg"/>
      </Products>
    </ActiveComposition>
    <ActiveComposition
    reference="{83D40617-DA85-4C76-B637-1FAB9F347022}">
      <Molecules count="1">
        <Molecule
        reference="{AC340A53-6611-425A-8A75-70E82F00B79C}"
        name="paracetamol" strength="160" maximumStrength=""
        strengthUnit="mg" perVolume="" perVolumeUnit=""/>
      </Molecules>
      <Products count="1">
        <Product
          reference="{AAEF6D44-C518-5DA1-E034-080020E1DD8C}"
          name="Acet supp 160 mg"/>
      </Products>
    </ActiveComposition>
  </List>
</Result>
```

2.4.7 GGPI

The Global Generic Product Identifier is a GDDB concept that assigns an identifier to each unique set of Specific Items (See List Specific Item request) that appear in a product. Those products whose Items share the same Specific Items will be assigned the same GGPI.

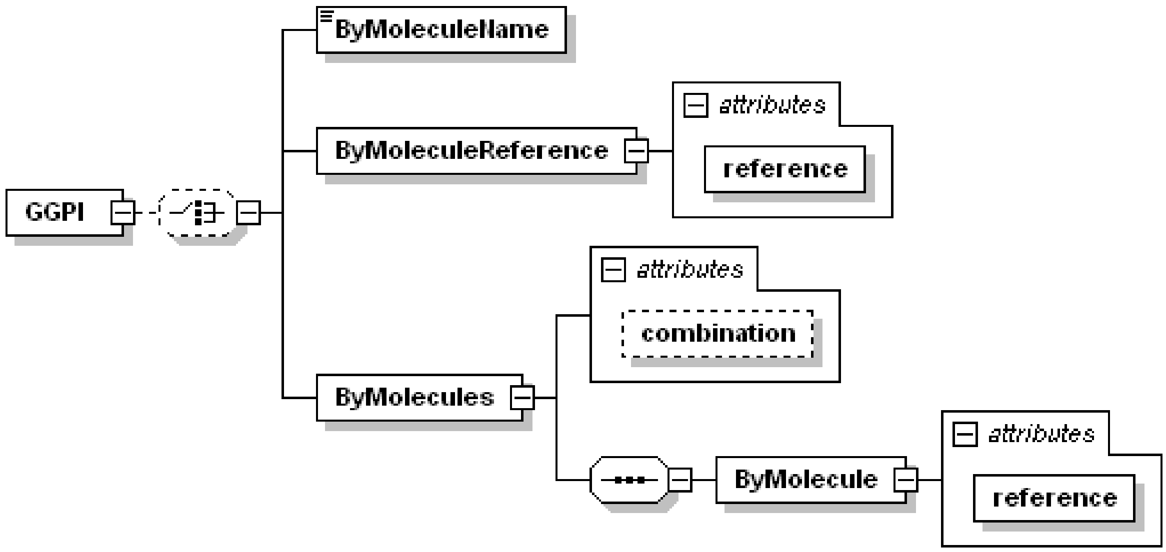


Figure 11 - List GGPI Request

2.4.7.1 ByMoleculeName

This behaves just the same as the previous ByMoleculeName requests described.

2.4.7.2 ByMoleculeReference

This behaves just the same as the previous ByMoleculeReference requests described.

2.4.7.3 ByMolecules

This behaves the same as all other ByMolecules requests. See the List Product request for details.

2.4.7.4 Blank criteria

This is the same as the previous requests described.

2.4.7.5 Result of GGPI List

This is a typical List result containing only a reference and name for each matching result.

```
<Result buildID="" copyright="" expiryDate="1/1/2015">
   <List total="3" from="0" to="3">
      <GGPI reference="" name=""/>
      <GGPI reference="" name=""/>
      <GGPI reference="" name=""/>
   </List>
</Result>
```

2.4.8 GenericItem

The Generic Item is a GDDB concept that identifies a set of active Molecules in a pharmaceutical Item with the specific Dosage Form of the Item.

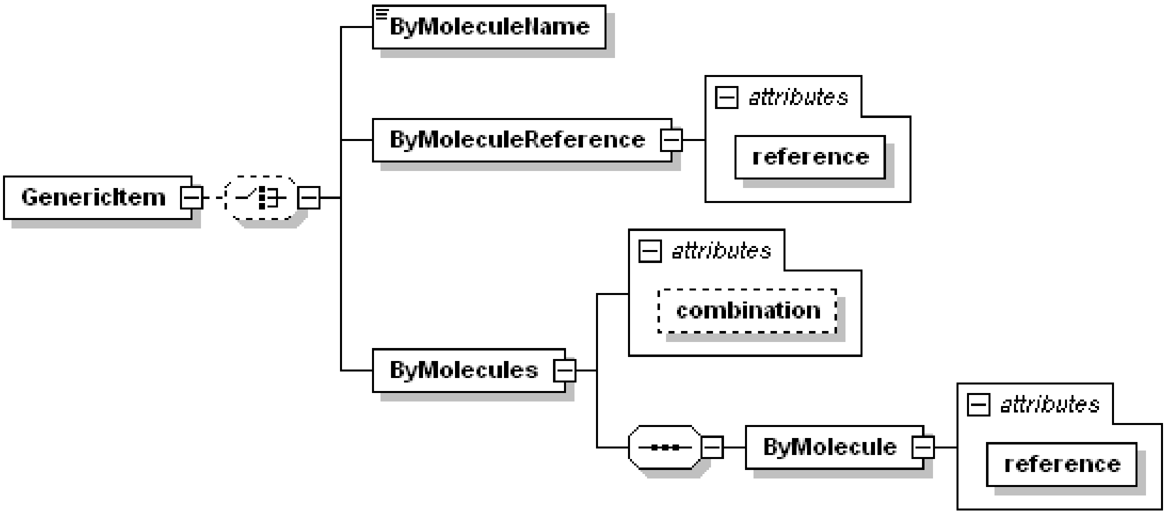


Figure 12 - List GenericItem Request

2.4.8.1 ByMoleculeName

This behaves just the same as the previous ByMoleculeName requests described.

2.4.8.2 ByMoleculeReference

This behaves just the same as the previous ByMoleculeReference requests described.

2.4.8.3 ByMolecules

This behaves the same as all other ByMolecules requests. See the List Product request for details.

2.4.8.4 Blank criteria

This is the same as the previous requests described.

2.4.8.5 Result of GenericItem List

This is a typical List result containing only a reference and name for each matching result.

```
<Result buildID="" copyright="" expiryDate="1/1/2015">
   <List total="3" from="0" to="3">
      <GenericItem reference="" name=""/>
      <GenericItem reference="" name=""/>
      <GenericItem reference="" name=""/>
   </List>
</Result>
```

2.4.9 SpecificItem

The Specific Item is a GDDB concept that is similar to Generic Item, but adds the strength/concentration information for each active Molecule contained in the Item.

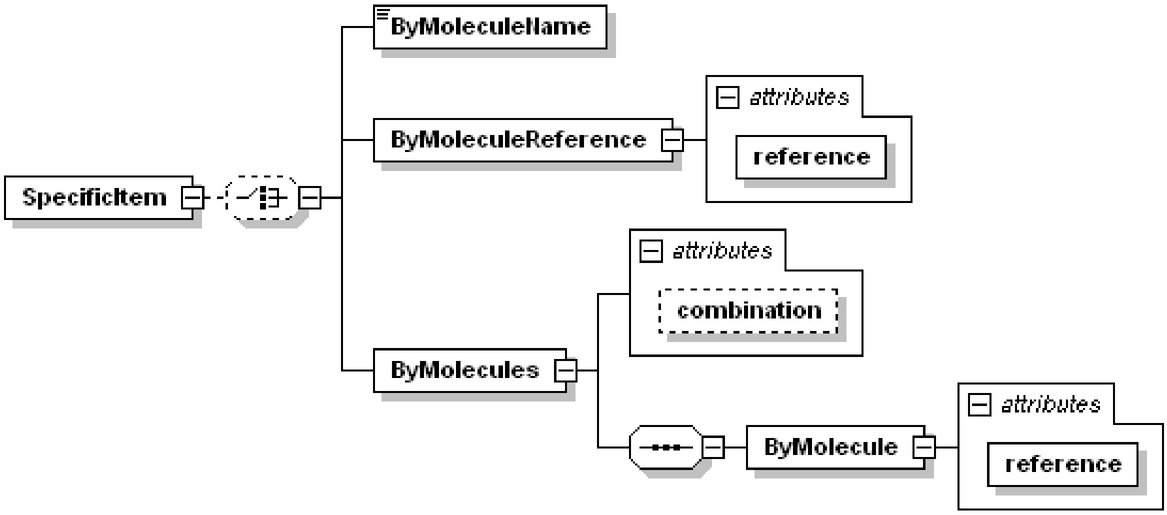


Figure 13 - List SpecificItem Request

2.4.9.1 ByMoleculeName

This behaves just the same as the previous ByMoleculeName requests described.

2.4.9.2 ByMoleculeReference

This behaves just the same as the previous ByMoleculeReference requests described.

2.4.9.3 ByMolecules

This behaves the same as all other ByMolecules requests. See the List Product request for details.

2.4.9.4 Blank criteria

This is the same as the previous requests described.

2.4.9.5 Result of SpecificItem List

This is a typical List result containing only a reference and name for each matching result.

```
<Result buildID=““ copyright=““ expiryDate=“1/1/2015”>
   <List total="3" from="0" to="3">
      <SpecificItem reference="" name=""/>
      <SpecificItem reference="" name=""/>
      <SpecificItem reference="" name=""/>
   </List>
</Result>
```

2.4.10 ATC (Anatomical Therapeutic Class)

This supports the Anatomical Therapeutic Class list published by the World Health Organization. Products are associated with these classes and this list request allows the user to find a desired ATC code, which in turn can be used to find Products.

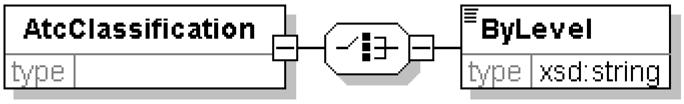


Figure 14 - List AtcClassification Request

2.4.10.1 ByLevel

An ATC Code from the 1st through 4th level is passed in as the value of the ByLevel element, or the ByLevel element is left empty. The request returns a list of all ATC Codes contained by the supplied code at the next level. For instance, if supplied the ATC code "A", "Alimentary Tract and Metabolism", the result contains the A01-A16 classes, but not the A01A, or A01B classes. The code is not case sensitive so a01aa is the same as A01AA. If the ByLevel element is empty, the result is the list of all the top-level codes representing the organ systems. If the supplied criteria is the wrong length for an ATC code, an error is returned in the ResultInformation. The ByLevel element also support a wildcard search, but only in top-level codes.

```
<Request>
   <List>
      <AtcClassification>
         <ByLevel>A01AA</ByLevel>
      </AtcClassification>
   </List>
</Request>
```

2.4.10.2 Result of ATC List

```
<Result buildID="" copyright="" expiryDate="1/1/2015">
   <List total="6" from="0" to="6">
      <AtcClassification
      code="A01AA01" name="Sodium fluoride" />
      <AtcClassification
      code="A01AA04" name="Stannous fluoride" />
      <AtcClassification
      code="A01AA51" name="Sodium fluoride, combinations" />
      <AtcClassification
      code="A01AA30" name="Combinations" />
      <AtcClassification
      code="A01AA03" name="Olaflur" />
      <AtcClassification
      code="A01AA02" name="Sodium monofluorophosphate" />
   </List>
</Result>
```

2.4.11 TherapeuticClass

The Therapeutic Class is a concept supplied by the individual country data providers to classify their own Products. These list requests support searching through the list of country-provided Therapeutic Classes.

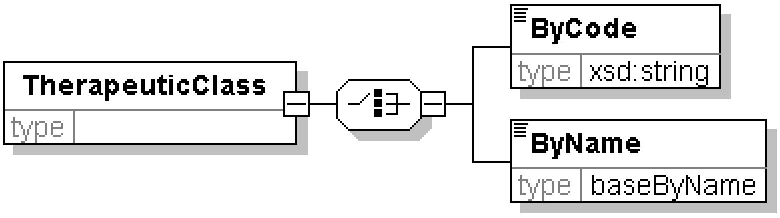


Figure 15 - List TherapeuticClass Request

2.4.11.1 ByCode

The value of the ByCode element is the code to search for. The search behavior is the same as for ByName requests, but the codes that identify Therapeutic classes are searched rather than their names.

2.4.11.2 ByName

This is a typical ByName search.

2.4.11.3 Result of TherapeuticClass List

The result consists of just a code and name for each matching Therapeutic Class.

```
<Result buildID="" copyright="" expiryDate="1/1/2015">
   <List total="3" from="0" to="3">
      <TherapeuticClass code="" parentCode="" name="" />
      <TherapeuticClass code="" parentCode="" name="" />
      <TherapeuticClass code="" parentCode="" name="" />
   </List>
</Result>
```

2.4.12 SubstanceClass

This request returns a list of substance classes based on either the name or the reference.

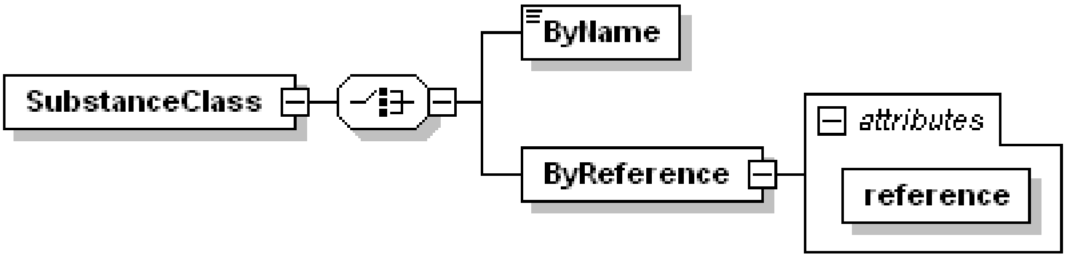


Figure 16 - List SubstanceClass Request

2.4.12.1 ByName

This behaves as all other ByName requests. A list of matching substance classes with their references will be returned.

```
<Request>
    <List>
        <SubstanceClass>
            <ByName>ch</ByName>
        </SubstanceClass>
    </List>
</Request>
```

2.4.12.2 ByReference

A substance class reference is passed into the request as the value of the reference attribute of the ByReference element. The substance class name that matches the reference will be returned.

```
<Request>
    <List>
        <SubstanceClass>
            <ByReference
            reference="{9F815675-B183-4C11-8126-EA03CDD7FD34}"/>
        </SubstanceClass>
    </List>
</Request>
```

2.4.12.3 Result of SubstanaceClass List

```
<Result buildID="" copyright="" expiryDate="1/1/2015">
   <List total="6" from="0" to="6">
      <SubstanceClass
      reference="{1F5D7AE4-8DFA-46ED-9D86-4CB0AA14C062}"
      name="Charcoal" />
      <SubstanceClass
      reference="{4ACE11FA-A70E-4766-BD32-B3AFF59B3C5F}"
      name="Chlorhexidine" />
      <SubstanceClass
      reference="{EDFC9733-54A5-494D-B449-FDB9FF0D8008}"
      name="Chlorobutanol" />
      <SubstanceClass
      reference="{A7BD4CA4-6029-5BB8-E034-080020E1DD8C}"
      name="Cholinergic agents" />
      <SubstanceClass
      reference="{30FEA752-F178-428B-B94B-7E498A0E99D9}"
      name="Chorionic gonadotrophins" />
      <SubstanceClass
      reference="{87AF732C-868E-494E-A021-14EBBAF599B8}"
      name="Chromium compounds" />
   </List>
</Result>
```

2.4.13 Health Issue

This request returns Health Issue entities based on the Health Issue name, reference or a Health Issue Code from a system such as ICD10 as available in the particular data file.

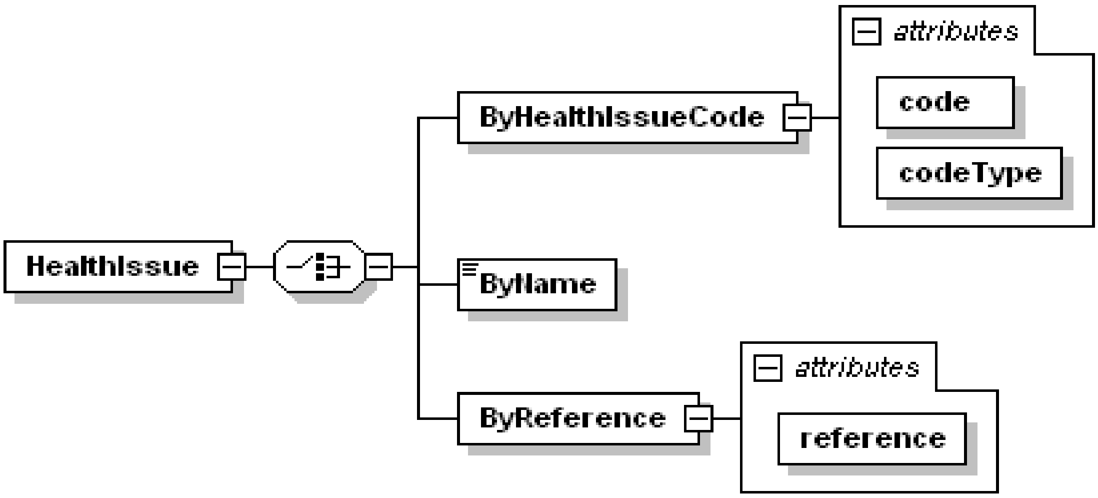


Figure 17 - List Health Issue Request

2.4.13.1 ByName

This behaves as the typical ByName request does.

```
<Request>
    <List>
        <HealthIssue>
            <ByName>hea</ByName>
        </HealthIssue>
    </List>
</Request>
```

2.4.13.2 ByReference

This behaves as the typical ByReference request.

```
<Request>
    <List>
        <HealthIssue>
            <ByReference
                reference="{466502E9-5F90-4A26-BE3C-A01348DF48A3}" />
        </HealthIssue>
   </List>
</Request>
```

2.4.13.3 ByHealthIssueCode

This allows retrieving the Health Issue that accounts for the condition identified by a particular Health Issue Code. The code itself is supplied in the code attribute and the type of that code, (meaning the coding system), is supplied in the codeType attribute. FastTrack supports having several coding systems present in one data file. The code and codeType values are case-sensitive and must be an exact match rather than simply matching just the first characters. Different data files only contain support for specific coding systems. If an unsupported coding system is requested, or there is no Health Issue found for the supplied code, the result is simply an empty list, where the total attribute of the List element is zero.

```
<Request>
    <List>
        <HealthIssue>
            <ByHealthIssueCode code="A01.4" codeType="ICD10"/>
        </HealthIssue>
    </List>
</Request>
```

2.4.13.4 Result of HealthIssue List

```
<Result buildID="" copyright="" expiryDate="1/1/2015">
    <List total="4" from="0" to="4">
        <HealthIssue reference="{55892C33-C0D0-40CA-97BE-FD460D3AF526}"
        name="Headache" />
        <HealthIssue reference="{D53A8C27-E0D9-4F3D-921D-9F45EEE0FD89}"
        name="Hearing loss" />
        <HealthIssue reference="{52990F2A-E337-466C-88D0-166D94D7D5F4}"
        name="Heart block" />
        <HealthIssue reference="{A5CFB283-FC0B-4DC4-8242-21F0F151F434}"
        name="Heart failure" />
    </List>
</Result>
```

2.4.14 HealthIssueCode

This request allows the end user to determine which codes from a given system are associated with a given GDDB Health Issue.

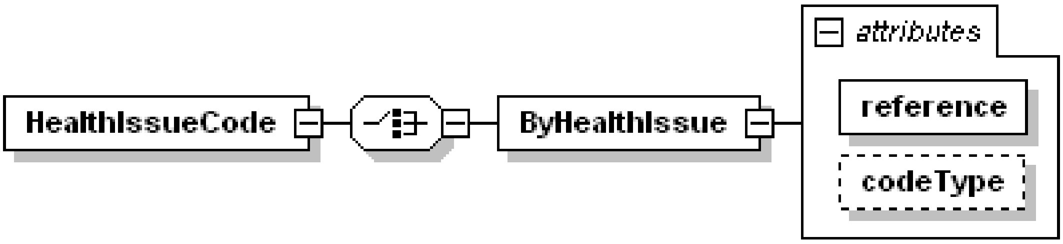


Figure 18 - List HealthIssueCode Request

2.4.14.1 ByHealthIssue

The desired GDDB Health Issue reference is supplied in the reference attribute. The desired code type of the resulting codes is supplied in the codeType attribute.

```
<Request>
  <List>
    <HealthIssueCode>
      <ByHealthIssue
          reference="{B38E1270-6B97-49C2-BE87-B8025420BFF9}"
          codeType="ICD10" />
    </HealthIssueCode>
  </List>
</Request>
```

2.4.14.2 Result of HealthIssueCode List

The result contains those codes that associate with the supplied GDDB Health Issue.

```
<Result buildID="" copyright="" expiryDate="1/1/2015">
   <List total="8" from="0" to="8">
      <HealthIssueCode code="F10"
      codeType="ICD10"
      name="Mental and behavioral disorders due to use of alcohol" />
      <HealthIssueCode code="F10.0"
      codeType="ICD10"
      name="Acute intoxication" />
      <HealthIssueCode code="Y91"
      codeType="ICD10"
      name="Evidence of alcohol involvement determined by level of
intoxication" />
      <HealthIssueCode code="Y91.0"
      codeType="ICD10"
      name="Mild alcohol intoxication" />
      <HealthIssueCode code="Y91.1"
      codeType="ICD10"
      name="Moderate alcohol intoxication" />
      <HealthIssueCode code="Y91.2"
      codeType="ICD10"
      name="Severe alcohol intoxication" />
      <HealthIssueCode code="Y91.3"
      codeType="ICD10"
      name="Very severe alcohol intoxication" />
      <HealthIssueCode code="Y91.9" codeType="ICD10"
      name="Alcohol involvement, not otherwise specified" />
   </List>
</Result>
```

2.4.15 MoleculeNameSource, Unit

These list requests also do not have any available criteria and their results are typical of a list result except that they are identified by a textual code rather than a reference. These code values also will appear in the results of various requests.

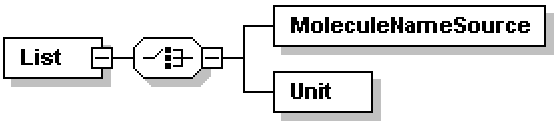


Figure 19 **- List Coded Lookup Values**

2.4.16 Filter-Select-Range

Deprecation Warning

The Filter element will be removed in a future version of FastTrack. Developers are advised to avoid its use.

The filter element can be added to any of the List requests to restrict the number of results that are returned.

This restricts the return list to results in the range defined by the attributes of the Range element. The "to" attribute identifies the first result to be returned and the "from" attribute points to one result past the last one to be returned. The results are indexed starting with zero and the number of results returned is determined by subtracting the "from" attribute value from the "to" attribute value. The following example would retrieve the first through the tenth result. To retrieve the last result in a list, set the range from one less than the total to the list total. If the list total is 1796, (meaning there are 1796 results indexed from 0 to 1795), then set from=1795 and to=1796. Also, if the Range filter is applied, the attributes of the List result element reflect the "from" and "to" attribute values as supplied in the request. For example, if there are only 4 results and the filter specifies from=0 and to=500, the result will say <List total="4" from="0" to="500" >.

```
<Request>
   <List>
      <Molecule>
         <ByName>G</ByName>
      </Molecule>
      <Filter>
         <Select>
            <Range from="0" to="10"/>
         </Select>
      </Filter>
   </List>
</Request>
```

2.4.17 Other List Lookup Values

There are other list lookup values, see following:

DSMRouteOfAdmin, RouteOfAdministration, CompanyType, DocumentType, Form, PackageIndentifierType, ProductIdentifierType

None of these requests have any available criteria and their results are all the typical list result containing only a reference and name for each element. These entities are supplied for informational purposes. Their reference values and names will appear in the results of other requests especially the various detail requests. From this list, only the DSMRouteOfAdmin and DocumentType references can be supplied back to FastTrack as criteria in requests.

The **DSMRouteOfAdmin** result is slightly different from the rest. It includes two additional boolean attributes that indicate which Decision Support Modules make use of that route of administration. The attributes are D2D (Drug to Drug) and D2H (Drug to Health Issue).

The **DocumentType** is used to filter for the <Content> Request. The example of DocumentType can be Brief Monograph, Full Monograph, Generic Monograph and etc.

The **CompanyType** is used in <Detail> of a Product. Eg: Manufacturer or Distributor of a Drug.

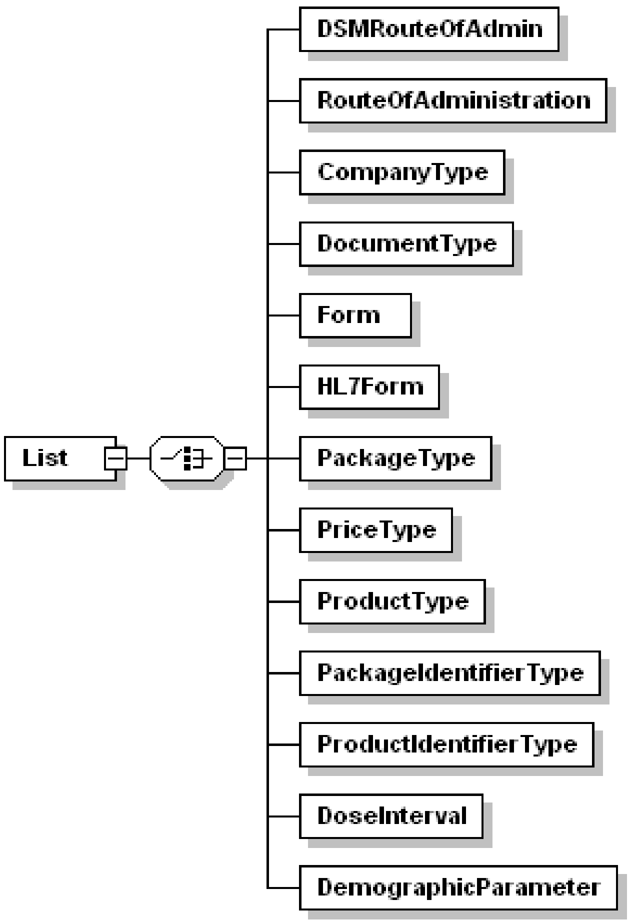


Figure 20 - List Lookup Values

The **PackageIndentifierType** & **ProductIdentifierType** are those that identify the Product of Package. An example of ProductIdentifierType in Taiwan is NHI Code.

The **RouteOfAdministration** describes how a drug is administered for example; **Intra-articular** or **Otic/Aural.**

The **Form** result list the dosage form of a drug. Eg: **Tablet** or **Syrup.**

2.5 Detail

2.5.1 Datafile

The Datafile Detail request supplies all the metadata for the current version of FastTrack and the currently initialized data file. This is a single request with no criteria. This result is returned upon initialization by the CreateServer method call. The request is as follows.

```
<Request>
   <Detail>
      <Datafile/>
   </Detail>
</Request>
```

The result is structured as follows.

```
<Result buildID="" copyright="" expiryDate="1/1/2015">
   <Detail>
      <Datafile countryCode="AU" currencyCode="AUD"
         dataReleaseDate="2008-08-15 08:40:00"
         dataSchemaVersion="5.0.3.1">
         <Copyright />
         <Languages>
            <Language code="en" name="English" />
         </Languages>
         <BuildID>1770</BuildID>
         <IssueDate>September 2008</IssueDate>
         <FastTrack version="5.4.0.0"
            buildDate="2008-11-20 09:23:05" />
      </Datafile>
   </Detail>
</Result>
```

2.5.2 Product

The Product Detail request is the most informative request in FastTrack. It allows the user to get full information about a Product and retrieve different details selectively.

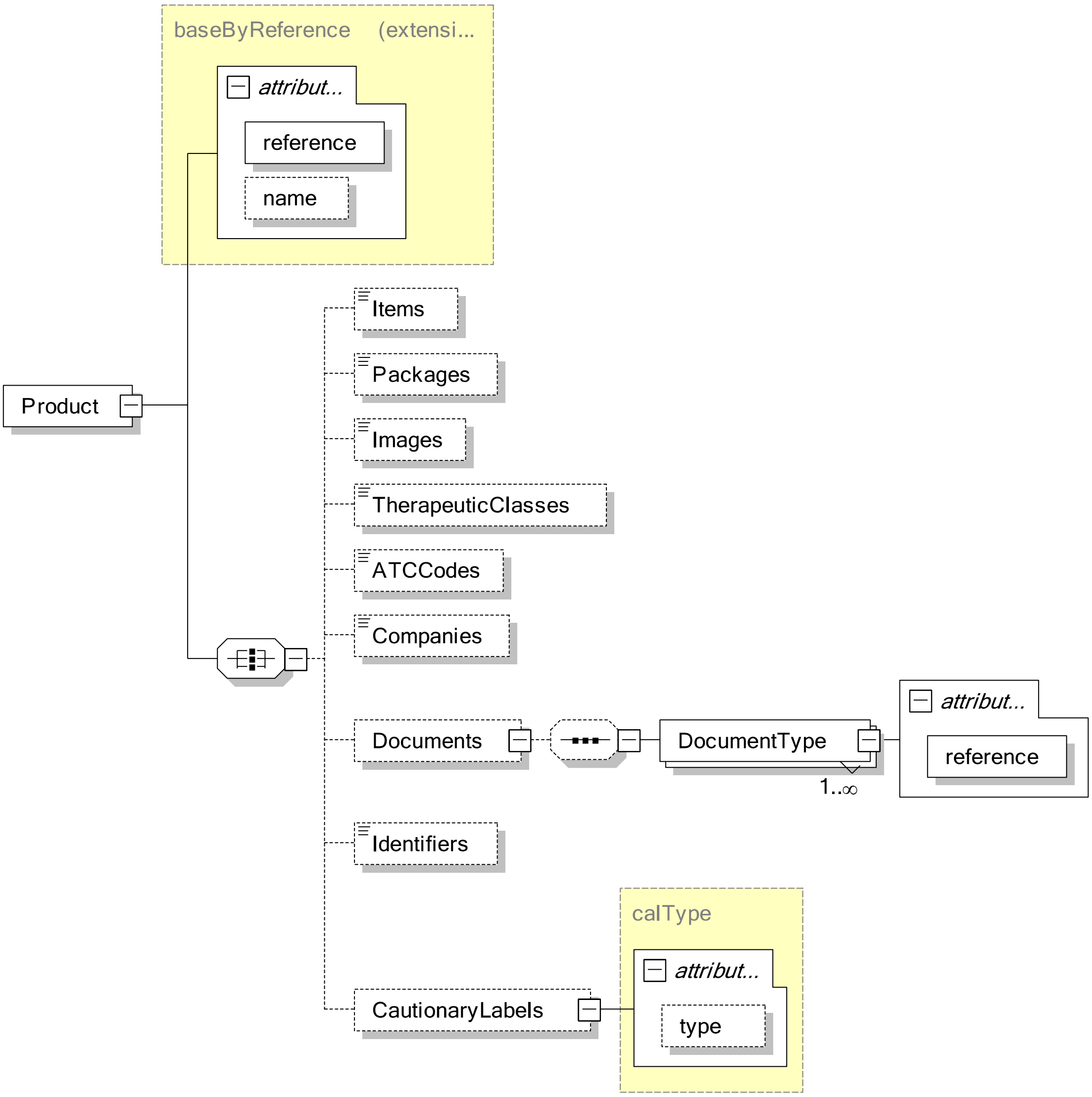


Figure 21 - Product Detail Request

The contents of the result depend on which elements are included in the request. For example, to retrieve only the list of Package references for a given product, issue the following request.

```
<Request>
    <Detail>
        <Product
            reference="{A633FF4F-399F-4938-8999-98CD52E7EBE1}">
            <Packages />
        </Product>
    </Detail>
</Request>
```

To retrieve the list of Package references and get the details of the Items contained by the Product, issue the following.

```
<Request>
   <Detail>
      <Product
         reference="{A633FF4F-399F-4938-8999-98CD52E7EBE1}">
         <Items />
         <Packages />
      </Product>
   </Detail>
</Request>
```

Any, none or all of the listed elements can be included in the request. They can be supplied in any order as indicated by the "all" relationship in the schema diagram. The Documents option is unique in that it allows additional filtering. Documents of several types may be associated with any given product. The DocumentType element controls which documents associated with this product will be returned in the result. The DocumentType element can be repeated to ask for several document types. If the DocumentType element is not included, then documents of all types will be returned.

2.5.2.1 Result of Product Detail

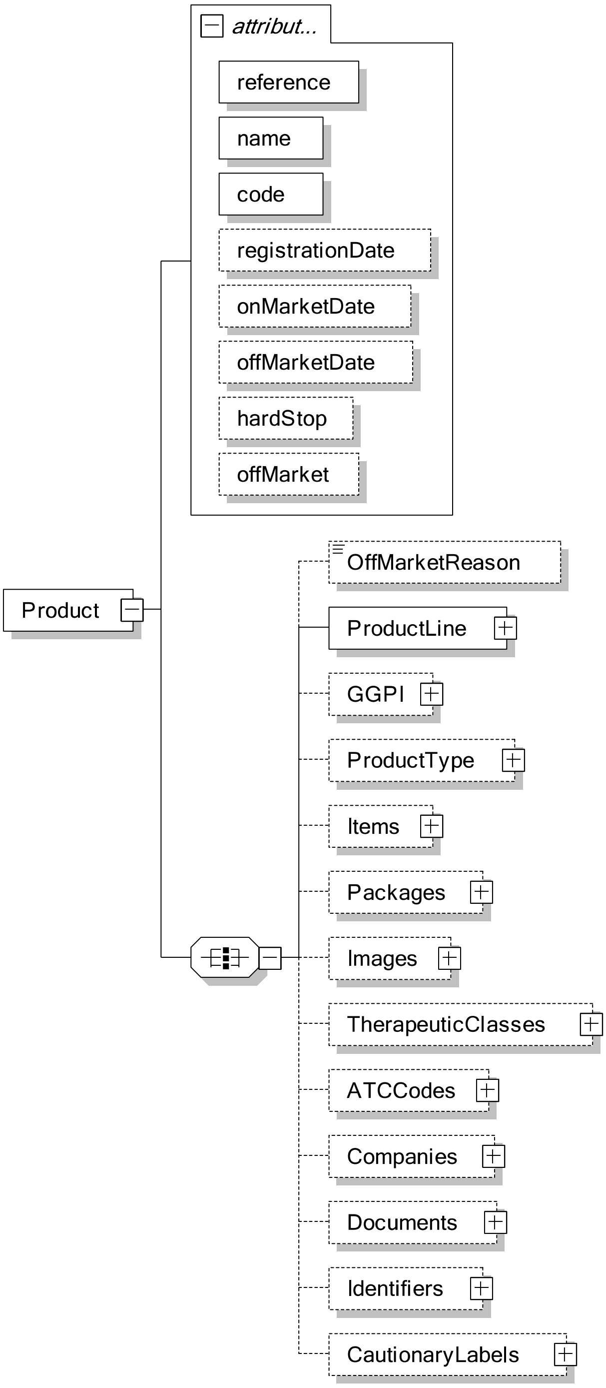


Figure 22 - Product Detail Result

The attributes of the Product element are

reference: the reference identifying this Product.

name: the Product's name.

code: a code identifying this Product uniquely within the country's dataset.

registrationDate: the date on which the Product was registered with the appropriate governing body in the country. This attribute will not be present if the data is not available.

onMarketDate: the date on which the Product became available for patient use. This attribute will not be present if the data is not available.

offMarketDate: the date on which the Product was no longer available. This attribute will not be present for any Products still on the market. If this is specified then a textual OffMarketReason may be supplied in the OffMarketReason child element.

hardStop: if this attribute is present, its value will be "True" and it indicates that the Product was recalled or dispensing of it needs to cease immediately for whatever reason.

The OffMarketReason element will only appear if that data is available and it only contains text content and no attributes. The ProductLine element always appears and contains reference and name attributes to identify the Product Line that this Product is a member of. The GGPI element appears if this Product has been assigned a GGPI. It will contain the reference and name of the GGPI associated with this Product. The ProductType element appears if that value is available for the Product and the element contains the reference and name of the Product Type. The values for the remainder of the elements will only appear if the same element is included in the request. The structure of the optional elements is described below.

2.5.2.2 Optional Results

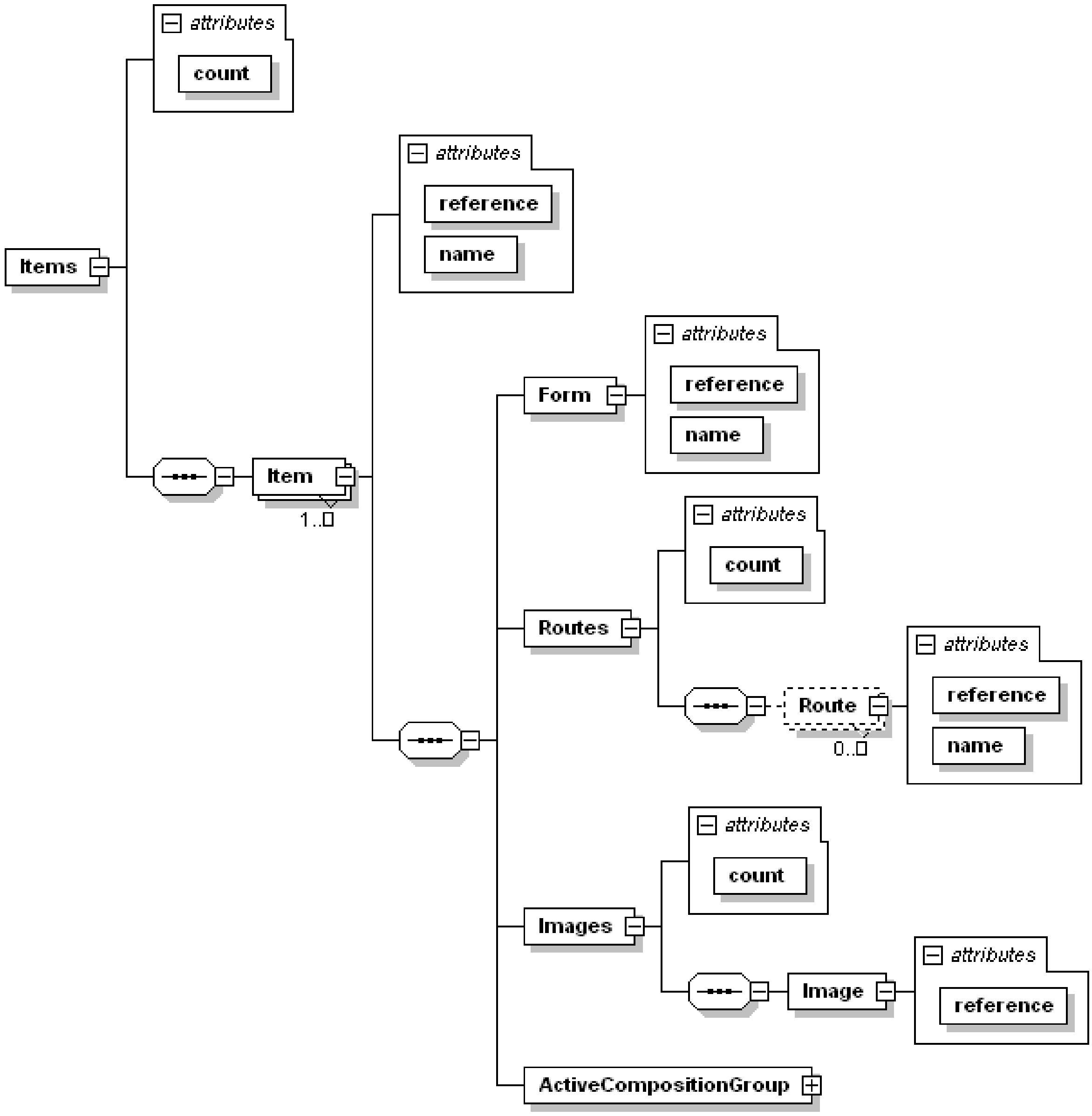


Figure 23 - Product Detail Items Section Result

The Items element contains one Item element for each Item in the Product. The Item element has reference and name attributes. Unlike other name attributes, the name attribute of the Item element may be blank. The Form element has reference and name attributes. The Routes element contains one Route element for each Route Of Administration available for this Item. The Route element contains reference and name attributes. The Images element contains one Image element for each Picture of this Item that is available in the .mrc file. (This Images structure also appears for the Product itself and for Package details.) The ActiveCompositionGroup element contains all of the information on the active Molecules of the Item.

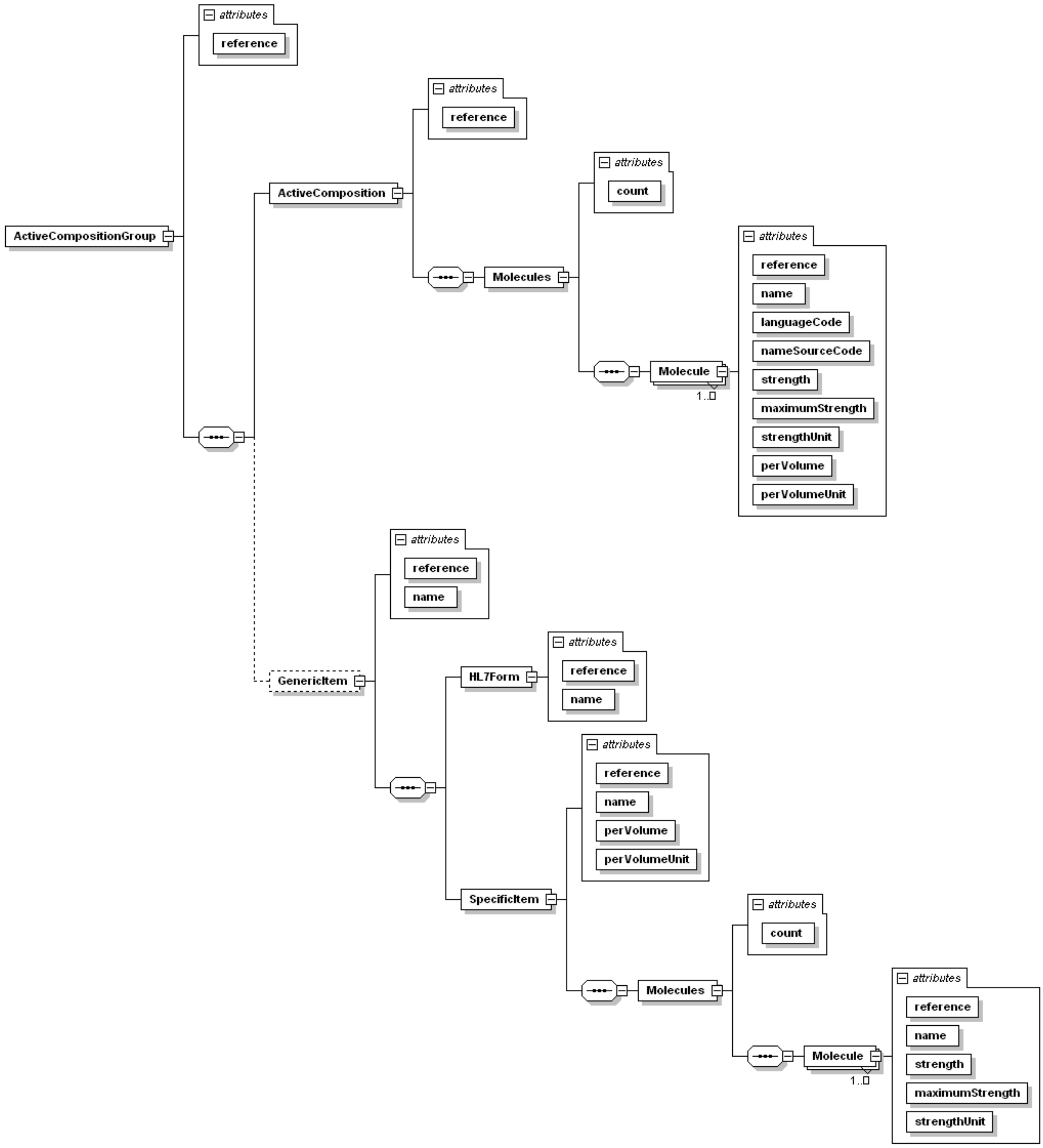


Figure 24 - Product Details Item Active Molecules Result

The ActiveCompositionGroup element contains only the reference attribute. ACGs do not have names. The ActiveComposition element identifies which Active Composition within that Active Composition Group is associated with this Item. This element contains only the reference attribute. The Molecules element contains one Molecule element for each Molecule in that Active Composition. Each Molecule element describes the strength of that Molecule in the item. The Molecule element may have text content containing additional strength/concentration information for that Molecule. The attributes of the element itself are:

reference: the reference value for this Molecule.

name: the name of this Molecule as described in this Product. This may not be the primaryName for the Molecule, but could be one of the synonyms.

languageCode: the languagecode for the language of this Molecule synonym.

nameSourceCode: the code identifying the standards organization supporting this name for this Molecule. This may be an empty string.

strength: the numeric value of the strength of the Molecule in the item. This may be an empty string. If the maximumStrength attribute has a value, then this is the lower bound of a range of strength.

maximumStrength: the numeric value of the upper bound of a range of strength. Usually this is an empty string.

strengthUnit: the Unit Code for the unit of measure of the numeric strength.

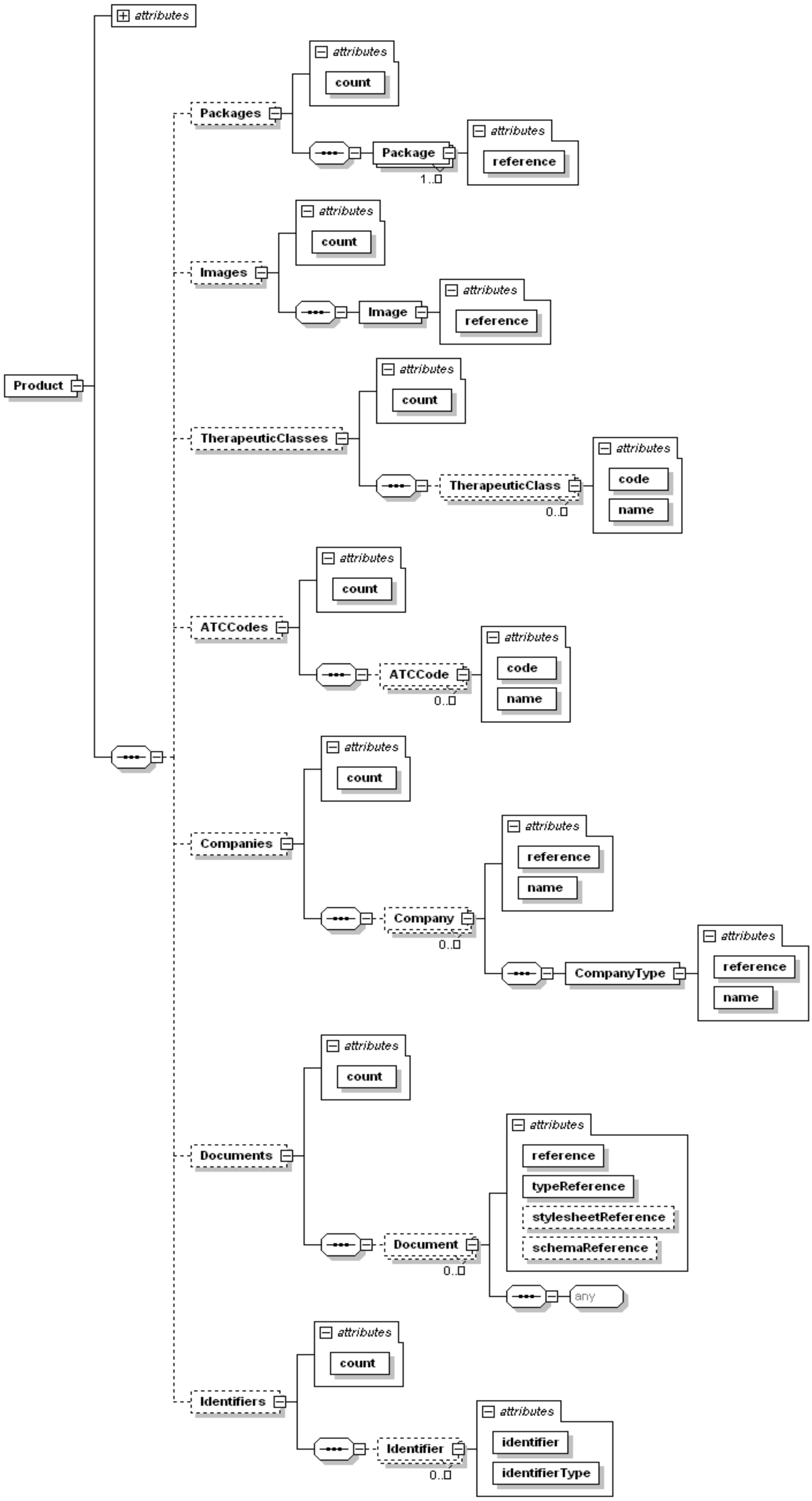
perVolume: the numeric value for the volume or mass containing the active Molecules.

perVolumeUnit: the Unit Code for the unit of measure of the numeric volume or mass. These Unit Codes can be referenced from the result of the List UnitCode request.

The GenericItem element is present if this Item has been assigned to a Specific Item. The GenericItem element contains a reference and name. This element contains the SpecificItem element identifying which Specific Item has been associated with the Item and it contains the HL7Form element with its reference and name attributes. The SpecificItem element also contains a reference and name. The names of the Generic Item and Specific Item are generally, but not always, combinations of the names of the Molecules with the form name and, for Specific Item, the Molecule strengths.

The form listed with the Generic Item comes from the HL7Form list. The Form element for the Item itself is from the Form list.

The Specific Item is also described by the molecules it contains along with their respective strengths or concentrations. The Molecule element within the SpecificItem contains attributes for the reference, moleculeName (the Primary Name of that molecule), the strength, maximumStrength, and strengthUnit code. The perVolume and perVolumeUnit attributes are part of the SpecificItem tag itself since those values are common for all molecules in the Specific Item.



attributes
Product
Packages
count
Package
reference
1..∞
Images
Image
TherapeuticClasses
TherapeuticClass
code
name
0..∞
ATCCodes
ATCCode
Companies
Company
CompanyType
Documents
Document
typeReference
stylesheetReference
schemaReference
any
Identifiers
Identifier
identifier
identifierType

Figure 25 - Product Details Other Optional Results

The Packages element contains one Package element for each different sized Package of this Product. The Package element contains only a reference attribute which can be used for Package Detail requests or Content ByPackage requests to get more information on the Package. There will always be at least one Package for every Product.

The Images element contains one Image element for each Picture available of the Product, or it is empty if the Product has no Images associated. The Image element contains only a reference attribute.

The TherapeuticClasses element contains one TherapeuticClass element for each Therapeutic Class associated with this Product, or it is empty if the Product has no Therapeutic Classes associated. The TherapeuticClass element has code and name attributes.

The ATCCodes element contains one ATCCode element for each ATC Code associated with this Product, or it is empty if the Product has no ATC Codes associated.

The Companies element contains one Company element for every Company associated with this Product. The Company element has only a name attribute and the child element CompanyType indicates the relationship the Company has with this Product such as Manufacturer, Distributor, Importer, etc.. The CompanyType element contains reference and name attributes.

The contents of the Documents element will match the results of the Content – Product result.

The Identifiers element contains one Identifier element for every Identifier assigned to this Product. The Identifiers themselves are just text strings and are supplied in the identifier attribute of the Identifier element. Each Identifier element also includes an identifierType attribute that gives the reference for the type of this identifier.

The CautionaryLabels element contains the list of warning labels associated with the Product. It is permitted to be empty, if the Product has no warning labels. Each CautionaryLabel element makes up one warning label.

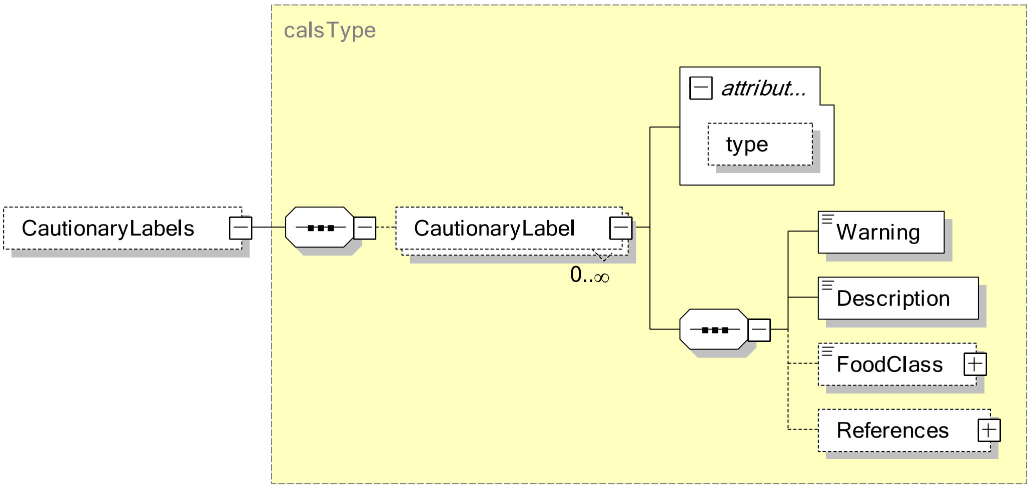


Figure 26 – Product Details CautionaryLabels Result

The CautionaryLabel element has a single attribute **type** which can be either “Food” or “ADR” (Adverse Drug Reaction). CautionaryLabel can contain four constituent elements:

Warning: The warning message for labeling purpose.

Description: A more detailed explanation of the warning, if available. May be identical to the warning message.

References: Literature references that support the warning.

FoodClass: For food related warnings only. This is a MIMS proprietary food classification.

2.5.2.3 Result Example (Multiple Items no GGPI)


```
<Result buildID=““ copyright=““ expiryDate=“1/1/2015”>
 <Detail>
  <Product reference="{B814B3B7-0896-4CA5-AAA3-6D881F8D9906}"
   name="Go Kit" code="35660101">
   <ProductLine reference="{C82DB8ED-DD82-4122-BBD3-E66C7214A762}"
   name="Go Kit"/>
   <Items count="2">
    <Item reference="{9D4C14C7-75B7-4BE4-E034-080020E1DD8C}"
    name="tablets">
     <Form reference="{C6E7CE69-1C3D-4B37-A428-07BE539AB226}"
     name="Tablet, Coated"/>
     <Routes count="1">
      <Route reference="{E5B64CDE-EBAE-4DBF-ABAC-64FAE8561DE1}"
      name="Oral"/>
     </Routes>
     <Images count="0"/>
     <ActiveCompositionGroup reference="{1A33E805-61E8-4FC9-97B8-
      327628C0E91D}">
      <ActiveComposition reference="{4DC46A6C-4B8F-45E1-B88F-
      0C3565CE431C}">
          <Molecules count="1">
              <Molecule reference="{9800ED75-F3ED-44D5-9F3A-
              2F7162AEFDD9}" name="bisacodyl" languageCode="en"
               nameSourceCode="INN" strength="5" maximumStrength=""
               strengthUnit="mg" perVolume="" perVolumeUnit=""/>
          </Molecules>
      </ActiveComposition>
     </ActiveCompositionGroup>
    </Item>
    <Item reference="{9D4C2050-333A-4D7D-E034-080020E1DD8C}"
     name="sachet">
     <Form reference="{B20CDE66-0C80-42D2-BF4D-A46F46C1DD93}"
     name="Powder, Effervescent"/>
     <Routes count="1">
      <Route reference="{E5B64CDE-EBAE-4DBF-ABAC-64FAE8561DE1}"
      name="Oral"/>
     </Routes>
     <Images count="0"/>
     <ActiveCompositionGroup reference="{6E5E6DEB-7293-4D57-8B0C-
     4A0D8360BADA}">
      <ActiveComposition reference="{3723CCE0-3A1E-4326-9A90-
      D121B21F2EE4}">
          <Molecules count="2">
              <Molecule reference="{D602AF95-34B8-420C-AB73-
              3E4524D58F5B}" name="citric acid" languageCode="en"
              nameSourceCode="" strength="14" maximumStrength=""
              strengthUnit="g" perVolume="" perVolumeUnit=""/>
              <Molecule reference="{17FEBA74-7B6F-4A81-9F46-
              2407E2D72086}" name="magnesium carbonate"
              languageCode="en" nameSourceCode="" strength="7.5"
```

```
              maximumStrength="" strengthUnit="g" perVolume=""
              perVolumeUnit=""/>
          </Molecules>
       </ActiveComposition>
      </ActiveCompositionGroup>
     </Item>
    </Items>
    <Packages count="1">
     <Package reference="{8986F808-9E47-48A4-ABC9-C5FF1733501E}"/>
    </Packages>
    <Images count="0"/>
    <TherapeuticClasses count="1">
     <TherapeuticClass code="130" name="Radiographic agents and
     bowel preparations"/>
    </TherapeuticClasses>
    <ATCCodes count="2">
     <ATCCode code="A06AB02" name="Bisacodyl"/>
     <ATCCode code="A06AD10" name="Mineral salts in combination"/>
    </ATCCodes>
    <Companies count="1">
     <Company reference="{68BD8D47-EB51-477D-9B8B-19A1E6B394CE}"
     name="Pharmatel">
      <CompanyType reference="{C0C5A20A-A63D-4B02-A997-
      B333AB6DF9F9}" name="Manufacturer"/>
     </Company>
    </Companies>
   </Product>
  </Detail>
</Result>
```

2.5.2.4 Result Example (Single Item with GGPI and Document)

```
<Result buildID="" copyright="" expiryDate="1/1/2015">
 <Detail>
  <Product reference="{5F23B5BB-6A70-4FBD-90D4-371DA3466FE8}"
   name="Panadol Tablets" code="03910104">
   <ProductLine reference="{A7E1424F-111D-4674-BA8C-162FA1FBD22C}"
    name="Panadol"/>
   <GGPI reference="{B3E5E9FF-0857-4B5E-E034-080020E1DD8C}"
    name="paracetamol 500mg Oral Tablet"/>
   <Items count="1">
    <Item reference="{9D4AE15F-1858-454A-E034-080020E1DD8C}"
     name="">
     <Form reference="{C6E7CE69-1C3D-4B37-A428-07BE539AB226}"
      name="Tablet, Coated"/>
     <Routes count="1">
      <Route reference="{E5B64CDE-EBAE-4DBF-ABAC-64FAE8561DE1}"
       name="Oral"/>
     </Routes>
     <Images count="0"/>
     <ActiveCompositionGroup
      reference="{96285061-FA52-497E-8708-F0FF749D4CAE}">
      <ActiveComposition
       reference="{52ED3587-A685-4849-92DB-693E6D3D766D}">
       <Molecules count="1">
        <Molecule
         reference="{AC340A53-6611-425A-8A75-70E82F00B79C}"
         name="paracetamol" languageCode="en"
         nameSourceCode="INN" strength="500" maximumStrength=""
         strengthUnit="mg" perVolume="" perVolumeUnit=""/>
       </Molecules>
      </ActiveComposition>
      <GenericItem
       reference="{B3E5E9FF-0858-4B5E-E034-080020E1DD8C}"
       name="paracetamol Oral Tablet">
       <HL7Form
        reference="{4704A731-292E-478E-B1F4-E8784F46D3C4}"
        name="Oral Tablet"/>
```

```
        <SpecificItem
         reference="{B3E5E9FF-0859-4B5E-E034-080020E1DD8C}"
         name="paracetamol 500mg Oral Tablet" perVolume=""
         perVolumeUnit="">
         <Molecules count="1">
          <Molecule
           reference="{AC340A53-6611-425A-8A75-70E82F00B79C}"
           name="paracetamol" strength="500" maximumStrength=""
           strengthUnit="mg"/>
         </Molecules>
        </SpecificItem>
       </GenericItem>
      </ActiveCompositionGroup>
     </Item>
    </Items>
    <Packages count="1">
     <Package reference="{D034D861-7492-45F6-B4B4-CEAA87D7EBD9}"/>
    </Packages>
    <Images count="0"/>
    <TherapeuticClasses count="1">
     <TherapeuticClass code="52"
      name="Simple analgesics and antipyretics"/>
    </TherapeuticClasses>
    <ATCCodes count="1">
     <ATCCode code="N02BE01" name="Paracetamol"/>
    </ATCCodes>
    <Companies count="1">
     <Company reference="{2522A5A9-4120-423E-B94E-53096E5522BC}"
      name="GlaxoSmithKline Consumer">
      <CompanyType
       reference="{C0C5A20A-A63D-4B02-A997-B333AB6DF9F9}"
       name="Manufacturer"/>
     </Company>
    </Companies>
    <Documents count="1">
     <Document reference="{A0DFBE6E-DFD8-68D8-E034-080020E1DD8C}"
      name="LocalProductData-03910104"
      type="{1D69392A-DB04-40F6-8F42-43A6AD91861B}">
      <LocalData rp="" qty="100" numberOfPacks="1"
       unitsPerPack="100" pbs="" pbsCode="" active="500"
       activeUnits="mg" perVolume="" perVolUnits=""
       indications=""/>
     </Document>
    </Documents>
   </Product>
  </Detail>
</Result>
```

The shaded portion in the preceding example shows the document content itself. This portion of the XML will vary in structure and content for different countries and different document types.

2.5.2.5 Result Example (Single Item with CautionaryLabels)


```
<Result buildID="2498" copyright="Copyright (c) 2014 MIMS"
expiryDate="2014-12-31">
  <Detail>
    <Product reference="{2A0130AB-A294-4EE9-BE57-F427AAAC2F0B}"
name="Pharmaniaga Simvastatin film-coated tab 20 mg" offMarket="false"
code="">
      <ProductLine reference="{A038DEA5-A837-4CF5-B102-49790CB01CC3}"
name="Pharmaniaga Simvastatin" />
      <GGPI reference="{B4356527-A133-482A-E034-080020E1DD8C}"
name="simvastatin 20mg Film Coated Tablet" />
      <ProductType reference="{AACBF624-2452-35B8-E034-080020E1DD8C}"
name="Brandname" />
      <CautionaryLabels>
        <CautionaryLabel type="Food">
          <Warning>Avoid grapefruit while taking this
medicine.</Warning>
          <Description>Avoid grapefruit while taking this
medicine.</Description>
          <FoodClass>Grapefruits</FoodClass>
        </CautionaryLabel>
      </CautionaryLabels>
    </Product>
  </Detail>
</Result>
```

2.5.3 Product By Identifier

The Product By Identifier Detail request behaves just like the Product Detail request except that an alphanumeric identifier and identifierType reference are supplied in the request instead of the Product reference. The identifier used must uniquely identify one product or the request will return with an error in the ResultInfo (described in Section 3 ResultInfo). Identifiers are text strings, allowing any Unicode character, and are case-sensitive.

```
<Request>
   <Detail>
      <ProductByIdentifier
         identifier="102.45.DC"
         identifierType="{DC156CD9-CA25-4D54-82A0-C60892753B73}">
         <Items />
         <Packages />
         ...
      </ProductByIdentifier>
   </Detail>
</Request>
```

2.5.4 Package

The Package Detail request is the only place in FastTrack to get data describing Packages. It has four optional elements for details.

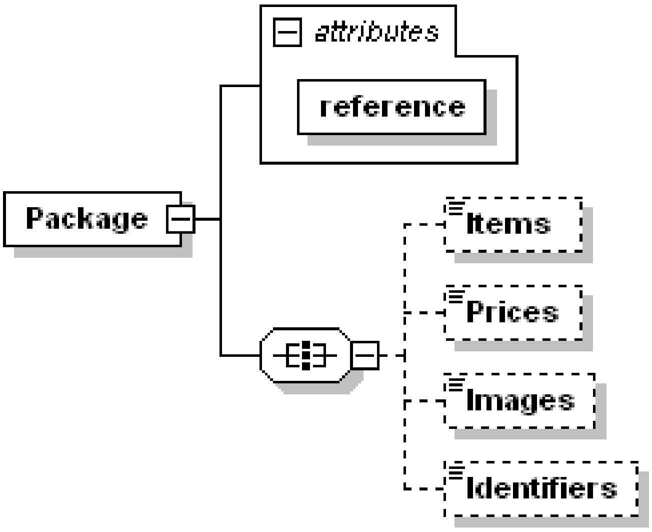


Figure 27 - Package Detail Request

The request XML looks like the following if all optional details are requested.

```
<Request>
   <Detail>
      <Package
      reference="{8986F808-9E47-48A4-ABC9-C5FF1733501E}" >
         <Items />
         <Prices />
         <Images />
         <Identifiers />
      </Package>
   </Detail>
</Request>
```

2.5.4.1 Result of Package Detail

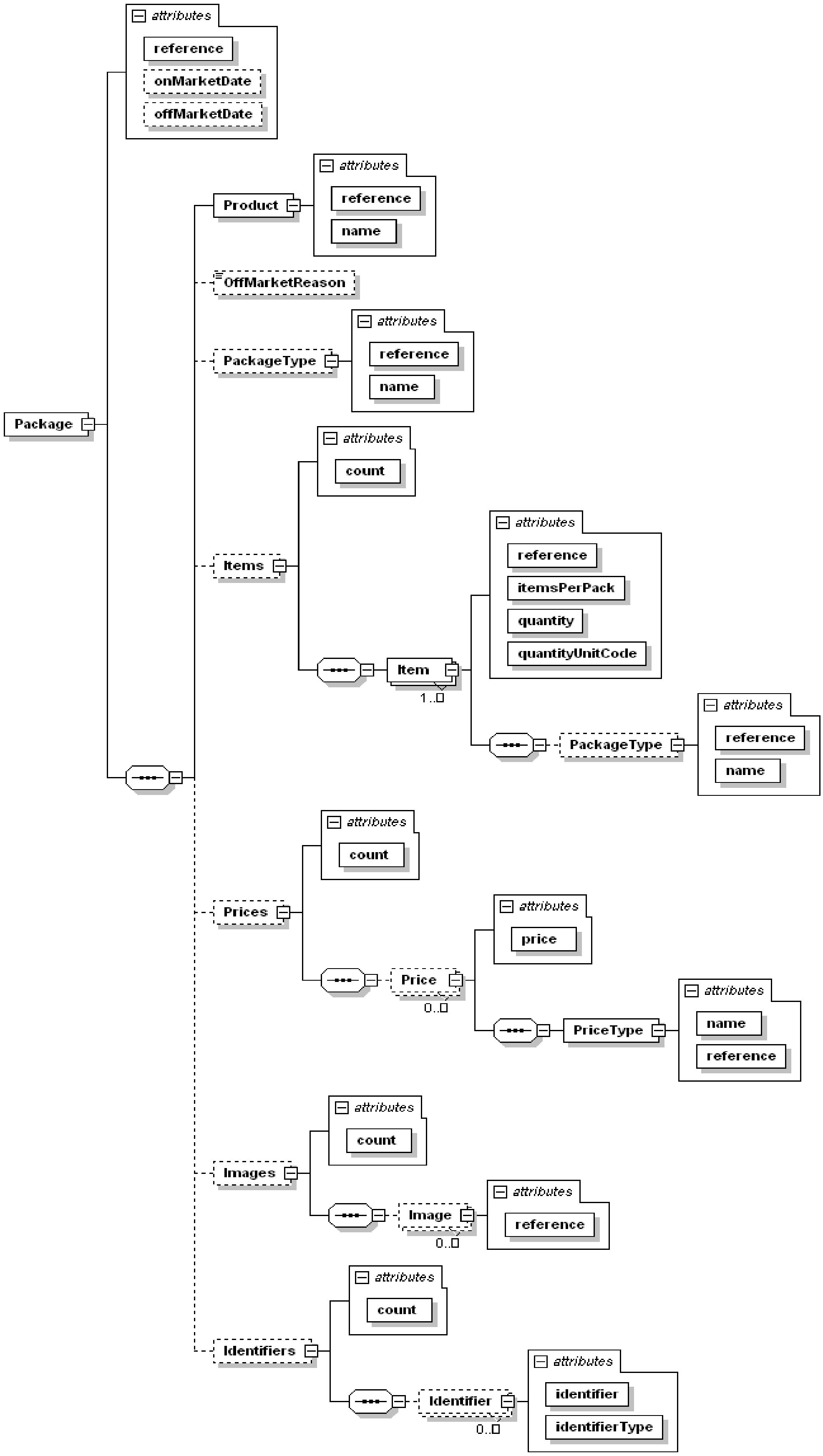


Figure 28 - Package Detail Result

The Package element has these attributes.

reference: the reference identifying this Package.

onMarketDate: the date on which this Package size became available. This attribute is not present if the data is not available.

offMarketDate: the date on which this Package was no longer available. This attribute is not present if the Package is still on the market.

The Product element identifies which Product this Package is for and it has reference and name attributes. The OffMarketReason element will appear only if there is information to display and it contains no attributes, only text describing the reason this Package went off market. The PackageType element will appear if there is data available. It identifies the type of Package this is (i.e. Box, Bottle, Vial, Bag, etc.). The PackageType element contains reference and name attributes. The Items optional element appears if it is requested and contains one Item element for each Item in the Product. The Item element has the following attributes

reference: the reference identifying this Item in the dataset. This can be used to cross-reference to the same Item reference in the Product Details results.

itemsPerPack: the count of how many of this Item are supplied in this Package. This will always be an integer value greater than zero.

quantity: the amount of the substance supplied by each Item of the Package. Items per pack is the "how many"; quantity is the "how much" of the Package. The quantity will be blank for items that are solid, countable forms such as tablets. It will be filled in with a value for forms that are measured such as liquids and powders.

quantityUnitCode: indicates the unit of measure for the quantity value. This will typically either be a mass or a volume unit. It references the Unit Codes from the List UnitCode result.

The Package Type may also be identified for each Item with the PackageType child element. This Package Type may differ from the Package Type for the Package itself. For example, the Package Type of the Package may be "box" while the Package Type for one of the Items in it is "bottle" and another Item has the Package Type of "tube." This would indicate that the package is a box that includes x bottles of the first Item and y tubes of the second Item where x and y are the itemsPerPack values for the respective Items. The values in the quantity and quantityUnitCode attributes would indicate the mass or volume contained by each tube and bottle, (assuming the forms of the items were liquids, creams, powders, etc.).

A Package may have several or no prices supplied for it. If there are no prices for the Package, the Prices element will be empty. Each Price supplied will have a price attribute that displays the actual price. Each price is in the currency identified in the datafile detail result. The PriceType child element has reference and name attributes that indicate the Price Type such as "wholesale", "retail", etc.

Any Pictures available for the Package will be identified in Image elements as children of the Images element. Each Image element will have a reference attribute.

The Identifiers element will contain one Identifier element for each Identifier assigned to this Package. The Identifiers themselves are just text strings and are supplied in the identifier attribute of the Identifier element. Each Identifier element also includes an identifierType attribute that gives the reference for the type of this identifier.

2.5.4.2 Result Example

```
<Result buildID="" copyright="" expiryDate="1/1/2015">
 <Detail>
  <Package reference="{8986F808-9E47-48A4-ABC9-C5FF1733501E}">
   <Product reference="{B814B3B7-0896-4CA5-AAA3-6D881F8D9906}"
   name="Go Kit Bowel Preparation Kit" />
   <Items count="2">
    <Item reference="{9D4C14C7-75B7-4BE4-E034-080020E1DD8C}"
    itemsPerPack="3" quantity="" quantityUnitCode="" />
    <Item reference="{9D4C2050-333A-4D7D-E034-080020E1DD8C}"
    itemsPerPack="1" quantity="" quantityUnitCode="" />
    </Items>
   <Prices count="1">
    <Price price="3.09">
     <PriceType reference="{AA69BA49-FE92-4FDF-808C-FDF1176F5560}"
     name="Consumer Price" />
      </Price>
    </Prices>
     <Images count="0" />
    <Identifiers count="1">
    <Identifier identifier="35660101"
    identifierType="{4F1F52AC-9469-4B25-B7C3-8B2232DFCC47}" />
    </Identifiers>
  </Package>
 </Detail>
</Result>
```

2.5.5 Package By Identifier

The Package By Identifier Detail request functions just like the Package Detail request except that an identifier and identifierType are supplied instead of a Package reference. The identifier used must uniquely identify one Package or the request will return with an error in the ResultInfo (described in Section 3: ResultInfo). Identifiers are text strings, allowing any Unicode character, and are case-sensitive.

```
<Request>
   <Detail>
      <PackageByIdentifier
      identifier="35660101"
      identifierType="{4F1F52AC-9469-4B25-B7C3-8B2232DFCC47}" >
         <Items />
         <Prices />
         <Images />
      </PackageByIdentifier>
   </Detail>
</Request>
```

2.5.6 ActiveCompositionGroup

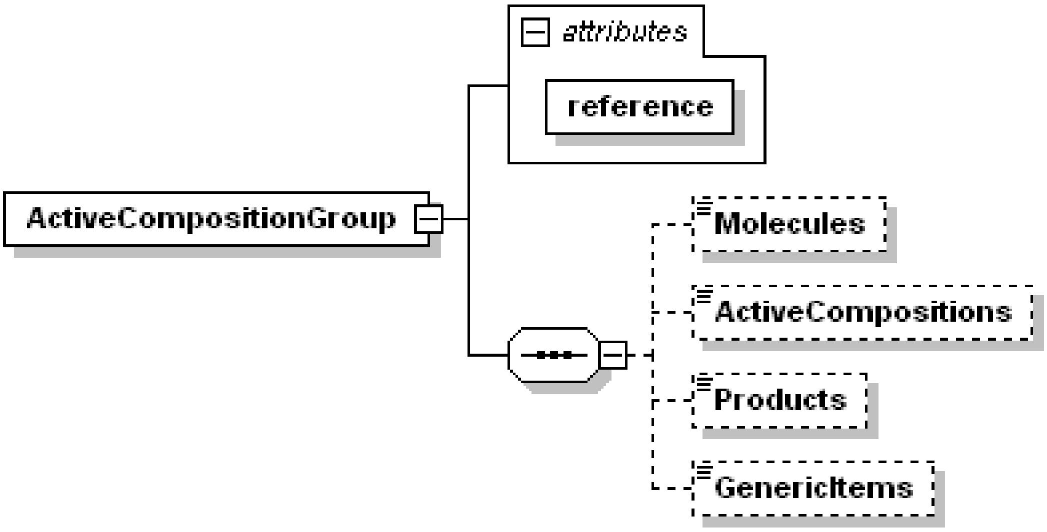


Figure 29 – ActiveCompositionGroup Detail Request

A full request would look like the following.

```
<Request>
    <Detail>
        <ActiveCompositionGroup
        reference="{9D4ADCC2-C618-2669-E034-080020E1DD8C}" >
            <Molecules />
            <ActiveCompositions />
            <Products />
            <GenericItems />
        </ActiveCompositionGroup>
    </Detail>
</Request>
```

2.5.6.1 Result of ActiveCompositionGroup Detail

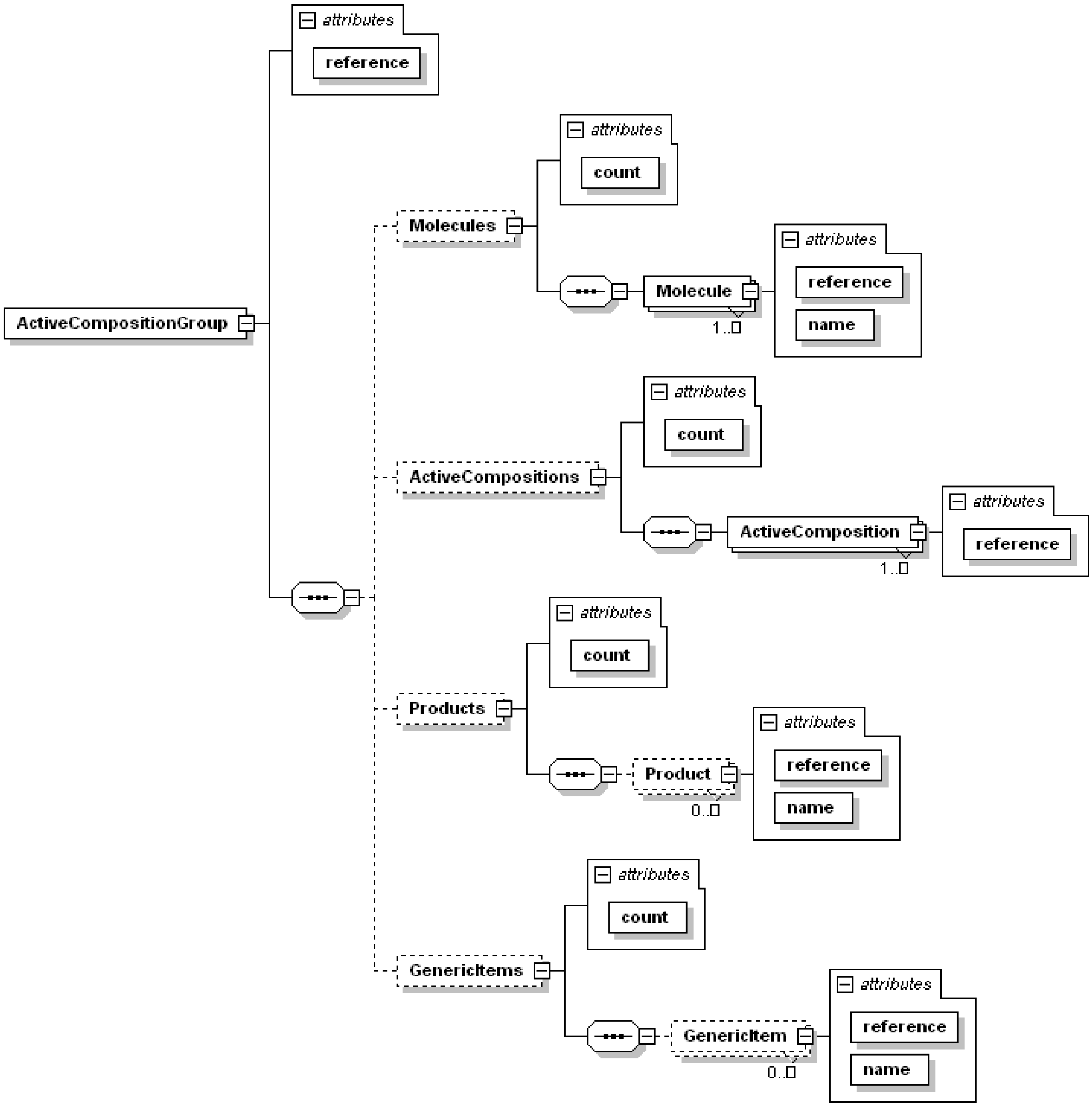


Figure 30 - ActiveCompositionGroup Detail Result

Each of the elements Molecule, Product and GenericItem have reference and name attributes. The ActiveComposition element has only a reference attribute. The Active Composition Group may have no Products associated with it if it is contained in the data file simply for backward-compatibility with older data files to provide continued support for Decision Support Modules. Also, this Active Composition Group may not have any Generic Items associated with it. The result list of Molecules, Active Compositions and Generic Items is the complete list of those entities that belong to the Active Composition Group. The list of Products are those Products where at least one of the Items in that product is assigned an Active Composition that is a member of this Active Composition Group.

2.5.7 ActiveComposition

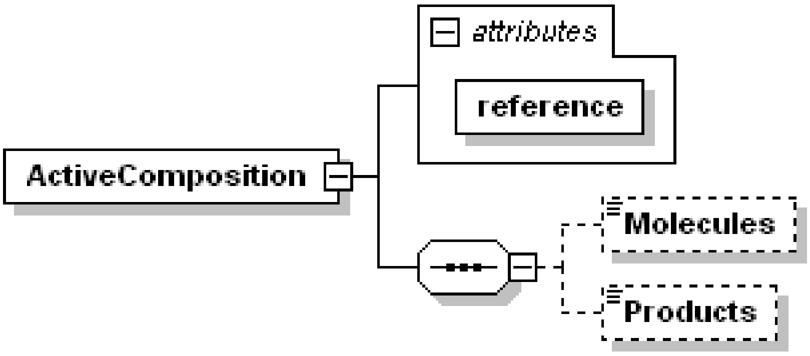


Figure 31 - Active Composition Detail Request

A full request would look like the following.

```
<Request>
   <Detail>
      <ActiveComposition
      reference="{E3D1C284-B49D-46B2-A111-4324F943EB2B}">
         <Molecules />
         <Products />
      </ActiveComposition>
   </Detail>
</Request>
```

2.5.7.1 Result of ActiveCompostion Detail

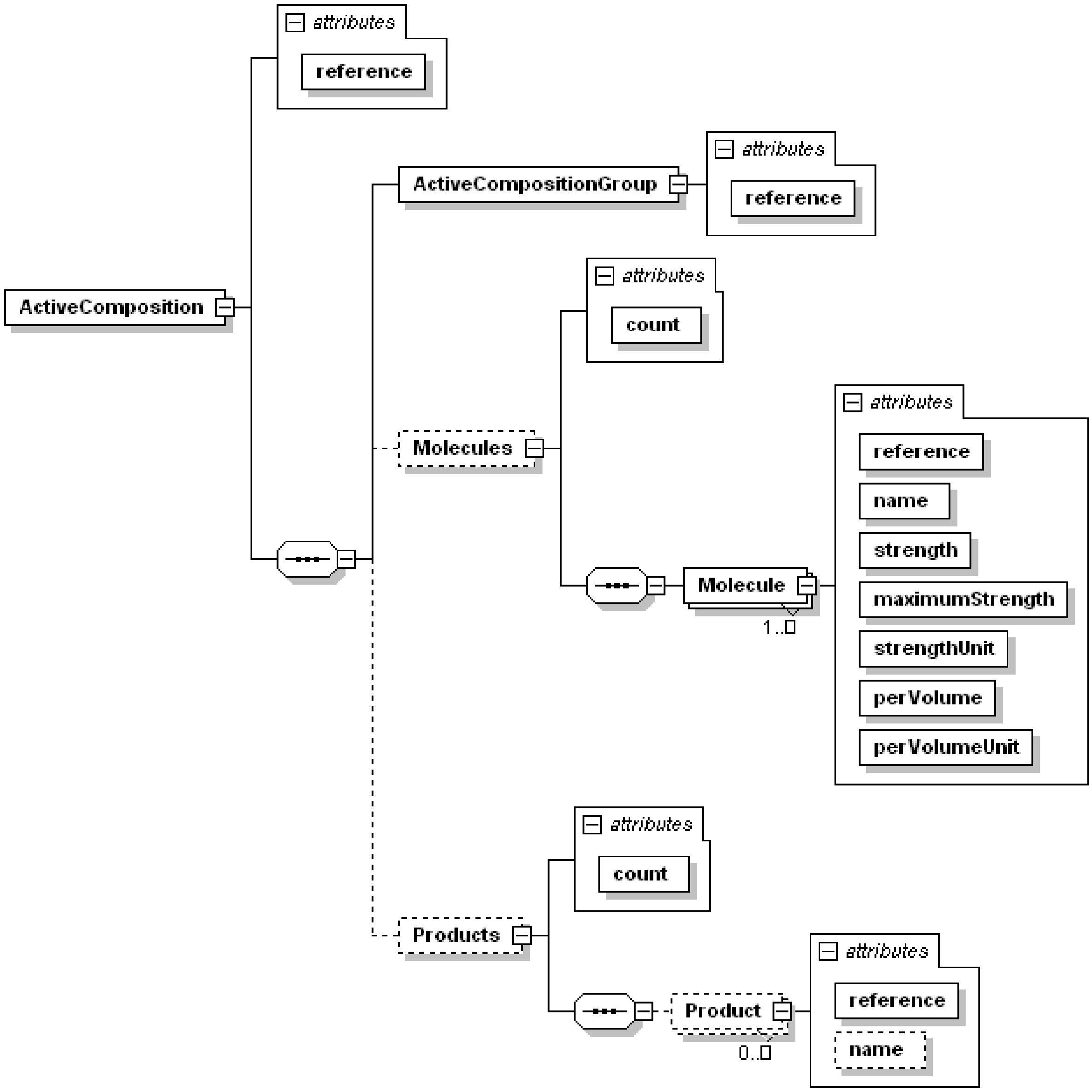


Figure 32 - ActiveComposition Detail Result

The ActiveComposition element contains only the reference attribute identifying the Active Composition requested. The ActiveCompositionGroup element contains a reference attribute identifying which Active Composition Group this Active Composition is a member of. The Molecules optional element contains one Molecule element for each Molecule in this Active Composition. The Molecule element may have text content with additional strength/concentration information and it has the following attributes.

reference: the reference identifying this Molecule.

name: the primary name for the Molecule.

strength: the numeric value of the strength of the Molecule in the Active Composition. This may be an empty string. If the maximumStrength attribute has a value, then this is the lower bound of a range of strength.

maximumStrength: the numeric value of the upper bound of a range of strength. Usually this is an empty string.

strengthUnit: the Unit Code for the unit of measure of the numeric strength.

perVolume: the numeric value for the volume or mass containing the active Molecules.

perVolumeUnit: the Unit Code for the unit of measure of the numeric volume or mass. These Unit Codes can be referenced from the result of the List UnitCode request.

2.5.8 GGPI

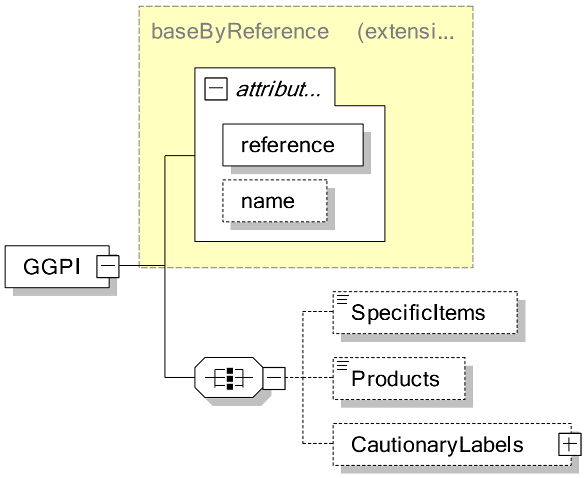


Figure 33 - GGPI Detail Request

A full request would look like the following.

```
<Request>
    <Detail>
     <GGPI reference="{588EEA60-19CA-445B-B7D2-9D1E50D7329E}">
        <SpecificItems />
        <Products />
     </GGPI>
   </Detail>
</Request>
```

2.5.8.1 Result of GGPI Detail

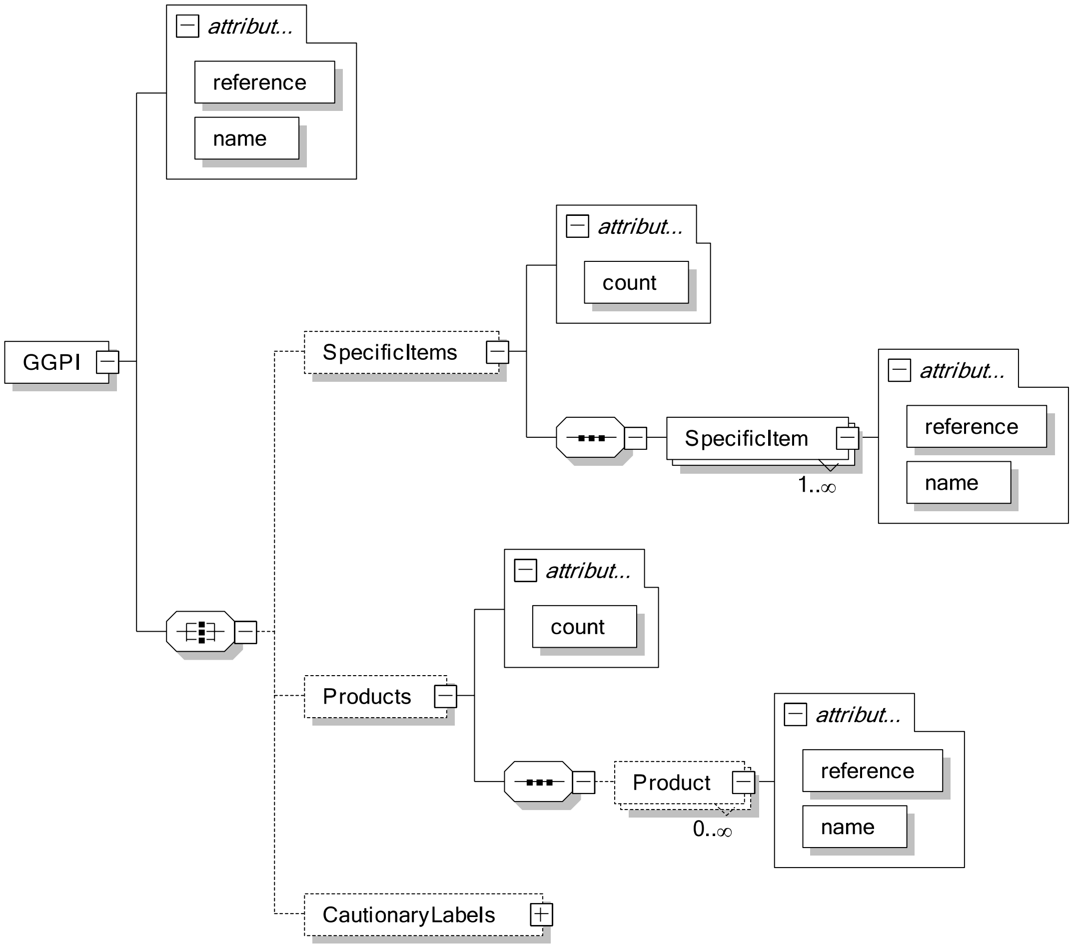


Figure 34 - GGPI Detail Result

The GGPI, SpecificItem and Product elements each contain name and reference attributes. The CautionaryLabels element is the same as that of Product Detail Result, please refer to section 2.5.2.2 for details.

2.5.9 GenericItem

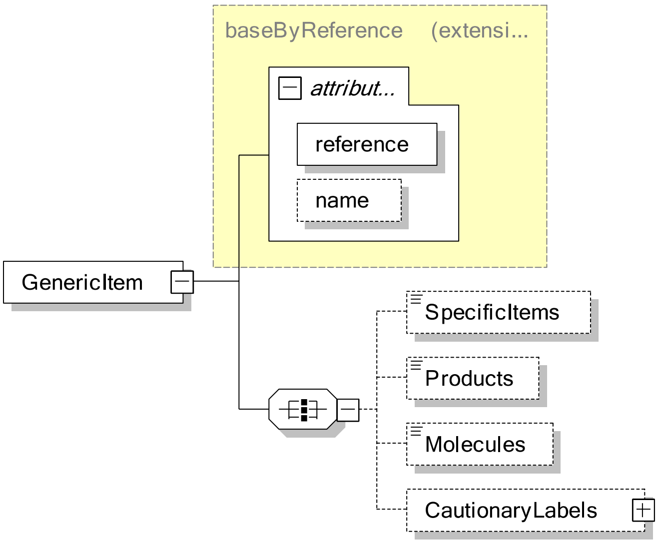


Figure 35 - GenericItem Detail Request

A full request would look like the following

```
<Request>
  <Detail>
    <GenericItem reference="{C0D8E1C7-BDDA-424A-E034-0003BA299378}">
      <SpecificItems />
      <Products />
      <Molecules />
    </GenericItem>
  </Detail>
</Request>
```

2.5.9.1 Result of GenericItem Detail

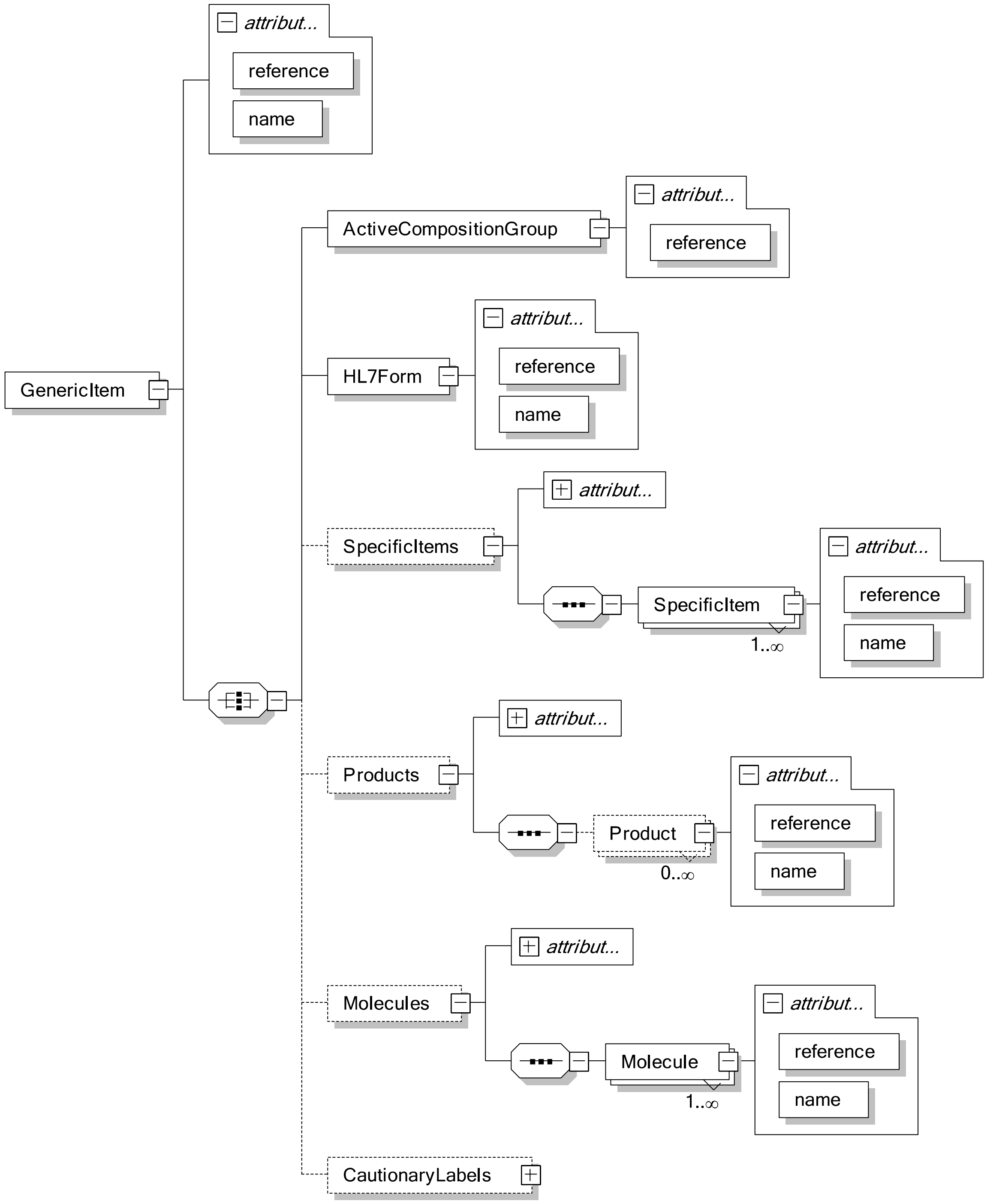


Figure 36 - GenericItem Detail Result

The GenericItem, HL7Form, SpecificItem and Product elements each contain name and reference attributes. The ActiveCompositionGroup element contains only a reference attribute and it identifies the "parent" Active Composition Group for this GenericItem. The Molecules making up the Generic Item are identified by the Active Composition Group. A Generic Item will always have at least one Specific Item, but it may not have any associated Products if this Generic Item is in the data file simply for backward-compatibility. The CautionaryLabels element is the same as that of Product Detail Result, please refer to section 2.5.2.2 for details.

2.5.10 SpecificItem

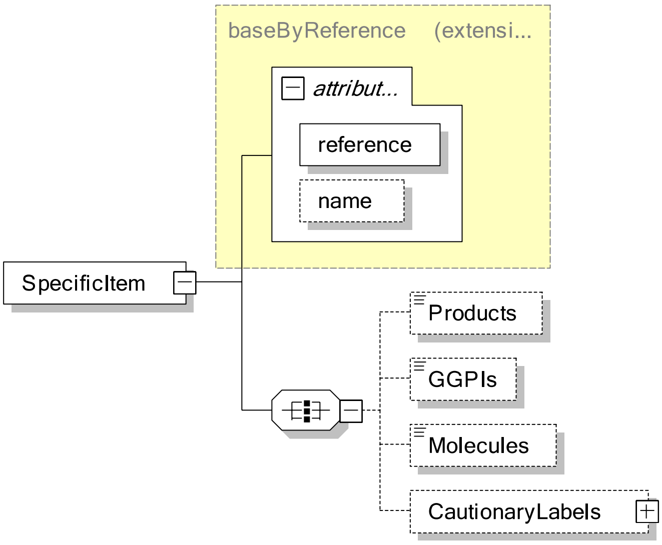


Figure 37 - SpecificItem Detail Request

A full request would look like the following

```
<Request>
    <Detail>
     <SpecificItem reference="{BE0617A4-3CCB-4233-B37F-E809BDEE20AB}">
        <Products />
        <GGPIs />
        <Molecules />
     </SpecificItem>
   </Detail>
</Request>
```

2.5.10.1 Result of SpecificItem Detail

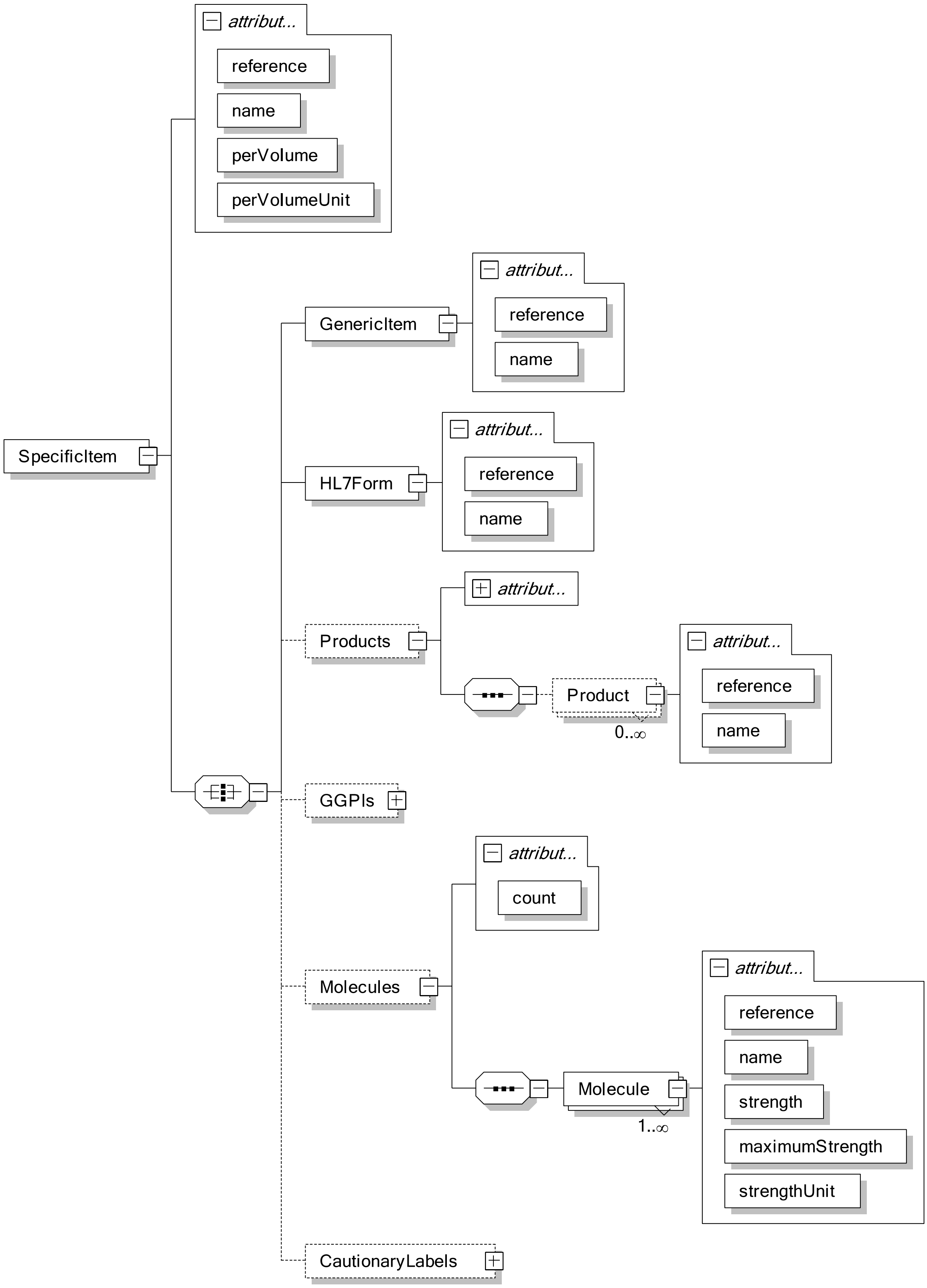


Figure 38 - SpecificItem Detail Result

The SpecificItem, GenericItem, HL7Form, GGPI and Product elements will contain name and reference attributes. The GenericItem element identifies the "parent" Generic Item for this Specific Item. A Specific Item will always be a member of at least one GGPI, but it may not have any Products associated if this Specific Item is in the data file simply to support backward-compatibility. The CautionaryLabels element is the same as that of Product Detail Result, please refer to section 2.5.2.2 for details.

2.6 Content

Content requests return either documents or images. XML Documents from FastTrack serve two purposes. They can contain documents that can be formatted to present to an end-user, or they can contain structured data that may be used programmatically to describe the associated entity in ways that are appropriate for electronic products in that country. Documents of types other than "Local Package" and "Local Product" can be associated with more than one entity. For example, one "Full Monograph" document can be associated with a Product Line and also with all of the Products contained by that Product Line. A single Consumer Information document may be associated with several Packages even if those Packages are for different products. Each Document is identified by its reference, so one Document reference may appear in the results of different Content requests. Each

Documents are returned as well-formed and valid XML data. Images are returned as binary data. Documents are the one place where the XML data structure of the FastTrack result will vary between different countries' data sets.

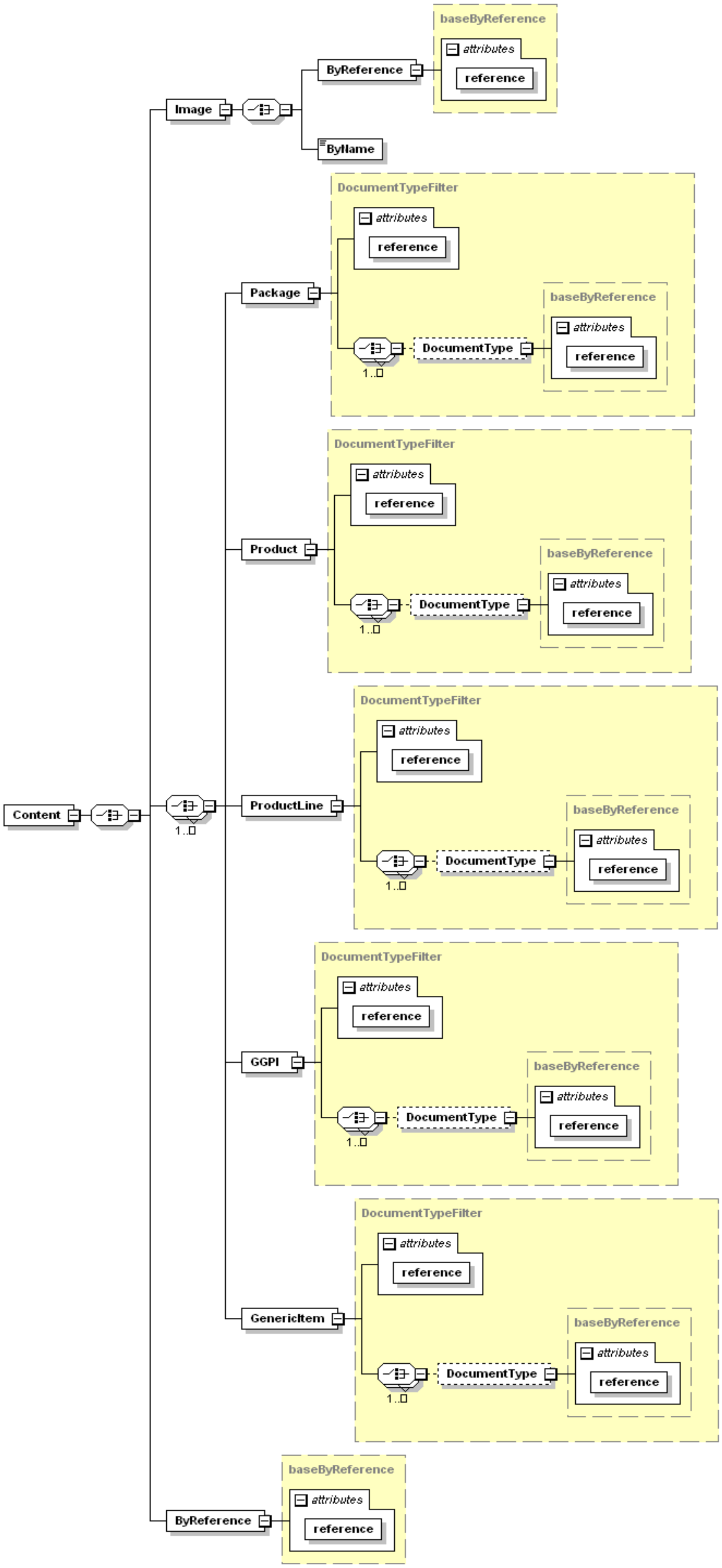

baseByReference
attributes
reference
ByReference
Image
ByName
DocumentTypeFilter
attributes
reference
Package
baseByReference
attributes
reference
DocumentType
1..∞
DocumentTypeFilter
attributes
reference
Product
baseByReference
attributes
reference
DocumentType
1..∞
Content
1..∞
DocumentTypeFilter
attributes
reference
ProductLine
baseByReference
attributes
reference
DocumentType
1..∞
DocumentTypeFilter
attributes
reference
GGPI
baseByReference
attributes
reference
DocumentType
1..∞
DocumentTypeFilter
attributes
reference
GenericItem
baseByReference
attributes
reference
DocumentType
1..∞
baseByReference
attributes
reference
ByReference

Figure 39 - Content Request Structure

Documents can be requested one at a time with the ByReference option, or one or several documents can be requested based on one or several entities in a single request. Images are requested with the RequestBINARY method of FastTrack by reference or by name. Image names may be retrieved from within some countries' document contents.

2.6.1 Package

This request returns a set of XML documents containing additional structured data pertaining to the identified Package and the country providing the data. The Package reference is passed in the request as the value of the reference attribute of the Package element. A list of Document Type references can be included in the request to limit the result to only Documents of those types. The possible Document Type references can be retrieved from the List Document Type request. Generally only documents of the "Local Package" type will be attached to Package entities, but any other document type could be included, such as "Package Insert", depending on the country data provider. Not all countries supply documents related to packages.

```
<Request>
   <Content>
      <Package
      reference="{3461726E-0F10-4CF7-B4D9-A2E345FFD378}">
         <DocumentType
            reference="{AB919A74-E33F-4C67-80B7-17089DE1402C}"/>
      </Package>
   </Content>
</Request>
```

2.6.2 Product

This request returns a set of XML documents containing additional structured data pertaining to the Product from the country providing the data. The Product reference is passed into the request as the value of the reference attribute of the Product element. A list of Document Type references can be included in the request to limit the result to only Documents of those types. The possible Document Type references can be retrieved from the List Document Type request. Generally only documents of the "Local Product" type will be attached to Product entities but, depending on the country data provider, any other document type could be included such as "Full Monograph" or "Abbreviated Monograph". Not all countries supply documents related to products.

```
<Request>
   <Content>
      <Product
      reference="{A633FF4F-399F-4938-8999-98CD52E7EBE1}">
         <Document
         type="{1D69392A-DB04-40F6-8F42-43A6AD91861B}"/>
         <Document
         type="{B0EED82E-B7DC-4F5B-8F92-8994D69CBA83}"/>
      </Product>
   </Content>
</Request>
```

2.6.3 ProductLine

This request returns a set of XML documents containing structured text and/or data pertaining to the Product Line from the country providing the data. The Product Line reference is passed into the request as the value of the reference attribute of the ProductLine element. A list of Document Type references can be included in the request to limit the result to only Documents of those types. The possible Document Type references can be retrieved from the List Document Type request. Generally documents of the "Full Monograph" and/or "Abbreviated Monograph" type will be attached to Product Line entities but, depending on the country data provider, any other document type could be included. Most countries supply documents related to Product Lines.

```
<Request>
    <Content>
        <ProductLine
        reference="{5515BEA6-2F34-452B-887E-5A8774A8597D}">
            <Document
            type="{9766B9FB-6A11-40E4-9B31-001B62ACE9F1}"/>
            <Document
            type="{47AB0ACB-BBBA-4401-A72B-C600855CA243}"/>
        </ProductLine>
    </Content>
</Request>
```

2.6.4 Image

The request submits the image reference or name and returns the image in binary form. The image reference is passed into the request as the value of the reference attribute of the ByReference element.

```
<Request>
    <Content>
        <Image>
            <ByReference
            reference="{FE479FC9-1B34-42F9-BA9E-49ACCABA0ACC}"/>
        </Image>
    </Content>
</Request>
```

The ByName option behaves just like the other ByName requests in FastTrack because it is not case-sensitive, however is behaves differently because the name must be a complete match rather than just matching the first part of the name.

```
<Request>
    <Content>
        <Image>
            <ByName>T0624301.gif</ByName>
        </Image>
    </Content>
</Request>
```

This request returns the Base-64 encoded image data enclosed within <ImageData>and </ImageData> tag. The meta information about the image like the dimensions of the image in pixels, it's MIME type and physical size in bytes, are returned as attributes of the ImageData tag

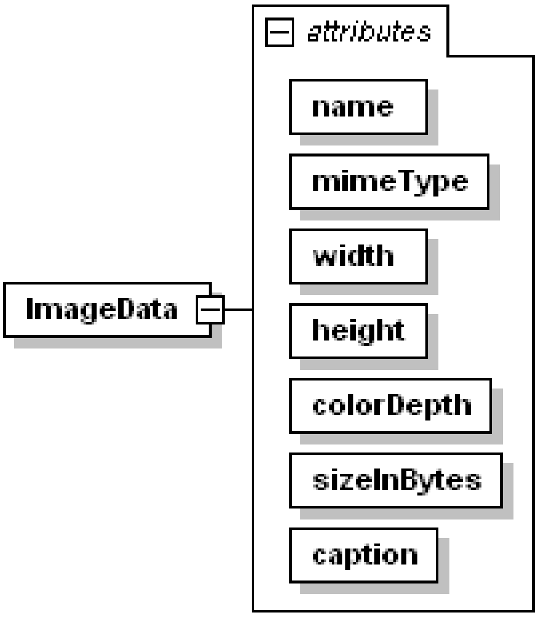


Figure 51 – Schema of ImageData element

```
<Result>

       <Content>

              <ImageData   name=""   mimeType=""   width=""   height=""   colorDepth=""
              sizeInBytes="" caption="">

              <!-- Base64 encoded image data -->

              </ImageData>

       </Content>

</Result>
```

2.6.5 Results

The XML schema for documents can be seen in the following figure. Each of the three elements (Package, Product and ProductLine) has a reference attribute whose value matches the request and contains a Document element for each returned Document. A well-formed XML document is contained within the Document element. The Document element contains up to four attributes: reference, typeReference, stylesheetReference, and schemaReference. The last two may not appear if there is no value to return. These four attributes identify the Document, the Document Type, the XSL stylesheet that can be used to display the document, and finally the XSD schema or DTD document that can be used to validate the Document respectively.

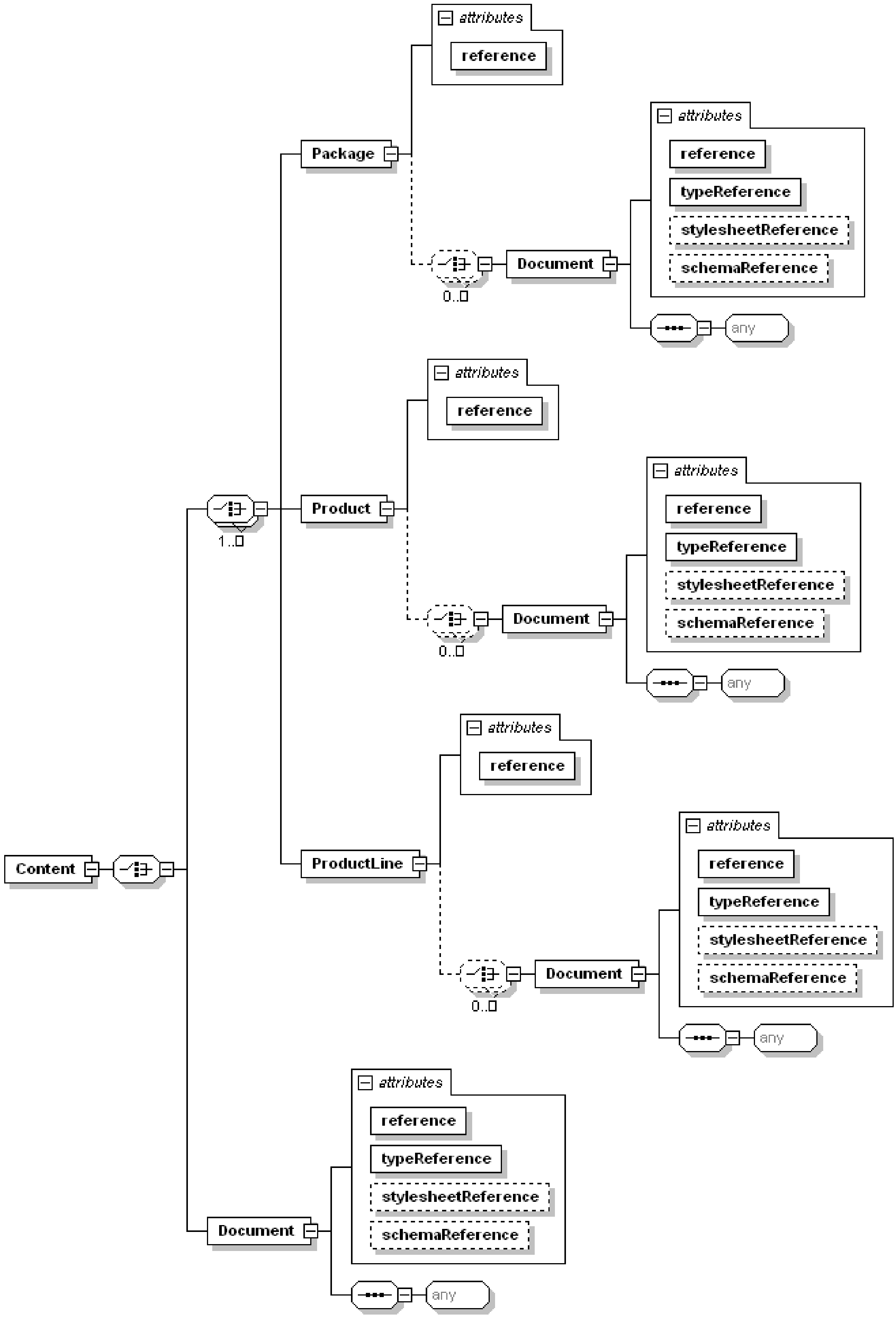


Figure 40 - Result Content Elements

2.6.6 Example

This example shows documents from two different entities returned in the same result.

```
<Request>
    <Content>
        <Package
        reference="{3461726E-0F10-4CF7-B4D9-A2E345FFD378}"/>
        <Product
        reference="{A633FF4F-399F-4938-8999-98CD52E7EBE1}"/>
    </Content>
</Request>
```

The shaded portions indicate the document contents themselves.

```
<Result buildID="" copyright="" expiryDate="1/1/2015">
    <Content>
        <Package reference="{3461726E-0F10-4CF7-B4D9-A2E345FFD378}">
            <Document
            reference="{AD12252F-9881-6779-E034-080020E1DD8C}"
            typeReference="{AB919A74-E33F-4C67-80B7-17089DE1402C}">
                <Pack prodCode="56" formCode="1" packCode="1"
                packSort="1" packDetails="(120)" pbsCode="1157X"
                pbs="PBS/RPBS" rp="5" noOfPacks="1" qty="120"
                authCode="" restCode="" price="" bpp="$1.94"
                specContrib="" tgp="">
                        <Composition active="200" activeUnits="mg"
                        perVolume="" perVolUnits="" unitVolume=""
                        unitVolUnits="" />
                        <Rx prescriptionOnly="Yes">S4</Rx>
                        <Price dispensingFeeIncluded=""
                        dangerousDrugFeeIncluded="" s3RecordFee="" />
                        <PBSPrice pbs_price="21.48" />
                        <DosageAdministration da="Acute duodenal, gastric
ulcer: 800 mg at bedtime, or 400 mg morning and bedtime, or 200 mg 3
times daily and 400 mg at bedtime (4-8 wks); may incr to max 1.6
g/day. Maintenance recurrent duodenal ulcer: 400 mg at bedtime.
Maintenance chronic benign gastric ulcer: 400 mg at bedtime for less
than or equal to 1 yr. Zollinger-Ellison syndrome: 200 mg 3 times
daily and 400 mg at bedtime; may incr to max 2 g/day. GORD: 800 mg at
night or in divided doses for less than or equal to 12 wks;
symptomatic therapy of GORD: 200 mg up to 4 times daily for less than
```

```
or equal to 2 wks, max 800 mg/day. Scleroderma oesophagus: usually
1200 mg/day in divided doses. Renal impairment: reduce dose" />
             </Pack>
             <Package ProductId="560101" ProductCode="56"
             FormCode="1" PackCode="1" PackSort="1"
             noOfPacks="1" qty="120">
                    <Composition active="200" activeUnits="mg"
                    perVolume="" perVolUnits="" unitVolume=""
                    unitVolUnits="" />
                    <Rx prescriptionOnly="Yes" S11="">S4</Rx>
                    <PackDetails>200 mg [120]</PackDetails>
                    <PBS PBScode="1157X" PBStype="PBS/RPBS"
                    Repeats="5" BrandSubstitution="1">
                           <PBSprice DispensingFeeCode="RP"
                           DispensingFee="ready-prepared">
                           21.48</PBSprice>
                           <BrandPricePremium>
                           $1.94</BrandPricePremium>
                           <TGP TherapeuticGroupCode="3"
                           TherapeuticGroupName="H2 Receptor
                           antagonists" />
                    </PBS>
                    <DosageAdministration>Acute duodenal, gastric
ulcer: 800 mg at bedtime, or 400 mg morning and bedtime, or 200 mg 3
times daily and 400 mg at bedtime (4-8 wks); may incr to max 1.6
g/day. Maintenance recurrent duodenal ulcer: 400 mg at bedtime.
Maintenance chronic benign gastric ulcer: 400 mg at bedtime for less
than or equal to 1 yr. Zollinger-Ellison syndrome: 200 mg 3 times
daily and 400 mg at bedtime; may incr to max 2 g/day. GORD: 800 mg at
night or in divided doses for less than or equal to 12 wks;
symptomatic therapy of GORD: 200 mg up to 4 times daily for less than
or equal to 2 wks, max 800 mg/day. Scleroderma oesophagus: usually
1200 mg/day in divided doses. Renal impairment: reduce
dose</DosageAdministration>
             </Package>
          </Document>
       </Package>
       <Product reference="{A633FF4F-399F-4938-8999-98CD52E7EBE1}">
          <Document reference="{A0DFBE60-7413-68D1-E034-080020E1DD8C}"
          typeReference="{1D69392A-DB04-40F6-8F42-43A6AD91861B}">
             <LocalData rp="" qty="100" numberOfPacks="1"
             unitsPerPack="100" pbs=""
             pbsCode="" active="600" activeUnits="mg"
             perVolume="" perVolUnits="" indications="" />
          </Document>
       </Product>
    </Content>
</Result>
```

2.7 PatientProfile

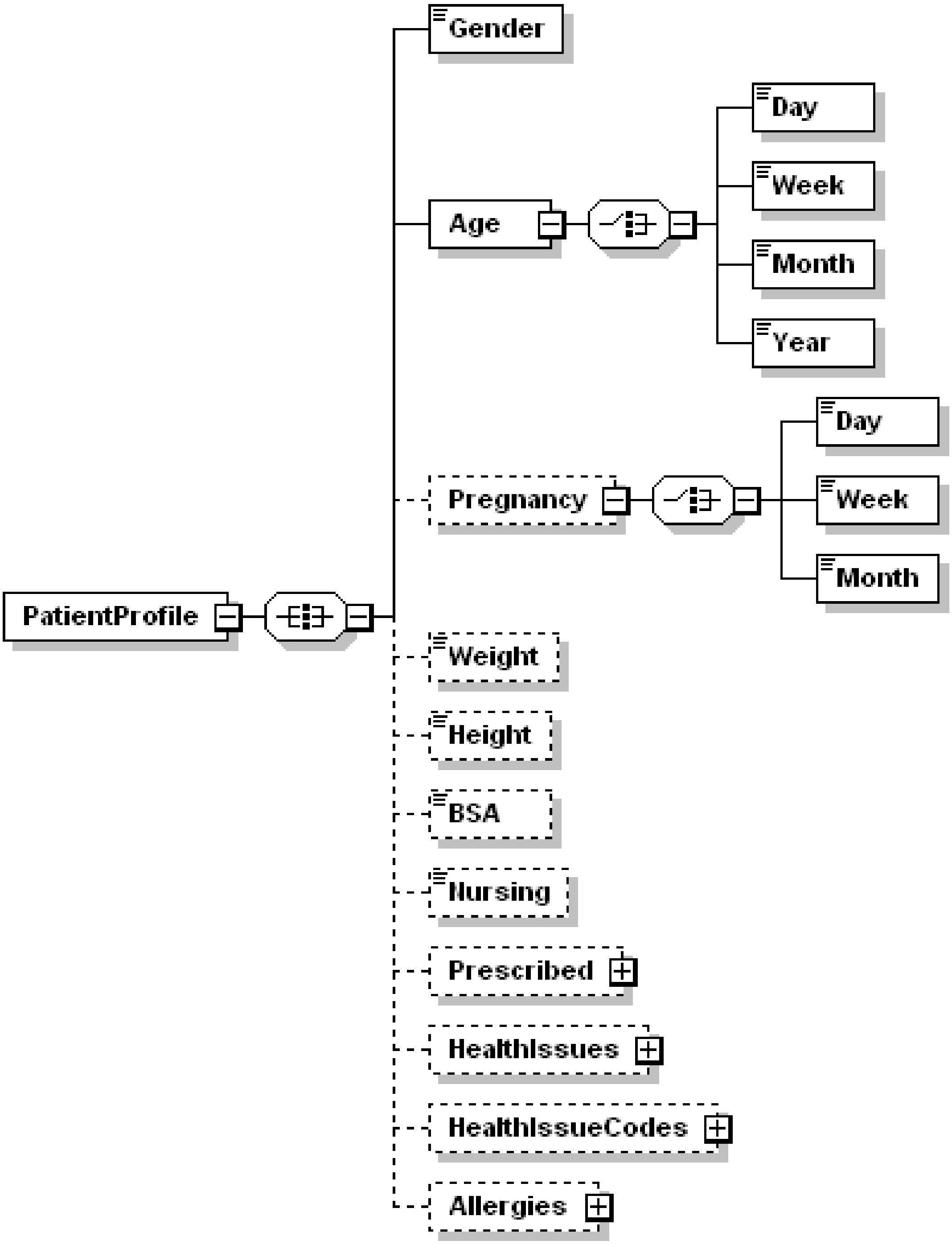


Figure 40 – Patient profile schema

The PatientProfile section is for presenting the patient's demographic profile and medical history to the Interaction check module.

2.7.1 Gender

Specifies the patient's gender, permitted values are "M" or "F".

2.7.2 Age

Specifies the patient's age. The patient's age can be expressed in terms of four time units (Day, Week, Month and Year). At least one or more time units must be present. Only integer values are accepted.

2.7.3 Pregnancy

Specifies that the patient is currently with child. The gestation period can be express in terms of three time units (Day, Week, Month). At least one or more time units must be present. Only integer values are accepted. Based on the gestation period, the system will determine which trimester the pregnancy is in.

```
<Request>
   <Interaction>
      <Prescribing>
         .
         .
         .
      </Prescribing>
   </Interaction>
   <PatientProfile>
    <Gender>F</Gender>
    <Age>
      <Year>32</Year>
    </Age>
    <Pregnancy>
      <Month>2</Month>
       <Week>3</Week>
       <Day>5</Day>
    </Pregnancy>
   </PatientProfile>
</Request>
```

2.7.4 Weight

Specifies the patient's body weight, in kilograms (kg), as a decimal number.

2.7.5 Height

Specifies the patient's height, in centimeters (cm), as an integer.

2.7.6 BSA

Specifies the patient's Body Surface Area, in square meters (m^2).

2.7.7 Nursing

Specifies that patient is currently breastfeeding. Valid values are "true" or "false". This parameter only applies if the patient is female and at least 11 years of age.

2.7.8 Prescribed

The Prescribed element is optional and it must contain at least one of the six entities in order to produce a result. More than one of the elements may be included and each element may be included multiple times in any order (e.g., two Products, three GGPIs, etc.). The entities passed in the request in the Prescribed element will be checked for interactions against each other and all of the items passed in the Prescribing, Allergies, HealthIssues and HealthIssueCodes (for both parent elements Interaction and PatientProfile). The drugs will be returned in the result in the order they are presented in the request with those in the Prescribing element (if it will set in Interaction) coming before those in the Prescribed element.

Each of the drugs supplied in Prescribed elements are identified by their reference in the reference attribute of the respective element. The ActiveComposition and ActiveCompositionGroup elements have one additional optional attribute for the name. Any name supplied by the user in this attribute will be returned in the result for those entities since they do not have names of their own.

2.7.9 HealthIssueCodes

The HealthIssueCodes element is optional. If it is included, it contains HealthIssueCode elements which include a Health Issue Code and the coding system it is from. The HealthIssueCode element has code and codeType attributes.

2.7.10 Allergies

The Allergies element is optional. This element is added to indicate Substance Classes, Active Compositions, Active Composition Groups, Molecules, Products, GGPIs, Specific Items and/or Generic Items for which the patient has demonstrated an allergic response. These items are compared with elements from the Prescribing and Prescribed lists to determine if there is a potential for an allergic reaction.

For each of the possible entities, only the reference attribute of the respective XML element is supplied in the request. Active Composition and Active Composition Group elements also have an optional name attribute just as they do in the Prescribing and Prescribed lists. Any number of any type of element can be supplied in any order. The order of any possible allergy warnings in the result is independent of the order of the elements in the request and is described below.

```
<Request>
   <Interaction>
        .
        .
        .
   </Interaction>
   <Interaction>
      <Prescribed>
         <Product
         reference="{23F76FEC-965F-465A-A5D5-5E957089E4EF}"/>
      </Prescribed>
      <HealthIssues>
         <HealthIssue
         reference="{8B4A41E9-3E7B-4CF2-9228-88E4F149242A}"/>
      </HealthIssues>
      <HealthIssueCodes>
         <HealthIssueCode code="A01.4" codeType="ICD10"/>
      </HealthIssueCodes>
      <Allergies>
         <SubstanceClass
         reference="{50227FDE-1699-4083-92FA-AA75F30A285D}"/>
         <ActiveComposition
         reference="{52ED3587-A685-4849-92DB-693E6D3D766D}" />
         <ActiveCompositionGroup
         reference="{96285061-FA52-497E-8708-F0FF749D4CAE}" />
         <Molecule
         reference="{AC340A53-6611-425A-8A75-70E82F00B79C}"/>
         <Product
         reference="{23F76FEC-965F-465A-A5D5-5E957089E4EF}"/>
         <GGPI reference="{79794E62-6226-48CF-9468-575544F3F0CA}"/>
         <SpecificItem
         reference="{B4AD97C9-3A09-3012-E034-080020E1DD8C}"/>
         <GenericItem
         reference="{0F6B4E5C-886F-45B4-A495-C95F90023C09}"/>
      </Allergies>
   </Interaction>
</Request>
```

2.8 Interaction

Information about how certain drugs may interact with other drugs or health conditions can be requested from FastTrack. The request includes a reference for the candidate drug(s) to be prescribed as well as the patient's profile, including drugs they are already taking, current health issues, and any drugs they are allergic to. The result will then include information about all drug to drug interactions among any of the drugs listed as well as drug to health issue precautions and allergy warnings. Duplicate therapy warnings are included automatically in all interaction results. FastTrack supports executing all of these in a single request, or performing specific checks one-by-one. The only element that is required in the request in order to get a result is at least one drug in the Prescribing element that indicates a candidate drug.

The drugs can be identified by an Active Composition, Active Composition Group, Product, GGPI, Generic Item or Specific Item. Allergies can be identified by a Substance Class, Active Composition, Active Composition Group, Molecule, Product, GGPI, Specific Item or Generic Item.

Information in the Prescribing and Prescribed elements can be further refined with RouteOfAdministration elements to identify the route(s) via which the drug is or will be administered. These Routes come from the DSMRouteOfAdmin (for reference attribute) and the RouteOfAdministration (for name attribute) list result. The XML Schema for Interaction request can be seen in the figure below.

Reference and name attributes of RouteOfAdministration element cannot use together for the same drug. RouteOfAdministration with name attribute can use only in singular for any drug item.

Whether or not the RouteOfAdministration element is included in the request has an impact on the contents of the result. Each RouteOfAdministration element will cause the corresponding Route element to appear in the result as a child of its respective drug regardless of whether there are any interactions to display for that Route.

If the RouteOfAdministration element is omitted, then all applicable routes will be checked for interactions. For Product/GGPI/GenericItem/SpecificItem, FastTrack will derive the RouteOfAdministration from the drug's formulation. For example, if the product is an 'Oral Tablet', FastTrack will only return Systemic interactions and not Topical ones. If the prescribing item is an ActiveComposition or ActiveCompositionGroup, FastTrack will check all routes.

For drug interactions, it is safer to let FastTrack figure out the RouteOfAdministration than explicitly specifying the RouteOfAdministration. For Dose Alert, it is mandatory to specify the RouteOfAdministration, since a drug can have a variety of uses with different dosing parameters.

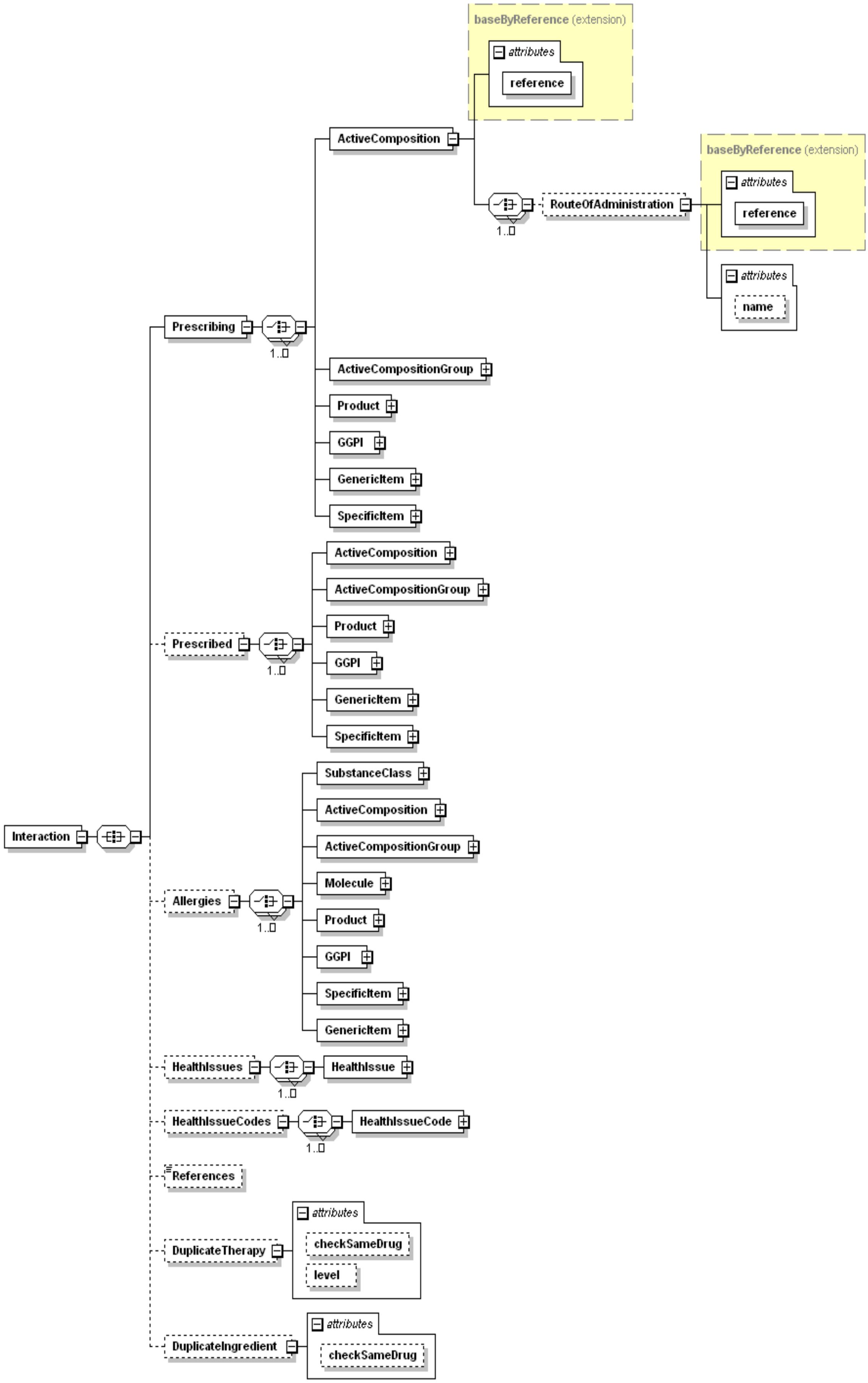

baseByReference (extension)
attributes
reference
ActiveComposition
1..∞
RouteOfAdministration
baseByReference (extension)
attributes
reference
attributes
name
Prescribing
1..∞
ActiveCompositionGroup
Product
GGPI
GenericItem
SpecificItem
ActiveComposition
ActiveCompositionGroup
Product
Prescribed
1..∞
GGPI
GenericItem
SpecificItem
SubstanceClass
ActiveComposition
Interaction
ActiveCompositionGroup
Molecule
Allergies
1..∞
Product
GGPI
SpecificItem
GenericItem
HealthIssues
1..∞
HealthIssue
HealthIssueCodes
1..∞
HealthIssueCode
References
attributes
checkSameDrug
DuplicateTherapy
level
attributes
DuplicateIngredient
checkSameDrug

Figure 41 - Interaction Request Schema

2.8.1 Prescribing and Prescribed

The Prescribing element is mandatory and it must contain at least one of the six entities in order to produce a result. More than one of the elements may be included and each element may be included multiple times in any order (e.g., two Products, three GGPIs, etc.). The entities passed in the request in the Prescribing element will be checked for interactions against each other and all of the items passed in the Prescribed, Allergies, HealthIssues and HealthIssueCodes. The options for what to supply in the Prescribed element are the same as for the Prescribing element and in fact, the two lists are combined internally before performing interaction checks. So anything listed in the Prescribed element will also be checked for interactions against other drugs, the supplied Health Issues, Health Issue Codes, Allergies. The drugs will be returned in the result in the order they are presented in the request with those in the Prescribing element coming before those in the Prescribed element.

Each of the drugs supplied in the Prescribing and Prescribed elements are identified by their reference in the reference attribute of the respective element. The ActiveComposition and ActiveCompositionGroup elements have one additional optional attribute for the name. Any name supplied by the user in this attribute will be returned in the result for those entities since they do not have names of their own.

```
<Request>
   <Interaction>
      <Prescribing>
         <Product reference="{83CF7D50-30B9-4D61-9C60-D1CF9FEBA7A9}">
            <RouteOfAdministration name="Oral"/>
         </Product>
      </Prescribing>
      <Prescribed>
         <Product
         reference="{4AE6AF0F-2890-4FBB-8F0D-ABFE307C1A71}">
            <RouteOfAdministration
            reference="{90075601-10D4-49DD-9292-9B884D9D3EF3}"/>
         </Product>
      </Prescribed>
   </Interaction>
</Request>
```

2.8.2 Allergies

The Allergies element is optional. This element is added to indicate Substance Classes, Active Compositions, Active Composition Groups, Molecules, Products, GGPIs, Specific Items and/or Generic Items for which the patient has demonstrated an allergic response. These items are compared with elements from the Prescribing and Prescribed lists to determine if there is a potential for an allergic reaction.

For each of the possible entities, only the reference attribute of the respective XML element is supplied in the request. Active Composition and Active Composition Group elements also have an optional name attribute just as they do in the Prescribing and Prescribed lists. Any number of any type of element can be supplied in any order. The order of any possible allergy warnings in the result is independent of the order of the elements in the request and is described below.

```
<Request>
   <Interaction>
      <Prescribing>
         <Product
         reference="{23F76FEC-965F-465A-A5D5-5E957089E4EF}"/>
      </Prescribing>
      <Allergies>
         <SubstanceClass
         reference="{50227FDE-1699-4083-92FA-AA75F30A285D}"/>
         <ActiveComposition
         reference="{52ED3587-A685-4849-92DB-693E6D3D766D}" />
         <ActiveCompositionGroup
         reference="{96285061-FA52-497E-8708-F0FF749D4CAE}" />
         <Molecule
         reference="{AC340A53-6611-425A-8A75-70E82F00B79C}"/>
         <Product
         reference="{23F76FEC-965F-465A-A5D5-5E957089E4EF}"/>
         <GGPI reference="{79794E62-6226-48CF-9468-575544F3F0CA}"/>
         <SpecificItem
         reference="{B4AD97C9-3A09-3012-E034-080020E1DD8C}"/>
         <GenericItem
         reference="{0F6B4E5C-886F-45B4-A495-C95F90023C09}"/>
      </Allergies>
   </Interaction>
</Request>
```

2.8.3 HealthIssues

Deprecation Warning

The HealthIssues element will be removed in a future version of FastTrack. Developers are advised to use HealthIssueCodes.

Health Issues are conditions that may have an impact on certain drugs or conditions that may be affected by the use of certain drugs. There are currently over 280 Health Issues in the database and include conditions like Hepatic Failure, Peptic Ulcer, Hypertension, etc.

The HealthIssues element is optional. The health issue reference(s) is(are) passed in the request as the value of the reference attribute of HealthIssue elements contained by the HealthIssues element. Any number of HealthIssue elements can be included in the request.

```
<Request>
   <Interaction>
      <Prescribing>
         <Product
         reference="{B360A3E4-3F6F-4D59-B173-CF0D5E0487C9}"/>
      </Prescribing>
      <HealthIssues>
         <HealthIssue
         reference="{8B4A41E9-3E7B-4CF2-9228-88E4F149242A}"/>
      </HealthIssues>
   </Interaction>
</Request>
```

2.8.4 HealthIssueCodes

The HealthIssueCodes element is optional. If it is included, it contains HealthIssueCode elements which include a Health Issue Code and the coding system it is from. The HealthIssueCode element has code and codeType attributes.

```
<Request>
   <Interaction>
      <Prescribing>
         <Product
         reference="{B360A3E4-3F6F-4D59-B173-CF0D5E0487C9}"/>
      </Prescribing>
      <HealthIssueCodes>
         <HealthIssueCode code="A01.4" codeType="ICD10"/>
      </HealthIssueCodes>
   </Interaction>
</Request>
```

2.8.5 References

The References element is included in the request to signal to FastTrack that they should be included with each interaction result. If the element is present in the request then each ClassInteraction element in the result will contain a References tag containing all the references supporting that interaction. References in this context does not mean the GUID references used elsewhere in this document. In this context, References are the sources of information where the interactions are documented in medical literature. There are five XML elements that may appear in any order within the References element. BookReference, JournalReference, WebReference, ElectronicReference and PackageInsertReference. Each of these has specific fields supplied. See the referencesType complex type in Response.xsd for further details. When the References tag appears in the result, it is the last element contained in the ClassInteraction element. References are structured in the same way for Drug-Drug and Drug-Health interactions.

```
<Request>
  <Interaction>
    <Prescribing>
      <Product reference="{7BA477CF-2DD5-4693-B477-A768E9113752}" />
      <Product reference="{0B3E259A-0C7F-4B1D-97E1-F2EB99773A69}" />
    </Prescribing>
    <References />
  </Interaction>
</Request>
```

2.8.6 DuplicateTherapy

FastTrack will return results for DuplicateTherapy by default

FastTrack DuplicateTherapy check is based on ATC (Anatomical Therapeutic Chemical Classification System). Two drugs are considered to be duplicate if they share the same ATC codes. The level of duplication is reflected in the "Warning Level".

Warning Level	**Definition**
1 (Most specific)	The drugs share the same chemical substance.
2	The drugs share the same chemical/therapeutic/pharmacological subgroup.
3 (Broadest)	The drugs share the same therapeutic/pharmacological subgroup.

```
<Request>
  <Interaction>
    <Prescribing>
      <GenericItem
         reference="{B3E5E9FF-0858-4B5E-E034-080020E1DD8C}"/>
      <GenericItem
         reference="{B49C9A6B-F98B-6199-E034-080020E1DD8C}"/>
    </Prescribing>
    <DuplicateTherapy/>
  </Interaction>
</Request>
```

2.8.7 DuplicateIngredient

DuplicateIngredient check complements DuplicateTherapy. Whilst DuplciateTherapy only checks via ATC code, the DuplicateIngredient check will check if two drugs shares the same ingredient. For example, if 'paracetamol oral tablets' and 'codiene-paracetamol oral tablets' are prescribed, DuplicateIngredient will alert that both drugs contain the same active ingredient 'paracetamol'.

```
<Request>
  <Interaction>
    <Prescribing>
      <GenericItem
         reference="{B3E5E9FF-0858-4B5E-E034-080020E1DD8C}"/>
      <GenericItem
         reference="{B49C9A6B-F98B-6199-E034-080020E1DD8C}"/>
    </Prescribing>
    <DuplicateIngredient/>
  </Interaction>
</Request>
```

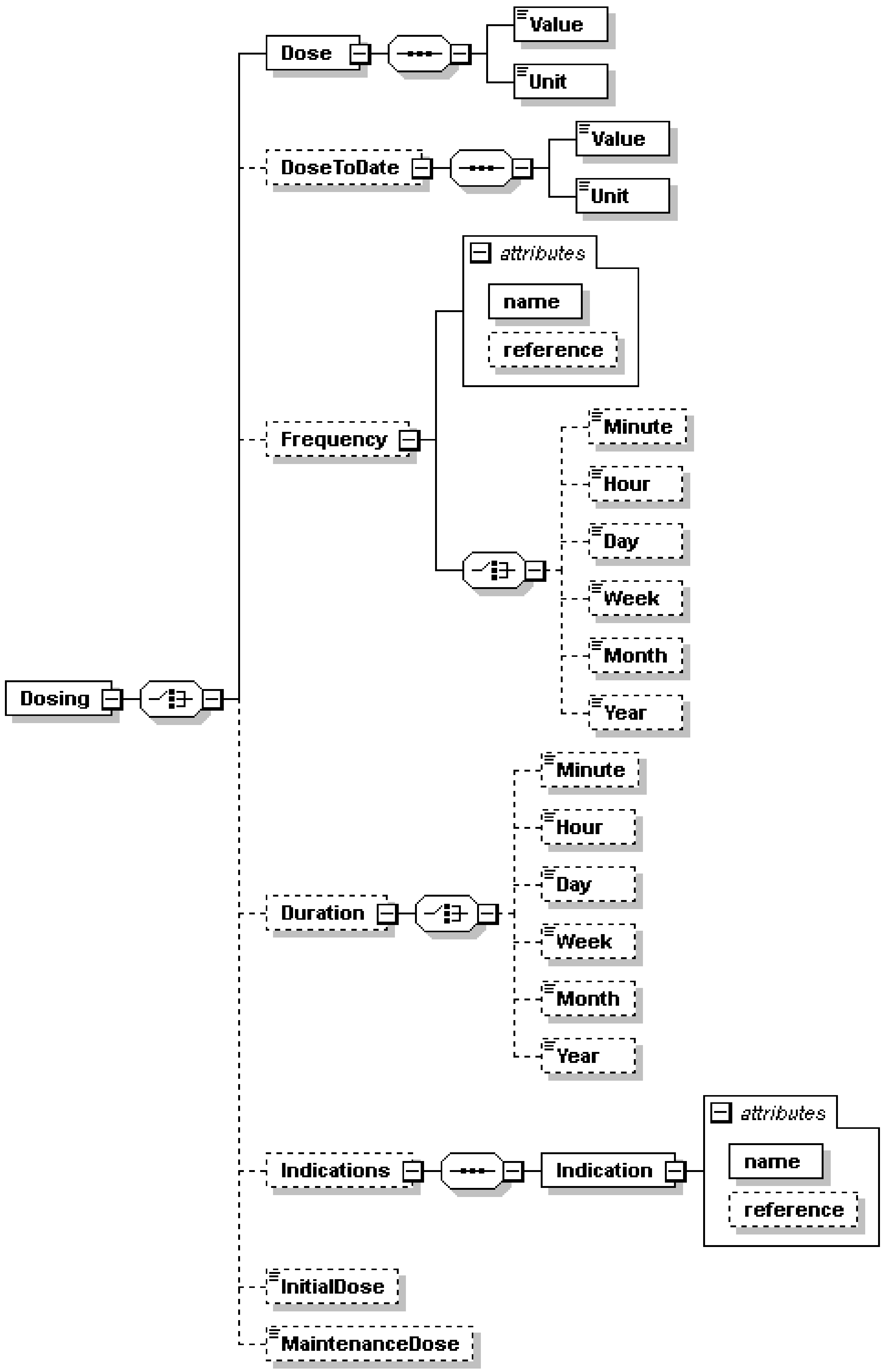
2.8.8 CautionaryLabels

If the CuationaryLabels element is present, FastTrack will return warning labels associated with each prescribing element. Please refer to section 2.5.2.2 for details about CautionaryLabels.

2.8.9 Dosing

For each prescribing element (Product, GenericItem, SpecificItem, GGPI, ActiveComposition, ActiveCompositionGroup) an additional Dosing element can be appended which can be use for dose checking of each prescribed element. Dose and DoseToDate contains Value and Unit child elements where Value to admit real positive number and Unit string value like "mg", "g", "ml", etc which concerted in current build data file. In addition to it dose structure contain frequency, duration, indications, initial and maintenance dose flags parameters for dose check. Child elements of Frequency and Duration to admit also real positive numbers. InitialDose and MaintenanceDose elements to admit only "true" or "false" values. Name and reference attributes of Frequency and Indication elements just using usual accepted values in data file and FastTrack API. To this effect Form element apply for checking of dose too.

```
<Request>
  <Interaction>
    <Prescribing>
      <Product reference="{7BA477CF-2DD5-4693-B477-A768E9113752}" >
        <RouteOfAdministration name="Inhalation" />
        <Form name="Metered Aerosol" />
        <Dosing>
           <Dose> <Value>100</Value> <Unit>mcg</Unit> </Dose>
           <Frequency name="daily" />
        </Dosing>
       </Product>
    </Prescribing>
      </Interaction>
</Request>
```


Dosing
Dose
Value
Unit
DoseToDate
Value
Unit
attributes
name
reference
Frequency
Minute
Hour
Day
Week
Month
Year
Duration
Minute
Hour
Day
Week
Month
Year
Indications
Indication
attributes
name
reference
InitialDose
MaintenanceDose

2.8.10 Interaction example

In the example below, a patient has already been prescribed the product Odrik and the active composition group containing Trimethoprim. The patient is going to be prescribed the product Alprim and an Active Composition containing Trimethoprim. The patient is allergic to Trimethoprim. The patient is elderly and suffering from renal failure.

Element	Name	Type	Identifier
Prescribed	Odrik	Product	{0B3E259A-0C7F-4B1D-97E1-F2EB99773A69}
Prescribed	Trimethoprim	Active Composition Group	{07FB1468-56DF-4ADD-A570-B88B567BF919}
Prescribing	Alprim	Product	{7BA477CF-2DD5-4693-B477-A768E9113752}
Prescribing	Trimethoprim	ActiveComposition	{18864A78-8A60-469E-AE29-5D5B27024E9F}
Allergies	Trimethoprim	Molecule	{7142A19F-EDE2-4A7C-A30B-06AB60B6B71F}
HealthIssue	Renal Failure	Health Issue	{2082ED86-1CF8-4A48-8EF0-0BF11DBC76D4}

2.8.10.1 Request XML

```
<Request>
   <Interaction>
      <Prescribing>
         <Product reference="{7BA477CF-2DD5-4693-B477-A768E9113752}"/>
         <ActiveComposition reference="{18864A78-8A60-469E-AE29-5D5B27024E9F}"
         name="Trimethoprim"/>
      </Prescribing>
      <Prescribed>
         <Product reference="{0B3E259A-0C7F-4B1D-97E1-F2EB99773A69}"/>
         <ActiveCompositionGroup
            reference="{07FB1468-56DF-4ADD-A570-B88B567BF919}"
            name="Trimethoprim"/>
      </Prescribed>
      <Allergies>
         <Molecule reference="{7142A19F-EDE2-4A7C-A30B-06AB60B6B71F}"/>
      </Allergies>
      <HealthIssues>
         <HealthIssue reference="{6E25924E-D899-4722-A7B1-5FB470638F3A}"/>
      </HealthIssues>
      <HealthIssueCodes>
         <HealthIssueCode code="N17" codeType="ICD10"/>
      </HealthIssueCodes>
   </Interaction>
</Request>
```

2.8.11 Result

While evaluating each requested drug in order, interactions are checked (and results returned) in this order

1) DrugAlert for all drugs listed in both Prescribing and Prescribed.
2) Drug to Health Issue (HealthAlert)
3) Drug to Health Issue code (HealthAlert)
4) AllergyAlert

DrugAlert are defined by a class of drugs interacting with another class of drugs. Drug-Health Issue interactions are defined by a class of drugs interacting with a Health Issue. These are a different set of drug classes from those used for DrugAlert checking. This is also a separate list of drug classes from those used for DrugAlert and HealthAlert checking. AllergyAlert are defined by grouping active molecules into Substance Classes. If two drugs contain Molecules from the same class and the patient is allergic to one of the drugs, the other one will have an allergy warning.

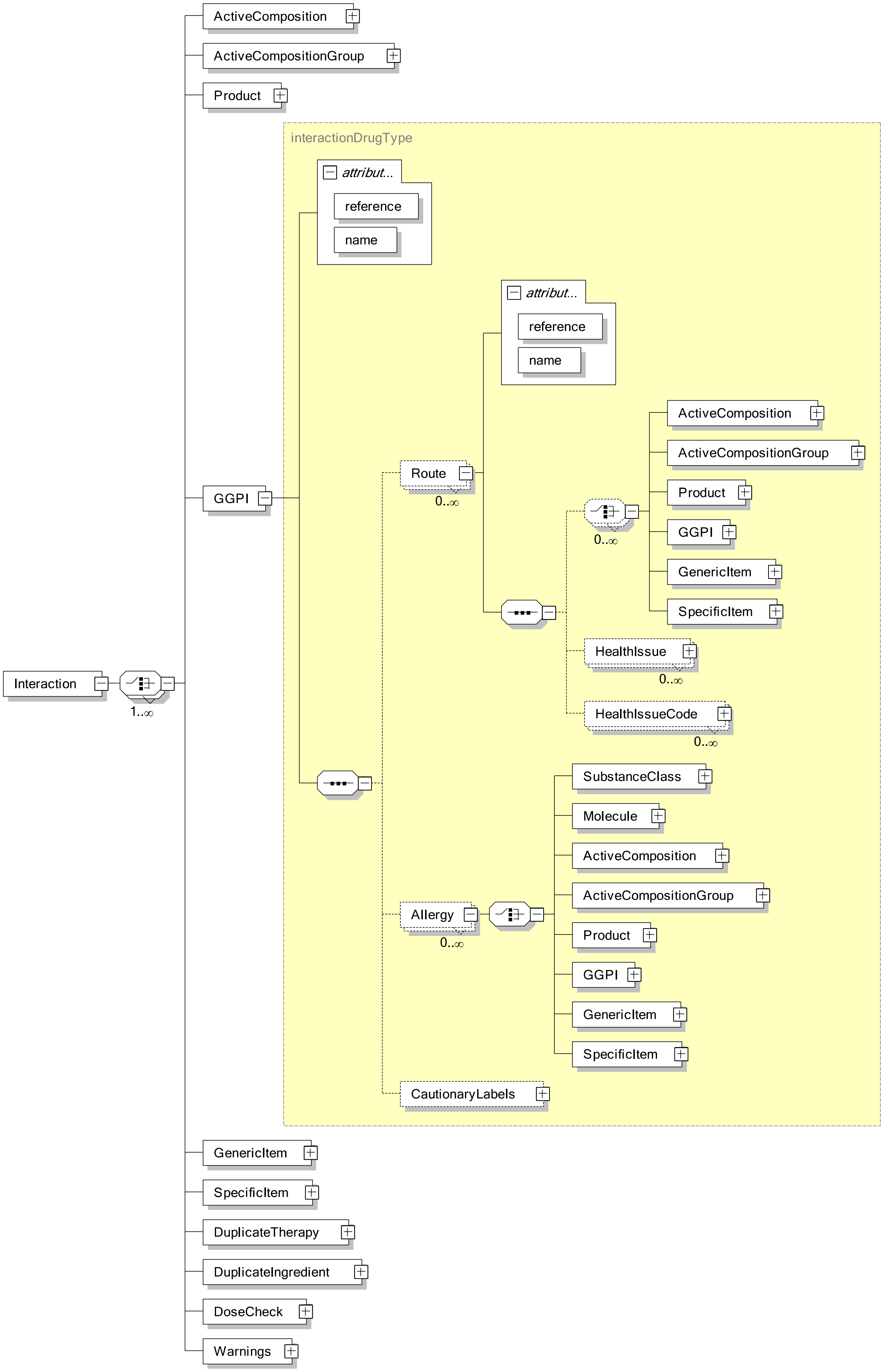

ActiveComposition
ActiveCompositionGroup
Product
interactionDrugType
attribut...
reference
name
attribut...
reference
name
ActiveComposition
ActiveCompositionGroup
Route
0..∞
Product
GGPI
0..∞
GGPI
GenericItem
SpecificItem
HealthIssue
0..∞
Interaction
1..∞
HealthIssueCode
0..∞
SubstanceClass
Molecule
ActiveComposition
ActiveCompositionGroup
Allergy
0..∞
Product
GGPI
GenericItem
SpecificItem
CautionaryLabels
GenericItem
SpecificItem
DuplicateTherapy
DuplicateIngredient
DoseCheck
Warnings

Figure 42 - Interaction Result

The shaded portion of the diagram is repeated and identical for the ActiveComposition, ActiveCompositionGroup, Product, GenericItem and SpecificItem elements.

The skeleton of the result from the above example request is displayed here. The specific sections will be described below.

```
<?xml version="1.0" encoding="UTF-8" ?>
<Result buildID="" copyright="" expiryDate="1/1/2015">
   <Interaction>
      <Product reference="{7BA477CF-2DD5-4693-B477-A768E9113752}"
name="Alprim">
          .
          .
          .
      </Product>
      <ActiveComposition reference="{18864A78-8A60-469E-AE29-
5D5B27024E9F}" name="Trimethoprim">
          .
          .
          .
      </ActiveComposition>
      <Product reference="{0B3E259A-0C7F-4B1D-97E1-F2EB99773A69}"
name="Odrik">
          .
          .
          .

      </Product>
      <ActiveCompositionGroup reference="{07FB1468-56DF-4ADD-A570-
B88B567BF919}" name="Trimethoprim">
          .
          .
          .

      </ActiveCompositionGroup>
   </Interaction>
</Result>
```

The first level of information (the child elements of the Interaction element) contains the list of drugs identified in the Prescribing and Prescribed elements of the request. Even if a given drug does not have any interactions defined for it, the element will appear at this level in the result. Each element at this level will have the same structure beneath it, so only the first Product result will be described.

```
<Product reference="{7BA477CF-2DD5-4693-B477-A768E9113752}"
   name="Alprim">
   <Route reference="{90075601-10D4-49DD-9292-9B884D9D3EF3}"
      name="Systemic">
      <ActiveComposition
         reference="{18864A78-8A60-469E-AE29-5D5B27024E9F}"
         name="Trimethoprim" />
      <Product reference="{0B3E259A-0C7F-4B1D-97E1-F2EB99773A69}"
         name="Odrik">
         <Route
            reference="{90075601-10D4-49DD-9292-9B884D9D3EF3}"
            name="Systemic">
            <ClassInteraction>
            .
            .
            </ClassInteraction>
         </Route>
      </Product>
      <ActiveCompositionGroup
         reference="{07FB1468-56DF-4ADD-A570-B88B567BF919}"
         name="Trimethoprim" />
      <HealthIssue
         reference="{2082ED86-1CF8-4A48-8EF0-0BF11DBC76D4}"
         name="Renal failure">
         <ClassInteraction>
         .
         .
         </ClassInteraction>
      </HealthIssue>
      <HealthIssueCode code=”A01.4" name="Paratyphoid fever,
unspecified" codeType="ICD10">
         <ClassInteraction>
         .
         .
         </ClassInteraction
      </HealthIssueCode>
   </Route>
   <Route reference="{AD02A5E3-6436-12B1-E034-080020E1DD8C}"
      name="Topical">
      <Product reference="{0B3E259A-0C7F-4B1D-97E1-F2EB99773A69}"
         name="Odrik">
         <Route
            reference="{90075601-10D4-49DD-9292-9B884D9D3EF3}"
            name="Systemic">
            <ClassInteraction>
            .
            .
            </ClassInteraction>
         </Route>
      </Product>
      <HealthIssue
         reference="{2082ED86-1CF8-4A48-8EF0-0BF11DBC76D4}"
         name="Renal failure"/>
      <HealthIssueCode code="A0.14" name="Paratyphoid fever,
unspecified" codeType="ICD10"/>
   </Route>
   <Route reference="{AD02A5E3-6436-12B1-E034-080020E1DD8C}"
      name="Oral">
      <HealthIssue
         reference="{2082ED86-1CF8-4A48-8EF0-0BF11DBC76D4}"
         name="Renal failure"/>
      <HealthIssueCode code="A01.4 " name="Paratyphoid fever,
unspecified" codeType="ICD10"/>
   </Route>
   <Allergy>
   .
   .
   </Allergy>
```

```
    <Allergy>
    .
    .
    </Allergy>
</Product>
```

This result displays an interaction between the product Alprim and the Product Odrik. It also indicates an interaction between Alprim and both the HealthIssue and HealthIssueCode, but only when the Alprim is administered Systemically. Also, an Allergy warning appears.

2.8.11.1 DrugAlert Result

Note that the same interaction appears for the Systemic and Topical routes under the product Alprim. This is because the active molecule, Trimethoprim, has been assigned to a drug interaction class for both of those routes of administration, and the respective drug class interacts with a drug class that has been assigned to the product Odrik. In this case, the active molecule in Odrik, Trandolapril, has only been assigned to the interacting drug class for the "Systemic" route, so only one Route element appears under that product's element in the result.

Only the Systemic and Topical routes will contain Drug-Drug interactions.

The drug-drug interaction definition itself, the portion between the first (and fourth) ClassInteraction tags, is displayed here.

```
<ClassInteraction>
   <PrescribingInteractionClass name="Trimethoprim"
      reference="{14AA0897-6EB3-4003-A3AF-262EDA2AE7A2}"
      description="Antibacterial of the diaminopyrimidine class.
      Trimethoprim reversibly inhibits the enzyme dihydrofolate
      reductase therefore interfering with the bacterial
      biosynthesis of nucleic acids and proteins. It is most
      commonly used for the treatment of urinary tract
      infections.">
      <PrescribingMolecule name="trimethoprim"
         reference="{7142A19F-EDE2-4A7C-A30B-06AB60B6B71F}" />
   </PrescribingInteractionClass>
   <InteractionClass
      name="Angiotensin converting enzyme (ace) inhibitors"
      reference="{97077317-8EB4-4EE1-8604-5654CF551B85}"
      description="Angiotensin II is a powerful vasoconstrictor
      which also causes the secretion of aldosterone,
      contributing to sodium and fluid retention and potassium
      loss. ACE inhibitors inhibit the enzymatic conversion of
      angiotensin I to angiotensin II. By inhibiting this
      conversion there is a decrease in the pressor substance
      angiotensin II, and also a decrease in the concentration of
      aldosterone found in blood and urine.">
      <Molecule name="trandolapril"
      reference="{C649F3E7-B7AF-4803-A277-417CB94D41E1}" />
   </InteractionClass>
   <Severity name="Moderate" ranking="4">Moderate</Severity>
   <Likelihood name="Rare">Rare</Likelihood>
   <Observation>
      <Abbreviated>increases toxicity of</Abbreviated>
      <Full>increases toxicity of</Full>
   </Observation>
   <Interaction>
      <Abbreviated>Severe hyperkalaemia has been noted in patients
        given trimethoprim (or co-trimoxazole) together with an ACE
        inhibitor.</Abbreviated>
      <Full />
   </Interaction>
   <Precaution>
      <Abbreviated>monitor serum potassium</Abbreviated>
      <Full>ask your doctor if blood potassium tests are
        needed</Full>
   </Precaution>
</ClassInteraction>
```


This shows that the first product, Alprim, is related to the drug interaction class "Trimethoprim" as indicated by the PrescribingInteractionClass element. This relationship is because Alprim contains the molecule "trimethoprim" as indicated by the PrescribingMolecule element. It then shows the classification of the interacting product, Odrik, in the InteractionClass element.

The Severity and Likelihood elements describe the interaction between these two drug classes. The ranking attribute value is larger for more severe interactions. The range is from 1- Not Clinically Significant to 5- Severe.

The Observation element provides text that forms a sentence when combined as Prescribing Interaction Class name, observation then Interaction Class name. So this result describes the interaction "Trimethoprim increases toxicity of Angiotensin converting enzyme (ace) inhibitors."

The Interaction element provides descriptive content about the nature of the interaction. Currently, the Full element will be empty and the Abbreviated element contains the only available description. In future releases, this description will move to the Full element and a truly abbreviated description will be in the Abbreviated element.

Drug Drug interactions also may provide a variable number of Precaution elements. Each Precaution element will contain an Abbreviated and a Full element.

2.8.11.2 Drug-Health Issue Result (HealthAlert)

After all drugs have been checked for interactions against the current drug, Interactions are checked for the Health Issues listed in the request. Regardless of the existence of an interaction with a specific Health Issue, each Health Issue in the request will be identified in the result as a child of each Route element for each top-level drug. If there is an interaction for that drug-health issue combination, the HealthIssue element will contain a ClassInteraction element.

Only the Systemic and Topical routes will contain Drug-Health Issue interactions.

This is the second pair of ClassInteraction tags in the example above and its contents are displayed here.

```
<ClassInteraction>
   <PrescribingInteractionClass name="trimethoprim"
      reference="{A0E1ECDC-7D4C-57C9-E034-080020E1DD8C}"
      description="Antibacterial of the diaminopyrimidine
      class.">
      <PrescribingMolecule name="trimethoprim"
      reference="{7142A19F-EDE2-4A7C-A30B-06AB60B6B71F}" />
   </PrescribingInteractionClass>
   <Severity name="Extreme Caution" ranking="2" />
   <Documentation name="Good" ranking="3">Good</Documentation>
   <Interaction>
      <Abbreviated>Care should be taken in patients with moderate to
        severe renal impairment and doses generally should be
        reduced. If the creatinine clearance is between 15 and 27 mL
        per minute, the normal dose may be given for 3 days but
        should then be reduced to one-half. If the creatinine
        clearance is below 15 mL per minute, half the normal dose
        should be employed from the start of the treatment period.
        Plasma concentrations should be monitored in patients with
        severe renal impairment.</Abbreviated>
      <Full />
   </Interaction>
</ClassInteraction>
```

Similar to the drug-drug interaction results, the drug-health issue interaction contains a PrescribingInteractionClass element that identifies which class the drug is assigned to (PrescribingInteractionClass) and which molecule in the drug is responsible for that assignment (PrescribingMolecule).

The interaction also contains a Severity and Documentation which use the same possible values and rankings as the drug-drug interaction results.

The Interaction element contains the same sort of information as the corresponding element in the Drug-Drug result, and similarly, the Full element is empty now, but will be populated in a future release.

2.8.11.3 Drug-Health Issue Code Result (HealthAlert)

This result is structured identically to the Drug-Health Issue result, and in fact, FastTrack translates the supplied Health Issue Code into a GDDB Health Issue reference before performing the interaction checks.

Only the Systemic and Topical routes will contain Drug-Health Issue Code interactions.

This is the sample content of the third pair of ClassInteraction tags.

```
<ClassInteraction>
   <PrescribingInteractionClass name="Paraldehyde"
        reference="{893B19AB-0A0D-4B53-9835-757DF554964A}"
        description="Sedative, hypnotic. ">
      <PrescribingMolecule name="paraldehyde" reference="{67F69F02-
        5B3E-4704-9B0E-0F0BA7084A92}"/>
   </PrescribingInteractionClass>
   <Severity name="Extreme Caution" ranking="2"/>
   <Likelihood name="Good" ranking="3">Good</Likelihood>
   <Documentation name="Good" ranking="3">Good</Documentation>
   <Interaction>
      <Professional>Paraldehyde is associated with pulmonary oedema
        and hyperventilation. In addition, unmetabolised paraldehyde
        is excreted via exhalation.</Professional>
      <Consumer/>
   </Interaction>
   <ConsumerNote>None</ConsumerNote>
</ClassInteraction>
```

2.8.11.4 AllergyAlert

Allergies are checked and returned in the following order regardless of the order of elements in the request.

1) to Substance Classes
2) to Molecules
3) to Active Compositions
4) to Active Composition Groups
5) to Products
6) to GGPIs
7) to Generic Items
8) to Specific Items

There are basically five possible structures for allergy warnings.

The first possible structure is when the request indicates an allergy to a Substance Class. In this case the result would be structured as follows.

```
<Allergy>
   <SubstanceClass
      reference="{9E74744A-51AE-1FB4-E034-080020E1DD8C}"
      name="Sulfonamides" >
      <PrescribingMolecule
         reference="{7142A19F-EDE2-4A7C-A30B-06AB60B6B71F}"
         name="trimethoprim"/>
   </SubstanceClass>
</Allergy>
```

The PrescribingMolecule element identifies which molecule in the current, (prescribing or prescribed), drug is in the class identified by the respective Substance Class from the Allergies portion of the request.

The second is from the example above, (the sixth and seventh missing blocks from the result skeleton). This is an allergy warning due to the patient being allergic to a molecule in the same substance class as some molecule in the current drug being evaluated.

```
<Allergy>
   <Molecule reference="{7142A19F-EDE2-4A7C-A30B-06AB60B6B71F}"
      name="trimethoprim">
      <SubstanceClass
         reference="{59E6DC7E-CD79-4629-8E98-42E1D56B6B06}"
         name="Trimethoprim" />
         <PrescribingMolecule
            reference="{7142A19F-EDE2-4A7C-A30B-06AB60B6B71F}"
            name="trimethoprim" />
      </SubstanceClass>
   </Molecule>
</Allergy>
<Allergy>
   <Molecule reference="{7142A19F-EDE2-4A7C-A30B-06AB60B6B71F}"
      name="trimethoprim">
      <SubstanceClass
         reference="{9E74744A-51AE-1FB4-E034-080020E1DD8C}"
         name="Sulfonamides" />
         <PrescribingMolecule
            reference="{7142A19F-EDE2-4A7C-A30B-06AB60B6B71F}"
            name="trimethoprim" />
      </SubstanceClass>
   </Molecule>
</Allergy>
```

In this example, there are two allergy warnings for the product Alprim, and both are due to the patient being allergic to the molecule trimethoprim. This is because trimethoprim is in both the Trimethoprim Substance Class and the Sulfonamides Substance Class. The PrescribingMolecule element identifies which molecule in Alprim is causing the allergy warning to appear.

The third possible structure is for allergies to drugs identified by elements other than Substance class or Molecule. In these cases, there may be two molecules involved in the allergy warning. One molecule from the drug the patient is allergic to and one molecule from the drug in the prescribing or prescribed lists. In this case, the result would be structured as follows.

```
<Allergy>
   <ActiveComposition
      reference="{18864A78-8A60-469E-AE29-5D5B27024E9F}"
      name="Trimethoprim"
      <SubstanceClass
         reference="{9E74744A-51AE-1FB4-E034-080020E1DD8C}"
         name="Sulfonamides" >
         <PrescribingMolecule
            reference="{7142A19F-EDE2-4A7C-A30B-06AB60B6B71F}"
            name="trimethoprim"/>
         <AllergyMolecule
            reference="{7142A19F-EDE2-4A7C-A30B-06AB60B6B71F}"
            name="trimethoprim"/>
      </SubstanceClass>
   </ActiveComposition>
</Allergy>
```

This example shows the same molecule twice, but the structure is the same for different Molecules in the same Substance Class. The AllergyMolecule element identifies which molecule in the "allergy" drug is in the substance class. The PrescribingMolecule element identifies which molecule in the prescribing or prescribed drug is in the substance class.

The fourth possible structure is to support "molecule self-check" allergy warnings. This is for checking allergies to drugs containing molecules that have not been assigned to a substance class, but are the same molecules as those being prescribed. For instance, if the patient says they have an allergy to kelp, then the request may indicate an allergy to that molecule. If they then are prescribed some product containing kelp, the allergy warning will be returned even though kelp is not assigned to any substance class. The structure of this kind of result is as follows

```
<Allergy>
   <Molecule reference="{7142A19F-EDE2-4A7C-A30B-06AB60B6B71F}"
      name="trimethoprim"/>
</Allergy>
```

The last possible structure is to support "molecule self-check" for allergies to drugs other than molecules. In this case, if a prescribing or prescribed drug contains the same molecule as an allergy drug and that molecule is not assigned to any substance class, the allergy result will be returned. This structure is as follows.

```
<Allergy>
   <ActiveComposition
      reference="{18864A78-8A60-469E-AE29-5D5B27024E9F}"
      name="Trimethoprim"
      <Molecule
         reference="{7142A19F-EDE2-4A7C-A30B-06AB60B6B71F}"
         name="trimethoprim"/>
   </ActiveComposition>
</Allergy>
```

It is important to note that there may be several Allergy nodes for a "single" warning. An Allergy node will appear for a drug for each molecule-substance class combination that applies. In the worst case, the molecules from the "Allergy" drug and the molecules from the "Prescription" drug are all different and they all participate in several overlapping substance classes. Maybe there are three molecules in the Allergy drug and two in the Prescription drug. If all five of these molecules are in one substance class, then there will be six Allergy nodes in the result. If all these molecules are each in two substance classes, then there will be twelve Allergy nodes.

2.8.11.5 DuplicateTheraphy

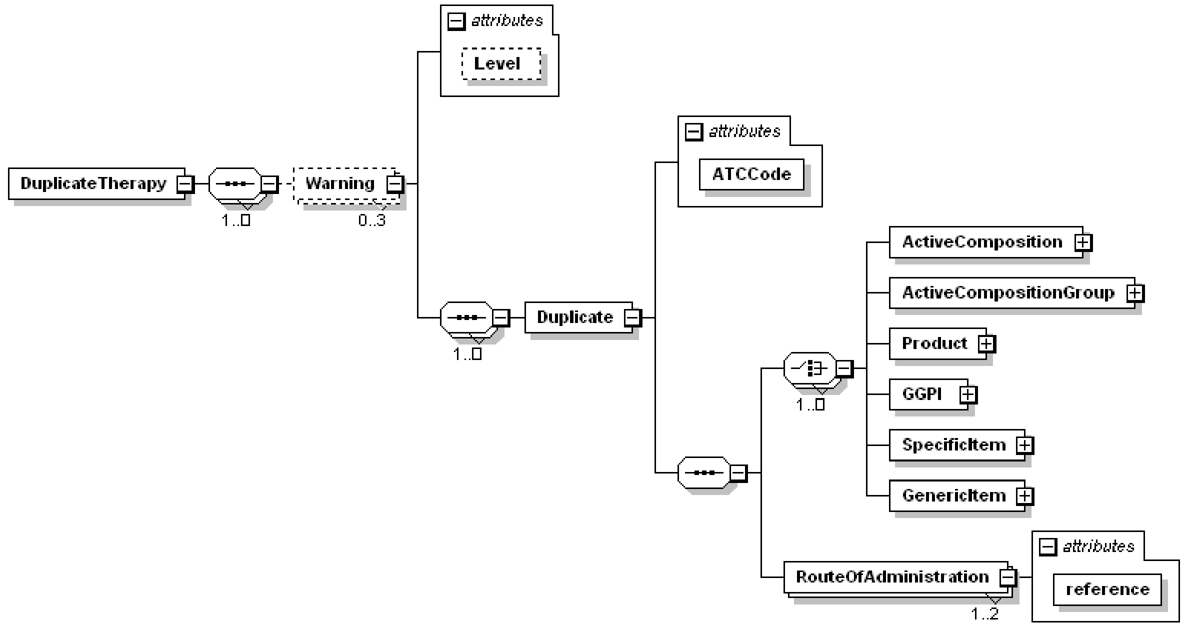


Figure 43 – Duplicate Therapy Result

In this particular example FastTrack return some DuplicateTherapy in interaction result:

```
<Result buildID="1905" copyright="MIMS" expiryDate="9/9/2012">
   <Interaction>
      <GenericItem reference="{F37499E7-F2AE-48FB-9C5E-64554FD46BBF}"
         name="warfarin Oral Tablet" />
      <GGPI reference="{B4356008-1D07-482E-E034-080020E1DD8C}"
         name="warfarin sodium 1mg Oral Tablet" />
      <DuplicateTherapy>
       <Warning Level="1">
        <Duplicate ATCCode="b01aa03">
         <GenericItem
            reference="{F37499E7-F2AE-48FB-9C5E-64554FD46BBF}" />
         <GGPI
            reference="{B4356008-1D07-482E-E034-080020E1DD8C}" />
         <RouteOfAdministration
            reference="{90075601-10D4-49DD-9292-9B884D9D3EF3}" />
        </Duplicate>
       </Warning>
      </DuplicateTherapy>
   </Interaction>
</Result>
```

2.8.11.6 DuplicateIngredient

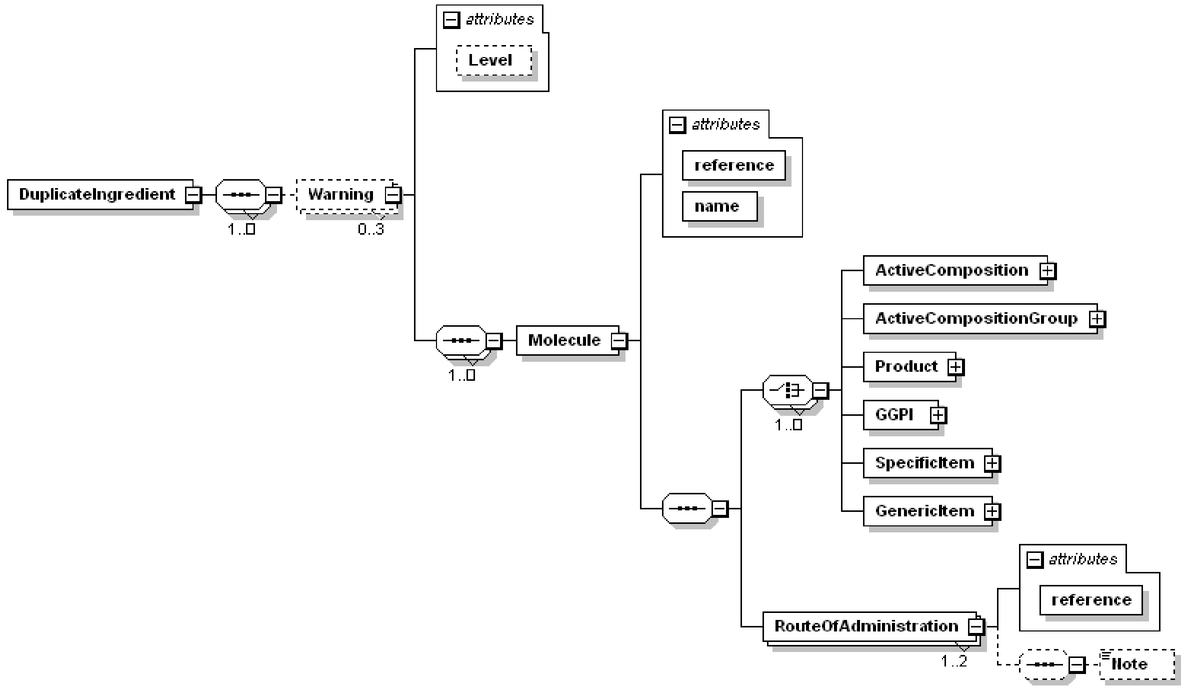


Figure 44 – Duplicate Ingredient Result

Using the same example as DuplicateTherapy, FastTrack returns the following DuplicateIngredient section in the interaction result:

```
<Result buildID="1905" copyright="MIMS" expiryDate="9/9/2012">
 <Interaction>
  <GenericItem reference="{F37499E7-F2AE-48FB-9C5E-64554FD46BBF}"
       name="warfarin Oral Tablet" />
  <GenericItem reference="{F37499E7-F2AE-48FB-9C5E-64554FD46BBF}"
       name="warfarin Oral Tablet" />
  <DuplicateTherapy />
  <DuplicateIngredient>
    <Warning Level="1">
      <Molecule reference="{4474F71F-26B0-411B-A3A2-16537132387F}"
           name="warfarin">
        <GenericItem
           reference="{F37499E7-F2AE-48FB-9C5E-64554FD46BBF}" />
        <GenericItem
           reference="{F37499E7-F2AE-48FB-9C5E-64554FD46BBF}" />
        <RouteOfAdministration
           reference="{90075601-10D4-49DD-9292-9B884D9D3EF3}" />
      </Molecule>
    </Warning>
  </DuplicateIngredient>
 </Interaction>
</Result>
```

2.8.11.7 Pregnancy, WOCBA and Lactation

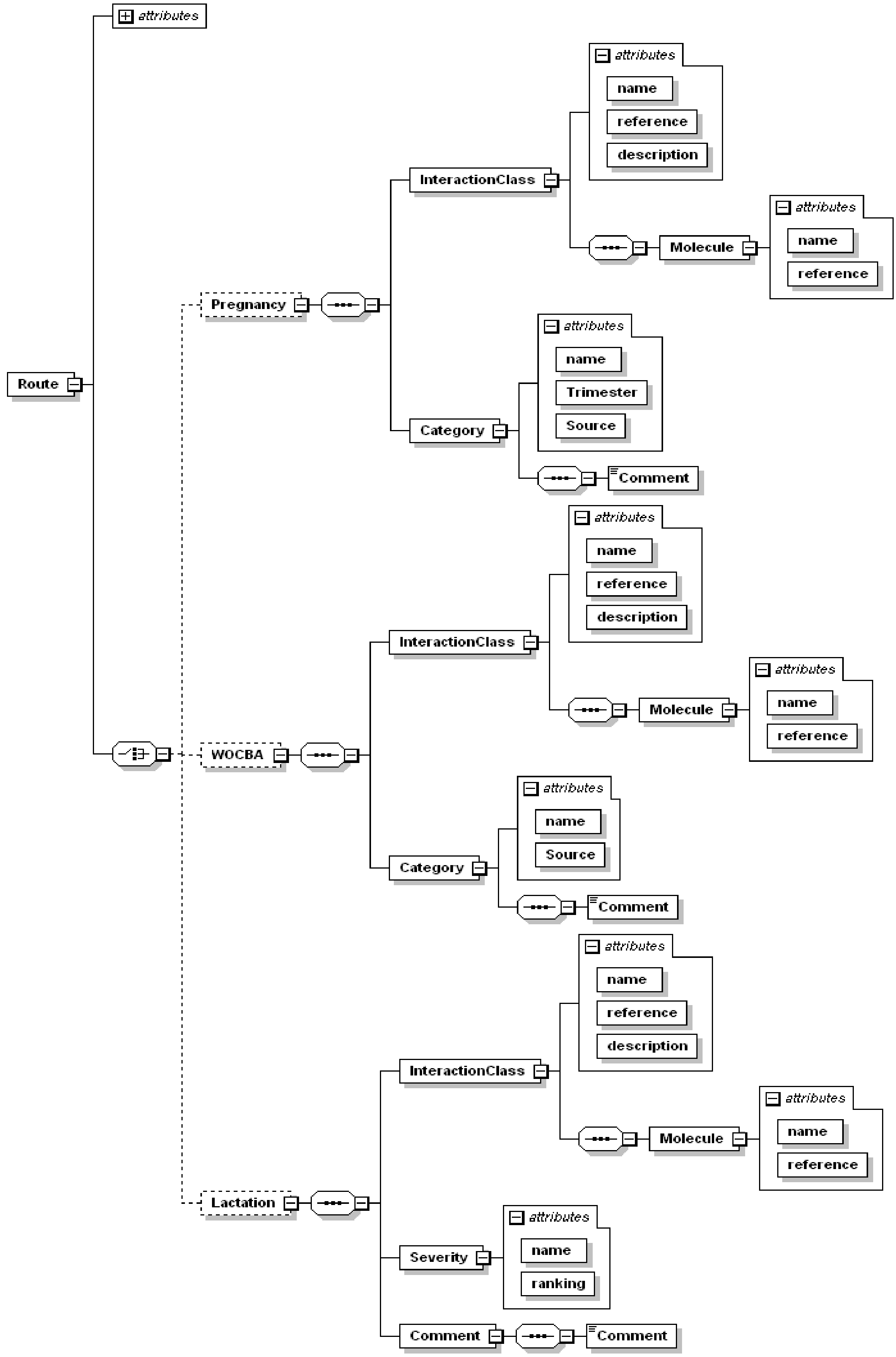


Figure 45 – Lactation, Pregnancy and Women of Childbearing Age Result

An example of FastTrack pregnancy, WOCBA and lactation alerts:


```
<Result buildID="1905" copyright="MIMS" expiryDate="9/9/2012">
  <Interaction>
    <GenericItem reference="{F37499E7-F2AE-48FB-9C5E-64554FD46BBF}"
     name="warfarin Oral Tablet">
      <Route name="Oral"
       reference="{AD02A5E3-6437-12B1-E034-080020E1DD8C}">
                  <Pregnancy>
                   <InteractionClass
                    name="Warfarin "
                    reference="{77350912-72DC-4387-9274-FC440FD7372A}"
                    description="Warfarin ">
                     <Molecule name="warfarin"
              reference="{4474F71F-26B0-411B-A3A2-16537132387F}"/>
                   </InteractionClass>
                   <Category name = "D"
                   Trimester = "1st"
                   Source ="ADEC">
                    <Comment>Warfarin has been associated with the
                           Development of a specific embryopathy
                           Following exposure at 6 to 9 weeks
                           postconception. Exposure following first
                           trimester of pregnancy can cause fetal
                           bleeding leading to CNS damage.
                           There is also an increased risk
                     </Comment>
                    </Category>
                  </Pregnancy>
                 <WOCBA>
                    .
                    .
                    .
                 </WOCBA>
                 <Lactation>
                    .
                    .
                    .
                 </Lactation>
                 <ClassInteraction>
                           .
                           .
                           .
                 </ClassInteraction>
          </Route>
       </GenericItem>
       .
       .
       .
    <DuplicateTherapy />
  </Interaction>
</Result>
```

2.8.11.1 DoseCheck and Warnings

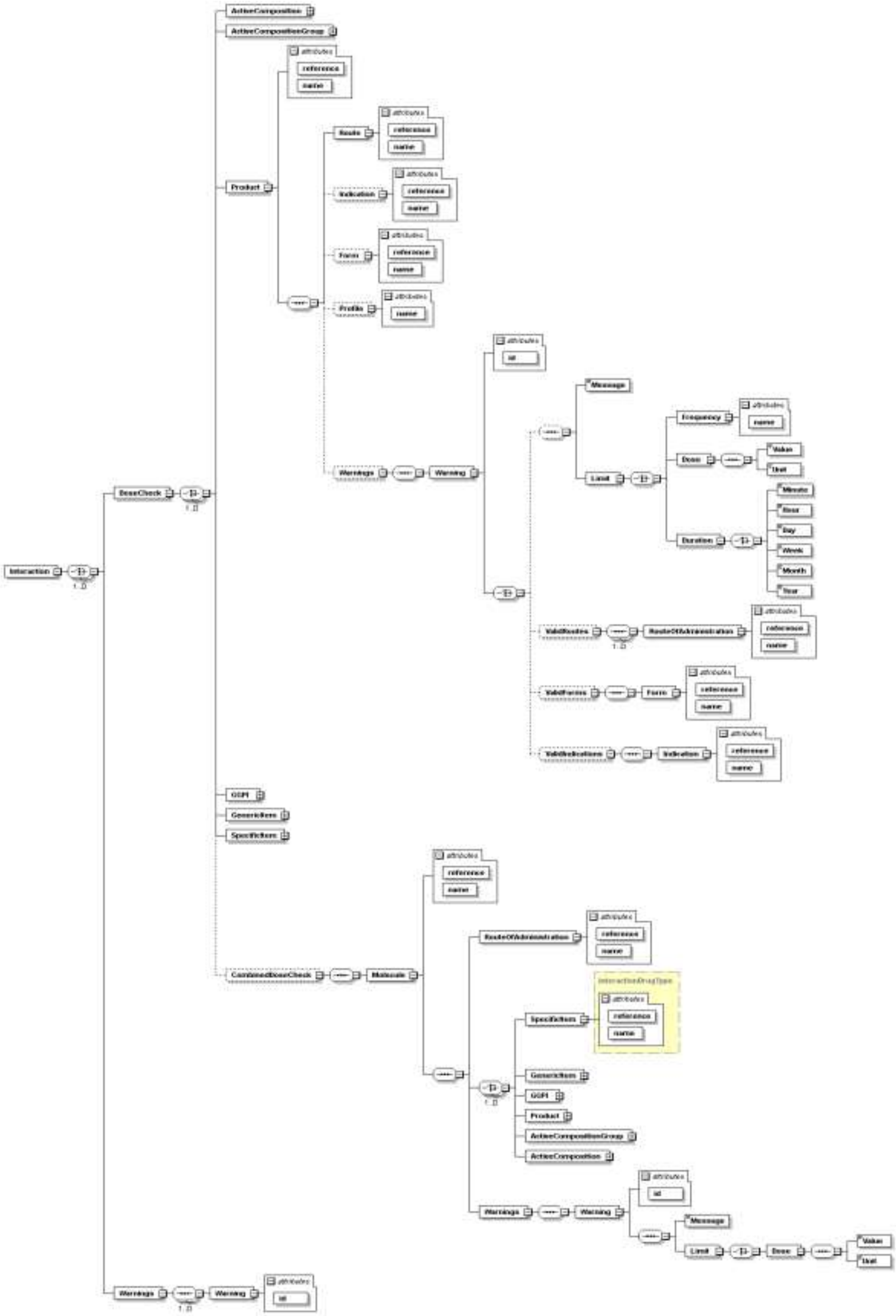


Figure 46 – DoseCheck and Warnings Result

Child Warnings element of DoseCheck in drug item appear in case if some dose check warning cases occur for current prescription during interaction. CombinedDoseCheck with Warnings element appear if two or more prescribed item has similar molecule ingredient and also occur if dose check warning cases.

Child Warnings element of Interaction appear in case if some FastTrack API warning cases occur for current situation per transaction.

Profile element shows name of profile which is matched to request.

This is example xml result for DoseCheck and Warnings elements:


```xml
<Result buildID="1905" copyright="MIMS" expiryDate="9/9/2012">
<Interaction>
    <Product reference="{C68A87FB-3372-4948-9A35-BF4A5BDE5315}"
name="Propecia tab 1 mg">
       .
       .
       .
    </Product>
    <DoseCheck>
      <Product reference="{C68A87FB-3372-4948-9A35-BF4A5BDE5315}"
name="Propecia tab 1 mg">
        <RouteOfAdministration reference="{E5B64CDE-EBAE-4DBF-ABAC-
64FAE8561DE1}" name="Oral" />
        <Profile name="finasteride[Adult ]" />
        <Warnings>
          <Warning id="IDS_DOSE_MAX_ABSOLUTE_WARNING">
            <Message>Overdose! Absolute maximum daily dose must be
equal or less than
 60 mg/m2</Message>
            <Limit>
              <Dose>
                <Value>60</Value>
                <Unit>mg/m2</Unit>
              </Dose>
            </Limit>
          </Warning>
          <Warning id="IDS_DOSE_ADJ_HEPATIC_IMPAIRMENT_WARNING">Refer
to PI for dose adjustment in hepatic impairment.
       </Warning>
    </Warnings>
      </Product>
    </DoseCheck>
    <DuplicateTherapy />
  </Interaction>
</Result>
```


2.8.11.2 Variations

It should also be clear that not all interactions need to be checked in one request. For instance, only the Drug-Allergy warnings can be retrieved by simply excluding all but one drug from the Prescribing element, and excluding the Prescribed, HealthIssues, HealthIssueCodes. Only those elements present in the request will be returned in the result as long as either the RouteOfAdministration element is included for the drugs, or there is some interaction information to return.